

भारत सरकार / GOVERNMENT OF INDIA
खान मंत्रालय / MINISTRY OF MINES
भारतीय भूवैज्ञानिक सर्वेक्षण
GEOLOGICAL SURVEY OF INDIA
मिशन -II / Mission – II
मध्य क्षेत्र / Central Region



राज्य इकाई : मध्य प्रदेश, जबलपुर / State Unit: Madhya Pradesh, Jabalpur
परियोजना: खनिज अन्वेषण / Project Name: Mineral exploration
मध्य प्रदेश के बालाघाट जिले में पश्चिमी उकवा खंड में मैंगनीज खनिजीकरण के निक्षेप पर
अंतिम प्रतिवेदन । (स्तर जी- 2)
धरातल पत्रांक ६४सी/०५
मद क्रमांक: ०५०/एम.ई./सी.आर./एम.पी./२०१६/४३
कार्य सत्र: २०१६-१७

Final report on general exploration for manganese mineralization in western block of Ukwa,
Balaghat district, Madhya Pradesh (Stage G-2)
Toposheet Nos. 64C/05
Field season: 2016-2017
Item no. 050/ME/CR/MP/2016/43

द्वारा
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केशव खंडेलवाल, भूवैज्ञानिक

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November 2017

मध्य प्रदेश के बालाघाट जिले में पश्चिमी उकवा खंड में मैंगनीज खनिजीकरण के निक्षेप पर अंतिम

प्रतिवेदन

**FINAL REPORT ON GENERAL EXPLORATION FOR MANGANESE
MINERALIZATION IN WESTERN BLOCK OF UKWA, BALAGHAT DISTRICT,
MADHYA PRADESH**

ITEM NO. 050/ME/CR/MP/2016/43

Field Season: 2016-17

By

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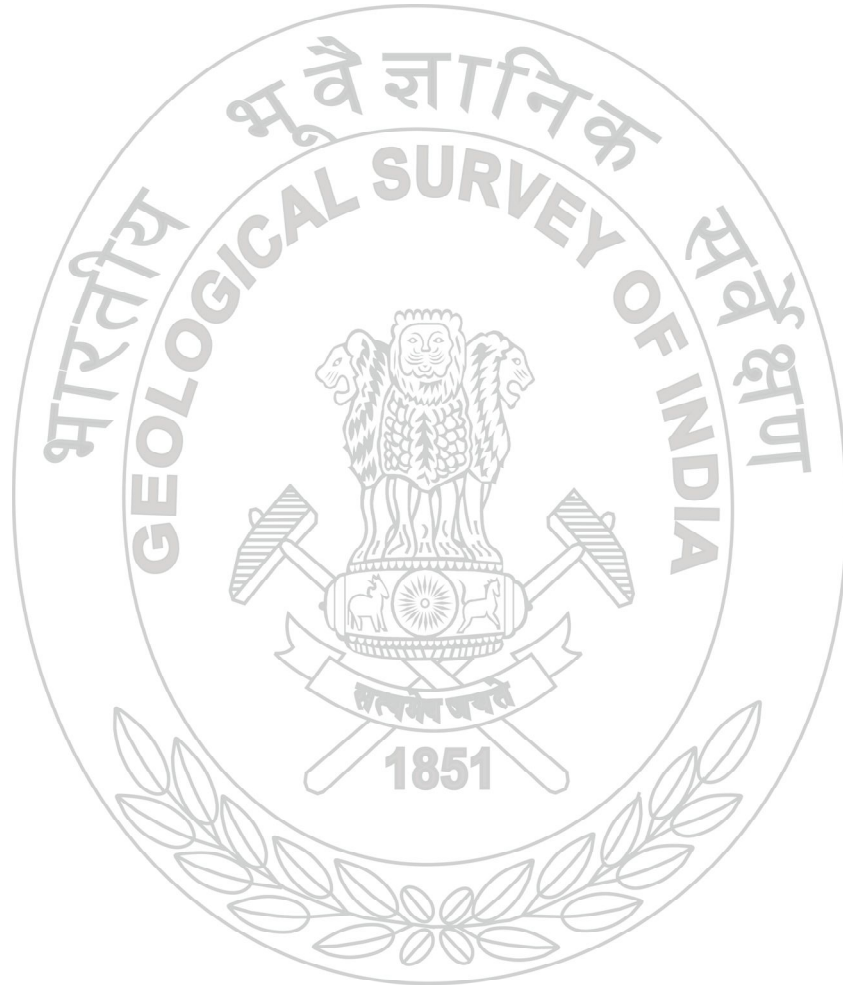
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हिन्दी सारांश

भारतीय भूवैज्ञानिक सर्वेक्षण के कार्य सत्र 2016-17 में मद संख्या 050/ME/CR/MP/2016/43 के अन्तर्गत उकवा बालाघाट में अन्वेषण का कार्य किया गया। कार्य का मुख्य उद्देश्य उकवा निक्षेप में मैंगनीज अयस्क का अयस्क रिजर्व बढ़ाने के लिए मूल्यांकन करना था। अन्वेषण कार्य में 1071.35 मी० वेधन किया गया तथा 51 सं० कोर नमूनें, 10 सं० पेट्रोकेमिकल नमूनें, 10 पेट्रोलोजिकल नमूने, 20 बी०आर०एस०, 10 सं० आर०ई०ई० नमूनें 5 सं० एक्स आर एफ नमूनें एकत्रित किया गया। चौधरी आदि 1979 एवं खान व फूलमारी आदि 1996 के द्वारा किये गये कार्यों का डाटा वर्तमान अन्वेषण कार्य के लिये उपयोग किया गया है एवं आगे उकवा अयस्क रिजर्व अनुमान के लिए किया जायेगा। निक्षेप में वर्तमान अन्वेषण में उत्पन्न डाटा के अलावा यह डाटा है।

अन्वेषण कार्य से पता चलता है कि इस क्षेत्र में प्रीकैम्ब्रीयन आयु की षिला ओल्डर मेटामोर्फिक नाईस व सौसर मेटासेडिमेन्ट्रीस के रूप में है। दोनों का क्षेत्रीय नतिलम्ब ENE-WSW से NE-SW है, व नति NW में मध्यम कोण में है। इस क्षेत्र में विरूपण के कम से कम चार चरण है। पुराने मेटामोर्फिक एवं सौसर के बीच में विवर्तनीकी सम्पर्क है। इस क्षेत्र में मैंगनीज खनिजीकरण सौसर सिनसेडिमेन्ट्री-स्ट्राटाबाउण्ड एवं सौसर संघ से जुड़ा है, संभवतः लोहांगी से मंसर तक है।

पूर्व कार्यानुसार उकवा निक्षेप 480 मी० पूर्वी अंत को छोड़कर 5400 मी० नतिलम्ब की दिशा में प्रमाणित है, अतः कार्य योग्य नतिलम्ब 4920 मी० है, पूर्वी क्षेत्र में अयस्क होराईजन 300 मी० तक गहरा है, जबकि अन्य में यह 200 मी० तक है। उकवा खदान में उपस्थित दृष्ट्यांश से डिप लेन्थ 700 मी० तक मापा गया है। अयस्क होराईजन की तलहटी नहीं देखी गई है। अयस्क होराईजन की मोटाई 15 सेमी० से 10 मी० तक है एवं औसतन मोटाई 3.58 मी है। हालांकि वर्तमान में किये गये वेधन में यह 0.15 मी० से 3.61 मी० पाई गयी है। लिथोलोजिकली, अयस्क होराईजन का बाह्य भित्ती एवं अतः भित्ती के साथ तीक्ष्ण सम्पर्क है। संरचनात्मक दृष्टि से अयस्क निक्षेप किसी भी जटिलताओं से मुक्त है। अयस्क निक्षेप केवल पिंच एवं स्वैल दिखलाता है। पिंचिंग उकवा निक्षेप के दोनो साईड नतिलम्ब की दिशा के साथ ही कुछ स्थानों में नति कि दिशा में है। कलमिनेशन एवं डिप्रेषन के कारण पिंचिंग मिलती है।

पूर्व में किये कार्यों के विप्लेषण परिणामानुसार उकवा निक्षेप में औसतन मैंगनीज 41% एवं फास्फोरस 0.23% है। निक्षेप के ज्यादातर भाग में Mn 35 to 45% मध्यस्तर से नीचे है। पश्चिमी ब्लॉक में फास्फोरस बढ़ता है तथा गहराई के साथ अन्य में भी बढ़ता है।

पश्चिमी ब्लॉक में वेधन कार्य 09/01/2017 की प्रारम्भ हुआ एवं 26/03/2017 को समाप्त हुआ। कुल 1071.35 मी० वेधन किया गया जिसमें 6 बोरहोल मेग्नीज होराइजन की द्वितीय एवं तृतीय लेवल पर इंटरसेक्ट करने के लिए किये गये। अधिकतम मोटाई 3.61 मी० के साथ, 6 बोर होल (UKBH-1,2,3,4,5,6) में कुल 10.01मी० मोटा अयस्क होराइजन मिला है एवं न्यूनतम मोटाई UKBH-1 में 0.15 मी० पाई गयी है। कुल 51 स. कोर नमूनें, 20 सं. बी.आर.एस. नमूने, 10 सं० आर ई ई नमूने, 10 सं० पी.एस. नमूनें, 10 सं० पी.सी.एस. नमूनें एवं 05 एक्स आर एफ नमूने एकत्रित किये गये एवं विप्लेषण किये गये। ओपन काष्ट खदान एवं अन्डरगाउन्ड खदान की अयस्क का भूसतह में व्यवहार देखने के लिए भ्रमण किया गया।

फिल्ड कार्य के दौरान रीजनल सेटप एवं संरचना समझने के लिए क्षेत्र के चारों ओर क्षेत्रीय भ्रमण किया गया। कोर लॉगिंग व भूभौतिकीय लॉगिंग की गई। 05 सं० जल नमूनें एकत्रित किये गये एवं विप्लेषण किये गये। ब्लॉक के सीमांकन के लिए DGPS से सर्वेक्षण किया गया।

मौजूदा ब्लॉकों में तत्वों के प्रकीर्णन के अध्ययन के लिये माईन एवं रोड सेक्सन में उपस्थित सभी षिलाओं से नमूनें एकत्रित किये गये हैं। बी.आर.एस. नमूनों में Mn, P, Si, Fe, Cu, Pb, Zn, Co, Ni, Cr. का विप्लेषण किया गया तथा कार्यान्तरण के पर्यावरण व प्रोटोलिथ जानने के लिए PCS का भी विप्लेषण किया गया। 50 सं० कोर नमूनें, 10 सं० पेट्रोकेमिकल नमूनें, 10 पेट्रोलोजिकल नमूने, 20 बी०आर०एस०, 10 सं० आर ई ई नमूनें एवं 05 सं० एक्स आर एफ नमूनें एकत्रित किये गए हैं। कोर BRS, PCS, PS, REE का परिणाम प्राप्त हो चुका है एवं XRF का परिणाम आना बाकी है।

BRS नमूनों में Mn का मान 56.241% से 3.11% है, फास्फोरस का 0.04 से 1.67%, आयरन का 14.196 से 1.31% तथा क्वार्टज के कारण Si का मान 41.03 से 2.832% तक है। Cu का मान 550 PPM से 55 PPM तक तथा एक नमूनें में यह 0.1% तक है। Cu, Pb, Zn, Ni, Co, एवं Cr एक जैसी आयनिक त्रिज्या के कारण सिमपैथेटिक संबंध दिखलाते हैं, तथा Pb, Zn, Ni, Co, Cr का मान Cu के समान है। (Annx-1)

कोर नमूनों को सम्पूर्ण अयस्क होराइजन से एकत्रित किया गया है (अयस्क एवं वेस्ट), एवं नमूनों का विवरण Annx-2 में संलग्न है। यहाँ Mn का मान 5.0494 से 39.7298% तक है। अयस्क रिजर्व गणना के लिये पुराने बोरहोलों का डाटा वर्तमान डाटा के साथ जोड़ा गया एवं ग्रेड 35.138% पाया गया है। इसके अनुसार पिछले कार्य में वर्णित Mn विषिष्ट गुरुत्वाकर्षण 3.55 का अयस्क रिजर्व गणना के लिए उपयोग किया गया है। क्रास सेक्सन से अयस्क रिजर्व 2.182 मिलियन टन एवं LV सेक्सन से 2.013 मिलियन टन अयस्क रिजर्व प्राप्त हुआ है।

SUMMARY

The exploration for manganese ore in Ukwa was carried out in pursuance of FSP Item No. 050/ME/CR/MP/2016/43, of the annual field season programme of the Geological Survey of India, Central Region for the year 2016-17. The objective of the investigation was assessment of manganese ore potential in Ukwa deposits for augmentation of ore reserves. The exploration involved drilling of 1071.35 m, core sampling of 51 nos, petrochemical sampling of 10 nos, petrological studies of 10 samples, bed rock sampling of 20 nos. REE 10 nos. and XRF 5 nos. The data of the exploration carried out by Choudhary et al., 1979 and Khan and Fulmari, 1996 were taken into consideration for planning the present exploration and have been utilised for estimation of resource and grade of the Ukwa deposit in addition to the data generated during present work.

The exploration work reveals that the area exposes rocks of Precambrian age in the form of older metamorphic gneisses and Sausar metasedimentaries. Both the sequences have a regional strike of ENE-WSW to NE-SW and dip at low to moderate angles towards NW. The area has undergone at least four phases of deformation. The contact between older metamorphics and Sausars is tectonic. The manganese mineralisation in the area is synsedimentary-stratabound and is associated with the Sausar Group of rocks, probably transgressing from Lohangi to Mansar Formations.

As per earlier workers data the Ukwa deposit has been proved to extend over a strike length of 5400 m leaving 480 m long thin ore horizon at the eastern end of the deposit, therefore the workable strike length measures 4920 m. The ore horizon continues at depth for more than 300 m vertically in the eastern part and more than 200 m in rest of the deposit. In terms of dip length it measures up to 700 m from the surface exposure in Ukwa mine. Bottoming of ore horizon has not been observed. Thickness of ore horizon ranges between <15 cm to about 10 m, the average being 3.58 m. However during current drilling, thickness of the ore horizon observed is ranging between 0.15 m to 3.61 m. Lithologically, ore horizon constitutes a bed having sharp contacts with the footwall and hanging wall rocks. Structurally the ore deposits are free from any complexities except pinch and swell structures. Pinching on the either ends of Ukwa deposit is

recorded along the strike. At places, along dip direction ore horizon shows pinching due to culmination and depression by interference of deformations.

According to earlier workers analytical results, the average Mn content of the Ukwā deposit is 41.23% and P is 0.23%. Major part of the deposit falls under medium grade ore having 35 to 45% Mn. Phosphorus content increases in Western block and also at depth in other blocks.

Drilling in western Ukwā block was started on 09.01.2017 and completed on 26.03.2017. A total of 1071.35 m of drilling was done in 6 nos. of boreholes were drilled at second and third level to know strike and depth continuity of manganese horizon. The Cumulative thickness of ore horizon (ore plus waste rock) intersected in six boreholes (UKBH-1, 2,3,4,5 & 6) was 10.01m. The maximum thickness of ore horizon was observed in UKBH-3(3.61 m) and minimum thickness in UKBH-1(0.15 m). Open cast as well as underground mines were visited to understand about sub-surface behaviour of body. The surrounding area was also investigated during field work along with regional traverses to understand tectonic as well as regional geological setup of area. Core Logging along with geophysical logging of each boreholes were done. DGPS survey was carried out to demarcate borehole points along with block boundary.

To study the dispersion of elements in current block, samples were collected from mine sections as well as from road sections to study their chemical composition of all lithologies available in area. Elemental analysis of Mn, P, Si, Fe, Cu, Pb, Zn, Co, Ni and Cr were analysed in BRS and whole rock analysis were carried out in PCS to understand about protoliths and environmental conditions of metamorphism. A total of 51 nos. core samples, 20 nos. BRS, 10 nos. REE, 10 nos. PCS and 5 nos. of XRF samples were analysed. A total of 5 nos. of water samples were also collected and analysed. Results of XRF are yet to receive.

BRS samples indicate Mn value ranging from 3.11% to 56.241%, with P value ranging in from 0.04 to 1.67 %. Values of Fe vary from 1.31% to 14.196%. Due to high amount of quartz Si value is also high and varies from 2.832% to 41.03%. The elements like Cu, Pb, Zn, Ni, Co, Cr show sympathetic relations due to similar ionic radius. The Cu values range from 55 ppm to 850 ppm with one sample showing value upto 0.1%. Core samples were collected from whole ore horizon (ore plus waste rock). Here value of Mn found to be varying from 5.0494 to 39.728%. 80 kg of Sample from core is submitted to IBM for beneficiation studies. Results of beneficiation studies are yet to receive.

For resource estimation; data of previous boreholes is combined with current data to find out the weighted average of current block and it is found to be 35.138%. Based on this grade specific gravity as described by previous workers on this range is 3.55 (90% of actual value considering 10% loss) is used and resource estimated is thus 2.182 mt by cross section method and 2.013mt by LV section method at 35.138% of Mn. Here resource estimation of current area is also determined by previous workers sections using their LV section and previous chemical values; based on which resource was found to be 3.597 mt at 40.34% of Mn (Annexure-VIII and IX and Plate-V). Category of the resource as per UNFC is 332.



INTRODUCTION

1. Introduction

The item titled “General Exploration for manganese mineralization in western block of Ukwa, Balaghat district, Madhya Pradesh.”(G-2) was taken up by Central Region, State Unit: Madhya Pradesh, Jabalpur during the Field Season 2016-17 vide FSP Item No. 050/ME/CR/MP/2016/43, with an objective of assessing of manganese ore potential in Ukwa deposits for augmentation of ore reserves. The exploration commenced in the month of August, 2016 (31.08.2016) and was completed in the month of April, 2017 (13.04.2017).

The main manganese producing belt in the country is associated with the Sausar Group of rocks of Precambrian age. The group is characterised by lower pelitic and upper psammo-pelitic sequences. The manganese ore horizon occurs in the lower part of the Sausars.

The rocks of the Sausar Group extend in northeast-southwest direction in Balaghat district M.P. in the east to west northwest-east southeast in Chhindwara district, M.P. in the west through Bhandara and Nagpur districts of Maharashtra, forming an overall arcuate shape with convexity towards south. The manganese bearing belt stretches over a length of 200 km and is about 25 to 30 km wide in its central part. (After Khan et al 1993-96)

The Ukwa deposit is located, almost, at the eastern end of the belt. The eastern part of the manganese belt, from Bharweli (Balaghat) to Ukwa comprises, mainly, rocks belonging to the lower sequence of the Sausar Group, which is characterised by a regionally uniform strike trend of NE-SW and low to moderate dips towards NW. The ore horizon has conformable relation with the enclosing rocks and can be traced intermittently from Bharweli to Ukwa through the occurrences at Kanhatola, Laugur, Ghondi and Jagantola. (After Khan et al 1993-96)

The Ukwa manganese deposit was discovered by P.N. Bose of Geological Survey of India in 1888- 89(GSI general reports). Geological Survey of India carried out regional exploration by means of large scale mapping, drilling, sampling etc. during 1970-71 to 76-77, in lease free areas

to the east and west of Ukwa deposit and also in contiguous areas of Bharweli deposit (Balaghat).

Choudhary et al, (1979) carried out exploration at Ukwa deposit. They drilled 41 vertical holes (GSU series) involving 5041 m of drilling and estimated gross reserves of 14.71 million tonnes of manganese ore of which 12.50 million tonnes constituted the recoverable ore over a strike length of 4670 m. Devarajan and Hakim (1992) carried out Second Generation Mapping in Bharweli-Ukwa area. Their work gives detailed accounts of geology, structure and metamorphism of the basement gneisses and overlying Sausar meta-sediments.

Khan and Fulmari (1996) carried out exploration work in the area in continuation of the exploration work completed earlier by Choudhary et al, (1979). Large scale mapping on 1:25,000 scale has been carried out in the area between Ukwa and Baputola to have an idea about regional geology and for locating the additional manganese ore horizons if any in the dip direction of Ukwa deposit. The boreholes drilled by Khan and Fulmari (1996) have been marked by UKH series and except UKH 4, all boreholes are drilled in 70° inclination towards southeast.

The Ukwa deposit falls under the jurisdiction of Rupjhar Police Station, Baihar tehsil, Balaghat district, M.P. It is located in the Survey of India toposheet nos. 64B/8 and 64C/5, and is bounded by North latitudes 21°56'30", 22°01'30" and East longitudes 80°23'00", 80°30'00".

For convenience, the entire Ukwa deposit has been divided into three blocks namely Ukwa Eastern, Central and Western blocks (Choudhary et al, 1979). In addition to above, the area between Ukwa western block and Jagantola open cast mine has also been covered by detailed mapping and the area named as Ukwa western extension block.

The eastern block extends over a strike length of 2000 m. The ore horizon crops out at around 650 m elevation at the western end of the block and gradually rises eastward reaching around 700 m at its eastern end. The central block extends over a strike length of 1550 m and forms the western continuation of Eastern block or middle part of the Ukwa deposit. The ore horizon crops out at 630 m elevation at the western end and rises to 650 m at its eastern end. The western block extends over a strike length of 1860 m and forms the western continuation of the Central Block. The ore horizon crops out at 600 and 630 m elevation at its eastern and western ends respectively.

Details of bore holes drilled by GSI, average thickness and percentage of Mn in different blocks of Ukwa are given below in table no. 1:

Block	Mn % (Probable Category 'A')	Average thickness of ore horizon	GSU series
Eastern Block	40.40	3.99 m	21
Central Block	43.14	3.51 m	07
<i>Western Block</i>	40.15	2.98 m	13
<i>Ukwa Deposit</i>		3.58 m	41

Table no. 1: Bore holes drilled by GSI and average thickness and percentage of Mn in different blocks of Ukwa. (After Khan et al; 1996)

Except two boreholes (UKH-11 & 12) drilled at the eastern extremity of the Ukwa deposit, all the boreholes drilled in the area by Geological Survey of India have intersected ore horizon.

Thickness of the ore horizon ranges from <15 cm to about 10 m. The aggregate thickness of manganese ore in Ukwa deposit varies from 0.16 m to 4.85 m, average being 1.76 m. The aggregate thickness of waste rock (i.e. manganiferous quartzite, quartzite with rare carbonate and schist intercalation) ranges from a few cm to about 6.00 m, the average being 1.82 m. The average thickness of ore horizon in Ukwa deposit is 3.58 m, of which 1.76 m (49.16%) constitutes the manganese ore and 1.82 m (50.84%) the waste rock. Thus ore horizon generally contains 50% ore and 50% waste rock. Ukwa deposit has been proved to extend over a strike length of 5400 m. Leaving the 480 m in long thin ore horizon at the eastern end of the deposit, the workable strike length measures 4920 m. The ore horizon has been tested to continue in depth for more than 300 m of vertical depth in the eastern part of the deposit and more than 200 m in rest of the deposit over major part of the strike length. In terms of dip length it measures up to 700 m from the surface exposure of ore horizon in Ukwa mine. Bottom of ore horizon has not been observed.

GSI carried out extensive work in the eastern block of Ukwa deposit by putting 30 boreholes (including GSU and UKH series). Most of the parts of central block fall in the leasehold area. In addition to GSI, Manganese Ore (India) Limited (MOIL) also carried out exploration by means of drilling of 7 boreholes and 11 boreholes in the eastern and central blocks respectively. Chemical and Ferro Alloy Ltd. (CFA) also drilled 11 nos. of bore holes in the central block.

In the western block, GSI only carried out exploration (as on September 2002) and drilled 24 nos. of boreholes which include GSU series and UKH series. The drilling has been carried out along section lines W-1 to W-6 placed perpendicular to the strike of the ore horizon. The spacing

of the section line is 190 m to 260 m. The calculated recoverable resource in the Western Ukwa block is 3.01million tonnes with average Mn of 40.34%. The ore is dark grey coloured, fine grained. Ore microscopic study reveals that the Mn ore comprises braunite, hollandite, bixbyte, hausmannite, haematite, pyrolusite and cryptomelane in varying proportion. Braunite is the most abundant mineral in the ore, followed by hollandite.

The borehole interval of work carried out by Khan and Fulmari (1996) was 260 m and efforts was made to get the uniform picture of the deposit upto 400 m level i.e. vertical depth of 200 m from the surface (600 m R.L. to 400 m R.L.).

With this background, the present work was proposed for 8 boreholes in the western block of the Ukwa deposit for a total of 800 m drilling on 130 m borehole interval during FS 2016-17. These boreholes were planned between the boreholes of GSU and UKH series. Of the 8 bore holes, 3 bore holes were planned to intersect the ore body at first level (around 50 m vertical depth), 3 bore holes were proposed to touch the ore body at second level (around 100 m vertical depth) and two boreholes were planned to intersect the ore body at third level (around 200 m vertical depth).

During the course of G-2 stage exploration in parts of Toposheet nos. 64C/5 in and around Ukwa area, an extensive study was done on Khan and Fulmari, 1996 previous work. Based on detailed map of Khan and Fulmari, 1996, profile sections and cross sections of previous boreholes along section lines were drawn and proposal of current 6 nos. of borehole were made considering profile section and depth of RL of intersection in previous boreholes and was approved by RMH-II. Field check and surveying for fixing borehole points and demarcation of MOIL boundary were done prior to initiation of drilling in western Ukwa block. Drilling in western Ukwa block was initiated on 09.01.2017 and completed on 26.03.2017. A total of 1071.35 m of drilling was done in 6 boreholes drilled at second and third levels to know strike and depth continuity of manganese horizon. The combined thickness of ore horizon intersected in 6 different boreholes (UKBH-1, 2,3,4,5 & 6) is 10.01m with maximum thickness of ore horizon within a single borehole is 3.61m (UKBH-3) and minimum thickness of ore horizon within a single borehole is 0.15 m (UKBH-1). A total of 51 nos. core samples, 20 nos. BRS, 10 nos. REE, 10 nos. PS. 10 nos. PCS and 5 nos. of XRF samples were collected and analysed. Open cast as well as underground mines were visited during field work to know about sub surface behaviour of body. The surrounding areas were also investigated during field work along

with regional traverses to understand structures as well as regional geological setup of area. Core Logging along with geophysical logging of each borehole (except UKBH-4) was done. 5 nos. of water samples were also collected and analysed. DGPS survey was carried out and demarcation of borehole points along with block boundary was done subsequently and these details are provided in further chapters.

Physical target achieved during FS 2016-17 along with field stay details and expenditure details are provided below in tabular form.

Item No:		050/ME/CR/MP/2016/43	
Target for FS 2016-17		Achievement	
1. Technological			
(a) Subsurface Exploration			
(i) Drilling (m)		800 m	1071.35 m
(ii) Core samples		50 Nos.	51 nos.
(iii) Bulk sampling for Beneficiation study		1 No. (250 kg of samples from all the mineralized zones of 1/4 th of borehole cores	80Kg of sample collected from Core was submitted to IBM. Result is still awaited.
(iv) Sampling for Bulk Density Study		As Necessary	Nil
2. Geochemical Survey			
a. BRS	20 Nos.		20 Nos.
b. REE	10 Nos.		10 Nos.
3. Petrographic and Mineragraphic Studies			
a. PS	10 Nos.		10 Nos.
b. PCS	10 Nos.		10 Nos.
c. XRF*	05 Nos.		05 Nos.
4. Chemical Analysis#			
(a) Core sample (SiO ₂ , Fe ₂ O ₃ , P ₂ O ₅ ,MnO, TiO ₂)	50 Nos.		51 Nos.
(b) BRS (Cu, Pb, Zn, Co, Ni, Cr, SiO ₂ , Fe ₂ O ₃ , P ₂ O ₅ , MnO, TiO ₂)	20 Nos.		20 Nos.
(c) PCS	10 Nos.		10 Nos.
(d) REE@	10 Nos.		10 Nos.

Chemical Laboratory, GSI, CR, Nagpur. *XRF Lab, GSI, Faridabad. @ Chemical Lab, GSI Bhopal.

Name of Officers	Field Days
1. Dr. Suresh Kumar, (Sr. Geologist)	114
2. Keshav Khandelwal,(Geologist)	133
Supervisory Officer A.K.Talwar, Director.	6

Total Expenditure in Rs.
210242/- (Wages)
38735/- (POL)
42032/- (OC)
2,91,009/- Total

2.2 Acknowledgement:

The authors express their sincere thanks to Shri A. K. Talwar, Director and HoO, SU: Jabalpur, M.P. for his constant guidance, important suggestions and encouragement in carrying out field work as supervisory officer. They are thankful to Shri N. Kutumba Rao, DG GSI, Shri V.P. Sable, Dy.D.G, (Ret.) for their guidance and facilities provided. They are thankful to A.S. Khan Director (Ret.) for his valuable guidance. They are thankful to H.N. Bawane, Dy.D.G (Fertilizer Minerals), CR,Nagpur for valuable support during his tenure as HoO of SU: Jabalpur for FSP 16-17. Special thanks to Vivek Kumar Maurya and Abhinav Om Kinker, Geologist SU: Jabalpur, M.P for their support in calculation of resource and Dr. Ashwini Raju in preparation of maps.

PROPERTY DESCRIPTION

3.1 Title of Ownership

Geological Survey of India, Central Region carried out this project under Mission-II, Ferrous Minerals, Mineral Resource Assessment (G-2 stage) (State Unit-Madhya Pradesh, Jabalpur) for the Field Season 2016-17 vide FSP Item Nos. 050/ME/CR/MP/2016/43.

3.2 Details of the area

3.2.1 Village name and district

The current study area belongs to Ukwa village, Roopjhar Tehsil, District Balaghat, M.P.

3.2.2 Survey of India Toposheet No.

The current study area falls under Survey of India Toposheet no. 64C/05.

3.2.3 DGPS Geo-coordinates.

DGPS co-ordinates of the block marked on plate-III having current as well as previous boreholes are as follows:

A 21° 58' 16.79"N	80° 26' 43.00" E
B 21° 58' 6.48" N	80° 26' 52.61" E
C 21° 58' 22.5" N	80° 27' 16.22" E
D 21° 58' 36.29" N	80° 27' 09.99" E

The co-ordinates are written in counter-clockwise manner. Area of block for augmentation is approximately 0.4 km². Here block marked on map (Plate-III) for augmentation of resource extends from section lines W-1 to W-4.

3.2.4 Cadastral details

The present block of drilling is a part of western Ukwa block which is covered by paddy fields till the month of October. For this reason the drilling work was initiated in month of January. Prior permission was taken by Gemko Kati out source drilling unit for drilling in the area. As first series of 3 boreholes fall under MOIL leasehold boundary therefore borehole proposal was

modified after field check. A total of 6 nos. of boreholes of second level and third level were drilled in Western Ukwā block during this study and all boreholes fall under paddy field.

3.2.5 Freehold/ Leasehold

The present drilling was performed in freehold area, however prior permission of land owner was taken by Gemko Kati outsource drilling unit before drilling.

3.2.6 Location and accessibility

The present study area is a part of Ukwā western block which falls in T.S. Nos. 64C/05 and bounded by latitude, 21°57'45"N to 21°58'21"N and longitude, 80°26'35"E to 80°27'14"E. The area of investigation falls under the jurisdiction of Rupjhar Police Station, Baihar tehsil, Balaghat District, MP. The Ukwā mine and town both are located 42 km ENE of Balaghat, on Balaghat-Baihar, all weather, asphalt road (SH-26). Ukwā is also well connected by roads with other important places and adjoining district headquarters like Rajnandgaon, Durg, Raipur and Mandla. The nearest railhead is Balaghat junction located on Jabalpur-Gondia narrow gauge line of the South Eastern Railways. The interior parts of the area on the either sides of the State Highway are connected with the network of all weather metalled road and fair weather kuchcha roads.

3.2.7 Climate

The area experiences varied climate. Summer commences from the end of March and continues up to middle of June, with May being the hottest month. During summer day temperature rises up to 45°C, but nights are pleasant. Winter is pretty cold with some times temperature dropping down to 3°C. Frosty conditions prevail during winter nights. Intermittent rains in winter are very common. December and January are the coldest months. Area receives rainfall from southwest monsoon with the July and August being the months experiencing maximum rains. The average annual rainfall is around 1537 mm.

3.2.8 Flora and fauna

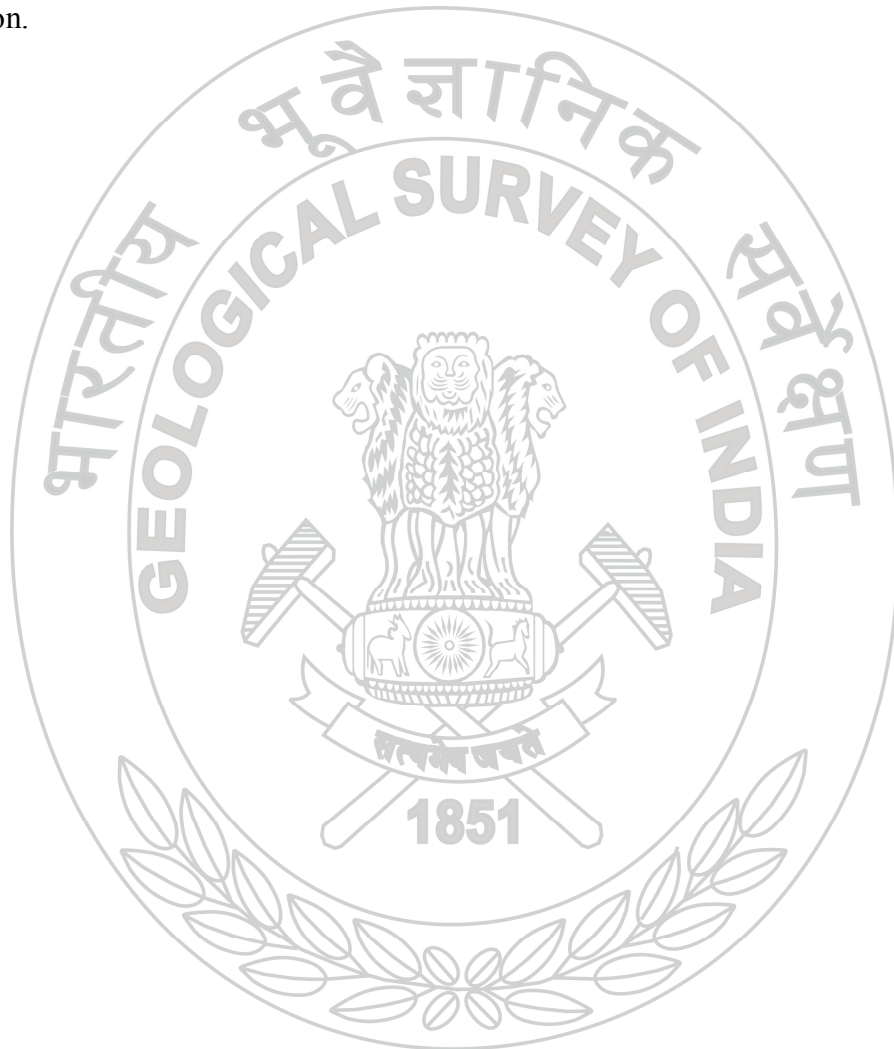
The area has forests of dry, mixed deciduous type with bamboo. Teak and Sal plantations are scattered. Some of the important flora found in the forests are Saja (*Terminalia tomentosa*), Dhaman (*Grewia tiliaefolia*), Teak (*Tectano grandis*), Sal (*Shorea robusta*), Bija (*Steroosperma suavelenns*), Dhure (*Anoaecissue latifolia*), Kumbhi (*Gareva arborea*), Tinsa (*Quoeina calberolodes*), Harra (*Terminalia chebula*), Behara (*Terminalia belirica*), Jamun (*Eugenia jambolana*), Tendu (*Diosplyros mclanoxyla*), Mahuwa (*Madhuka latifolia*) and Lendia (*Lagerstromia paulifora*).

The common wild animals found in the region are tigers, panthers, wild cats, hyena, wild dogs, bisons, sambars, wild boars, bears, barking deer and red and black-faced monkeys. The area around Ukwa is full of Cobras. Apart from the wild animals and reptiles, the area has various types of birds. The Kanha National wild life sanctuary is around 45 km northeast of Ukwa.

3.3 Infrastructure & Environment

The social infra structure in and around Ukwa area includes: Primary health center, Ukwa; Government Primary School, Jagantola; and Communication & Post-office, Rupjhar. Regarding host population, as of 2011 India census, Ukwa is a Census Town city in district of Balaghat, Madhya Pradesh. The Ukwa Census Town has population of 6,768 of which 3,335 are males while 3,433 are females as per report released by Census India 2011. Population of Children with age of 0-6 is 654 which is 9.66 % of total population of Ukwa (CT). In Ukwa Census Town, Female Sex Ratio is of 1029 against state average of 931. Moreover Child Sex Ratio in Ukwa is around 946 compared to Madhya Pradesh state average of 918. Literacy rate of Ukwa city is 84.33 % higher than state average of 69.32 %. In Ukwa, Male literacy is around 92.00 % while female literacy rate is 76.95 %. Kanha National Park is situated at 35 km NE of Ukwa. Kanha National Park is world famous national park and attracts lots of tourist for its beauty and presence of maximum numbers of tiger there. Topographically the area is located on the Baihar plateau and is drained by the tributaries of Wainganga River, namely, Nahra Nadi and its off shoots. The area around Ukwa is almost plain with very low south westerly gradient. The area between this plain and flat-topped Tipagarh hill range is quite rugged. The manganiferous horizon occurs as linear ridge striking ENE-WSW to NE-SW within the plain country of Ukwa, starting from Tipagarh and continuing

southwestward up to Jagantola. The highest point is 800 m at the fringe of Tipagarh range (NE corner of toposheet No. 64C/5) and lowest around 570 m along Nahra Nadi south of Dora (Khan et al; 1993-96). In ground conditions at present ridge of Mn has been replaced by open cast mines of MOIL. Ground water condition is quite good. The use of ground water is almost negligible in the area. Mica schist being the major litho unit in the country acts as good aquifer. Luxuriant growth of natural vegetation like palm and jamun trees indicates the shallow water table condition.



PREVIOUS EXPLORATION

4.1 Details of previous exploration

Bose P.N. of Geological Survey of India discovered the Ukwa manganese deposit in 1888-89. Fermor L.L. studied the important manganese deposits of the region in 1903-04 and proposed the name 'Gondite series' for the manganiferous suite of rocks in Chhindwara, Nagpur and Balaghat areas. Burton R.C. carried out Geological surveys in Balaghat district during 1911-12. Shukla K.D. alone in 1950-51, and Shukla K.D. and Anandalwar M.A. in collaboration with Stracjek J. A. of USGS carried out geological mapping in the area and paid special attention to the manganese ore deposits of the region (1,2,3,4,5 source - Shukla and Anandalwar, 1973).

MOIL (the then Central Provinces Manganese Ore Ltd.) carried out core drilling in Ukwa deposit at 19 sites during 1955-60. Geological Survey of India carried out regional exploration by means of large scale mapping, drilling, sampling etc. during 1970-71 to 76-77, in leasefree areas to the east and west of Ukwa deposit and also in contiguous areas of Bharweli deposit (Balaghat). The areas covered under the regional exploration are as follows:

1970-71 Jagantola-Ghondi, Dhamsua-Laugar and Laugar-Kanhatola sectors.

1971-72 Ukwa-Gudma block.

1972-73 Jagantola-Ghondi, Dhamsua-Laugar sectors and Balaghat-Aolajhiri block.

1974-75 Jagantola-Ghondi sector and NE extension Samnapur block.

1975-76 Dhamsua-Laugar, Laugar-Kanhatola sectors and Balaghat-Aolajhiri block.

1976-77 Dhamsua-Laugar and Laugar-Kanhatola sectors. Results of the above explorations are covered under unpublished reports of Geological Survey of India (Source -Choudhary et al; 1979).

Exploration by regional drilling for manganese ores in Ukwa-Gudma Block (included in above list) was carried out by Akolkar (1974). Based on the results of two regional boreholes, he indicated possible reserves of the order of 0.525 million tonnes of low grade, medium to high phos ore, of which 0.263 million tonnes constituted as recoverable one.

Exploration by drilling in Jagantola-Ghondi and northeast extension of Samnapur sectors, forming SW and NE extension of Ukwa deposit respectively were carried out by Bhoskar and Rao (1975). In Jagantola-Ghondi sector, they indicated the aggregate 'insitu' and inferred reserves of 2.08 million tonnes, of which 0.86 million tonnes constituted as recoverable ore containing 30 to 38 percent manganese. In Samnapur extension sector the manganese ore was more or less gonditic.

Ukwa deposit was also explored by Choudhary et al. (1979). They drilled 41 vertical boreholes involving 5041 m of drilling and estimated gross reserves of 14.71 million tonnes of manganese ore of which 12.50 million tonnes constituted the recoverable ore over a strike length of 4670 m. Pal and Deshmukh (1974) studied the thin siltstone bands associated with the manganese ore horizon. They concluded that the siltstones were of sedimentary origin and were probably present in metamorphic shadow zone.

Devarajan and Hakim (1992) carried out Second Generation Mapping in Bharweli-Ukwa area. Their work gives detailed accounts of geology, structure and metamorphism of the basement gneisses and overlying Sausar metasediments.

Khan and Fulmari (1996) carried out exploration and drilling in continuation of the exploration work completed earlier by Choudhary et al. (1979) and for deeper level drilling the section lines used by Choudhary et al; was extended further northwestward by them. They completed 4574.22 m of drilling and carried out 51 sq km large scale mapping on 1: 25,000 scale and 5.18 sq km detail mapping on 1:2000 scale. Large scale mapping on 1:25,000 scale has been carried out in the area between Ukwa and Baputola to have an idea about regional geology and for locating the additional manganese ore horizons if any in the dip direction of Ukwa deposit.

Detailed mapping (1:2000) of entire Ukwa open cast mine, adjoining foot wall and hanging wall rocks were covered prior to drilling and for convenience the entire area has been divided into three blocks namely Ukwa Eastern, Central and Western blocks. They have also reported that the average Mn content of the Ukwa deposit is 41.23% and P is 0.23%. Major part of the deposit falls under medium grade ore having 35 to 45% Mn. Phosphorus content increases in Western block and also at depth in other blocks. The deposit has estimated gross ore reserves of 16.23 million tonnes which would involve mining of 29.07 million tonnes of material consisting of ore and waste rock i.e. the run of mine. The recoverable reserves (90% of gross reserves) come to 14.61 million tonnes representing 50.26% of run of mine, of which 11.84 million tonnes fall

under Probable 'A', 1.37 million tonnes under 'Probable' and 1.40 million tonnes under 'Possible' category. The favourable geological and structural set up of the ore horizon may lead to easy minability of ore, but the high phosphorous content at depth could be a discouraging factor.

4.2 Details of Geophysical Investigation

Gaonkar S.G. and Das A.K. carried out geophysical surveys utilizing gravity, magnetic (V.F.) and self potential methods in Ukwa (21°58'30", 80°28'W) area, Balaghat district, M.P. for finding out extension along strike for the Ukwa manganese deposit in the soil covered gap area between Ukwa deposit and Jagantola trench in the Western Block and to the east of the Eastern Block.

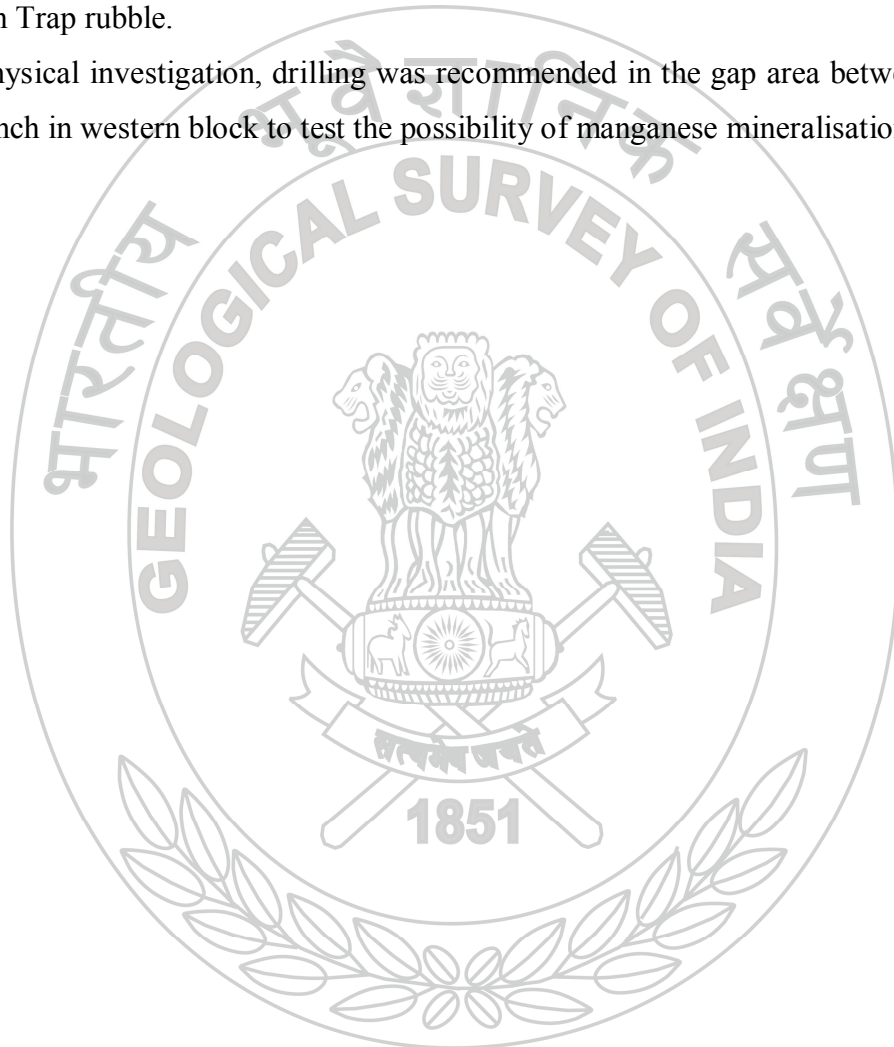
The strata bound manganese bodies are located between the mica schist and the phyllites metasedimentaries. The metasedimentaries are overlying the granite gneiss basement. The bedding and the foliation trends are along the ENE-WSW direction with moderate dips of about 25° to NW. The manganese horizon is exposed over a strike length of about 5 km and the thickness of the horizon ranges from 0.27 m to 6.1 m.

Two major magnetic anomalies, in the form of linear bands with strike length of about 3.5 km each and magnitude in the range of 100 to 400 gammas were picked up in the western block of this area. The width is about 25 and 100 m for the southern and the northern anomalies respectively. The southern magnetic high was recorded over the main Ukwa mine trench, where the ore body with average width of 3 m is exposed at the surface. The anomaly shows a dominant positive high with absence of negative low to the north. This shape is attributed to the northward dip of the ore body. The gravity surveys closely corroborate the results of magnetic surveys in this area. Two linear gravity anomalies corroborate with the observed linear magnetic anomalies. The geophysical surveys indicate strike continuation of the exposed manganese mineralisation under soil cover, from Ukwa trench in the east to Jagantola trench in the west, for over 1.6 km. Presence of another gravity-magnetic linear anomaly in the soil covered area to the north of the ore body exposed in the main trench indicates interesting possibilities for the presence of manganese mineralisation. S.P. surveys were also carried out in this area. The surveys have brought out feeble but definite low values of the order of 20 to 80 mv and corroborate with the gravity magnetic response over the manganese ore body exposed in trench and also support its extension to the west over a strike length of 1.6 km. No S.P. response was

recorded over the northern gravity magnetic anomaly band. This could be due to the thick soil cover and the causative body being at greater depth.

A magnetic peak with narrow width was picked up over manganese pit on Tr. TU4 in the eastern block. However this anomaly was not found to continue over the next traverse located at an interval of 300 m to east. From the appearance of rapid fluctuations in magnetic values on traverses laid further to the east in this block, it is inferred that Sausar formations have gone under Deccan Trap rubble.

As per geophysical investigation, drilling was recommended in the gap area between Ukwa and Jagantola trench in western block to test the possibility of manganese mineralisation in this area.



GEOSCIENCE INVESTIGATION

5.1 Regional Geology

The main manganese producing belt in the country is associated with the Sausar Group of rocks of Pre-cambrian age. The group is characterized by lower pelitic and upper psammo-pelitic sequence. The manganese ore horizon occurs in the lower part of the Sausars.

The rocks of the Sausar Group extend in northeast-southwest direction in Balaghat district, MP., in the east to west northwest-east-southeast in Chhindwara District, in the west through Bhandara and Nagpur districts of Maharashtra, forming an overall arcuate shape with convexity towards south. The manganese bearing belt stretches over a length of 200 km and is about 25 to 30 km wide in its central part. The Ukwa deposit is located, almost, at the eastern end of the belt.

The eastern part of the manganese belt, from Bharweli (Balaghat) to Ukwa, comprises, mainly, rocks belonging to the lower sequence of the Sausar Group, which is characterized by a regionally uniform strike trend of NE-SW and low to moderate dips towards NW. The ore horizon has conformable relation with the enclosing rocks and can be traced intermittently from Bharweli to Ukwa through the occurrences at Kanhatola, Laugur, Ghondi and Jagantola.

The low-lying areas around Ukwa, bordering the plateau, predominantly comprise rocks of Sausar Group. The Sausar rocks occur in the areas from southern vicinity of the existing manganese mine pits to more than 15 km north of Ukwa mines. The areas south of mine pits expose older metamorphics in the form of gneisses and associated rocks. A solitary thin bed of pale gritty clay belonging to Lameta beds, unconformably overlying the Mansar schist of Sausar group is recorded in Ghondi mine section. The Deccan Traps occur at higher levels forming plateau. The contact between Sausars and Deccan traps occurs around 700 m above m.s.l. The traps are capped by about 25 m thick laterite. Stray laterite mounds are also seen over older metamorphics and Sausars. The laterite contains pockets and lenses of bauxite.

The generalized succession established by Shukla and Anandalwar (1973) around Ukwa region is presented in below Table 5.1.

Pre-Cambrian Sausar Group	Formation	Brief description of rock types
	Recent and Sub - Recent	Sandy soils and alluvium
	Tertiary	Laterite and laterite gravels
	Post Sausar Intrusives	Pegmatite and vein quartz, Granite (?) Basic dykes and orthogneisses
	Bichua Formation	Marbles, Calc silicate granulite
	Junewani Formation	Quartz-biotite schist with garnet, Kyanite, staurolite
	Chorbaoli Formation	Quartz-biotite schist with garnet and Kyanite, micaceous-flaggy quartzites, quartz schist
	Mansar Formation	Muscovite schist, quartz-muscovite schist, quartz-sericite-muscovite schist with garnet
	Lohangi Formation	Pink calcitic marbles (impersistant)
	Sitasaongi Formation	Quartzite, quartz-muscovite schist and conglomerate. Felspathic grits and conglomerate
Archaean (Unclassified)	Older metamorphics	Ortho and paragneisses, composite gneisses and granite, bands and lenses of amphibolite, epidiorite, quartzite etc, with pegmatites and quartz veins

Table: 5.1 - Generalised geological succession of Ukwa area (after Shukla and Anandalwar, 1973).

Only a part of the above succession is exposed in the area around Ukwa where exploration work has been carried out. The older metamorphics are exposed in the area south of Ukwa mine. The authors could not locate the presence of Sitasaongi Formation either on surface or in subsurface in drill hole sections. Mansar Formation occupies the major part of the area and perhaps the litho unit exposed around Nagjhola and east of Khursurtola in Nalira Nadi belongs to Chorbaoli Formation. All the litho units have ENE-WSW regional trend with low to moderate northwesterly dip around Ukwa, but there is a general tendency of steepening as one goes further north.

5.1.1 Geological set up

Khan and Fulmari, 1996 carried out exploration an area of 51 sq.km and mapped around Ukwa on 1:25,000 scale with a view to have better understanding of the geological set up and to find out additional ore horizon, if any. The area covered by large scale mapping exposes the older metamorphics and mainly pelitic schists with manganese ore horizon. The succession examined in the area is indicated in below table 5.2.

Pre-Cambrian Sausar Group	Recent to sub-Recent	Soil and alluvium
	Tertiary	Laterite
	Post Sausar	Pegmatite and vein quartz
	Chorbaoli Formation (?)	Garnetiferous quartz-muscovite-biotite schist ± Kyanite
	Mansar Formation	Garnetiferous muscovite schist
		Garnetiferous sericite schist/phyllite
		Garnetiferous quartz muscovite schist with intercalations of thin quartzite
		Sericite schist. Manganese ore horizon, Interbanded Mn ore and manganiferous quartzite.
	Lohangi Formation (?)	Phyllite with calcareous bands
Tectonic contact		
Archaean	Older metamorphics	Granite gneiss with enclaves of biotite schist, amphibolite, intruded by K-granite, pegmatite and vein quartz

Table: 5.2 – Geological Succession in Ukwa-Baputola area (Khan and Fulmari 1996).

5.2 Detailed geological exploration

Khan and Fulmari, 1996 carried out detailed mapping of 5.18 sq.km area in and around Ukwa on 1:2,000 scale. The area covered by detailed mapping exposes ENE-WSW trending ridges and mostly paddy fields. Present area is a part of Ukwa western block having an area of approx. 0.4 Km² (Plate-III).

5.2.1 Description of Rock types

5.2.1.1 Older Metamorphics

The older metamorphics occur in the area to the immediate south of the mine pit and they continue down south beyond the area of exploration. The unit is represented by ortho and paragneisses which contain enclaves of amphibolite and biotite schist (Fig. 5.1). The amphibolite enclaves are rare in the area adjacent to the Ukwa mine. Their intensity increases further south. These enclaves are more susceptible to lateritisation. The laterite cappings occurring in the area south of Samnapur are perhaps developed on these enclaves. Biotite schist occurs as thin bands and lenses up to 50 cm in thickness within the gneisses. The rock is well foliated, and bands are stretched along the regional foliation of the gneiss.

The gneisses are banded and augen type composed mainly of quartz, plagioclase, K-feldspar and biotite with subordinate muscovite and chlorite (Fig 5.5). At places gradation is observed between gneisses and the biotite schist, amphibolite and biotite granite, from which they have been derived.

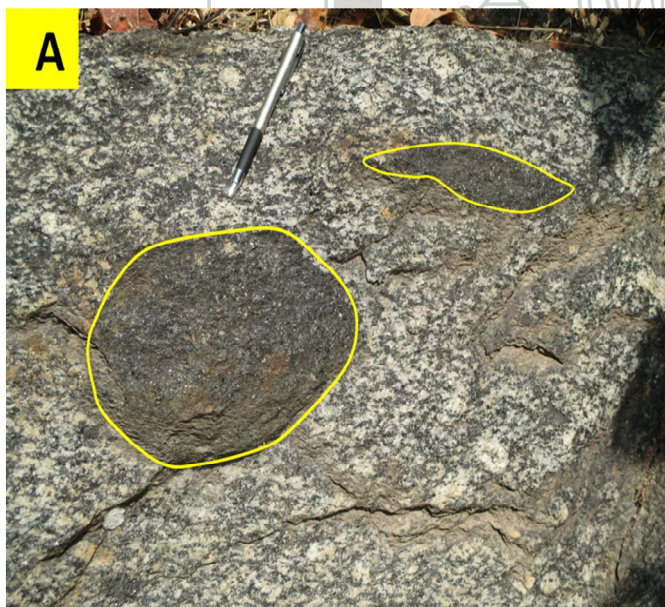


Fig 5.1A: Older metamorphic enclaves(biotite schist)exposed on road section near Gangulpara. T.S. no. 64C/05

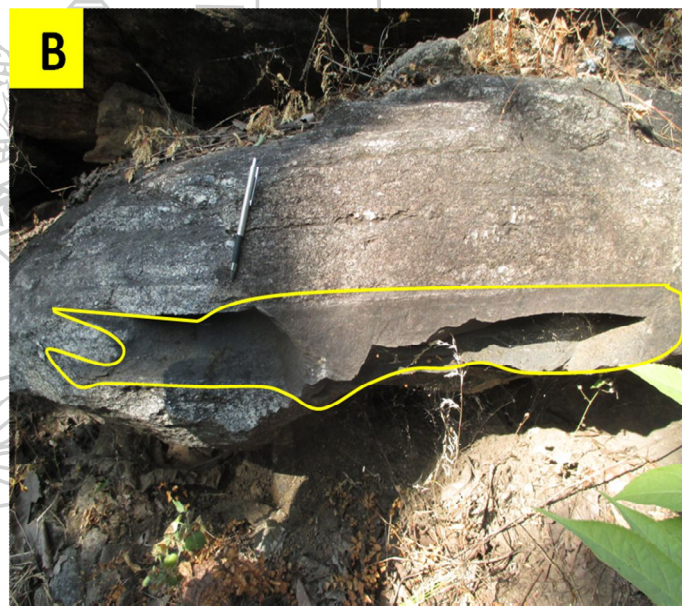


Fig 5.1B: Older metamorphic enclaves(biotite schist)exposed on road section near Laugur. T.S. no. 64C/05

The gneisses are intruded by fine grained pink granite, aplite, pegmatite and vein quartz. These intrusives are concentrated along the contact between gneisses and overlying metasedimentaries. The pegmatite and vein quartz are seen in both the gneisses and metasedimentaries, but no traces of granite and aplite have been recorded in metasedimentaries on the surface or in 60 numbers of holes drilled by Geological Survey of India. Hence, these granite and aplite are either part of the older metamorphics or have been emplaced just at the beginning of Sausar orogeny, as they have been involved in the tectonic movement taken place between the gneisses and metasedimentaries. The gneisses and granite both are, at places, quite rich in magnetite. Magnetite occurs as fine disseminations or mm thin stringers along the fractures. Pyrite also, occurs in the similar fashion either in association with magnetite or without it. In general gneisses are light grey in colour but wherever they are intruded by potassic granite they show pinkish grey to pink colour.

5.2.1.2 Phyllite/ Schistose rock

Metasedimentaries, the lower most unit occurring in the area is phyllite with calcareous intercalations. The phyllite is sericitic and chloritic. At places it grades into sericite schist. The calcareous intercalations in this unit are represented by quartz-sericite/muscovite-calcite schist. The presences of calcareous intercalations indicate that this unit belongs to probably Lohangi Formation. This unit is quite persistent forming foot wall rock for manganese ore horizon as evident during drilling. The exposures of phyllite/calcareous phyllite unit are seen in mine pit of MOIL adjacent to the working area. At places, the calcareous unit downward grades into garnetiferous calcite-quartz-chlorite-muscovite schist. Occasionally sericitic quartzite interbands are also noticed in this unit. The calcareous intercalations are commonly recorded in footwall rock, but very often they are also noticed interbanded with the lower part of ore horizon, thus, perhaps, indicating the gradational relationship between Lohangi and Mansar Formations. The general trend of the units are $N60^{\circ}E-S60^{\circ}W$ with 25° to 35° dip towards NW (Fig. 5.2).

The footwall phyllitic/schistose rock represents muscovite-quartz-biotite schist. The schistose fabric is defined by preferential alignment of micas, highly flattened quartz and elongate chlorite as observed with the help of hand lense. Muscovite is the most abundant mineral, followed by quartz (Fig. 5.3).

Petrographic study of schist reveals that the rock represents a quartz-muscovite-biotite-calcite-zoisite schist with accessory opaques, tourmaline, apatite and monazite. The rock is compositionally banded with alternate layers rich in micas and quartz. Calcite is more concentrated in the coarser quartz-rich layers in calc-schist while Qtz is dominating in Qtz-Muscovite schist. Chloritic as well as sericitic variant of schist is also present in the area. The rock is rich in quartz with lesser abundance of muscovite and zoisite. The precursor rock appears to be calcareous shale with thin sandstone partings.



Figure 5.2A: Calcareous phyllite exposed on road section in SW of Laugur showing kinks in S1

Figure 5.2B: Calcareous phyllite exposed on road section in SW of Laugur showing S1 and joints parallel to S1

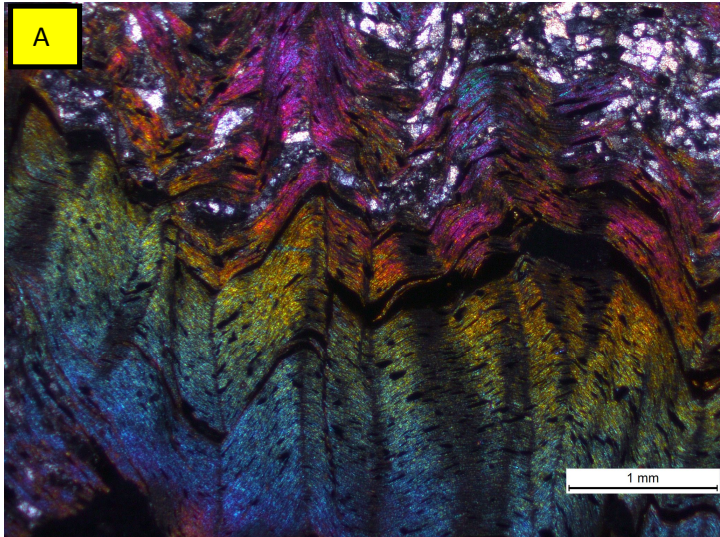


Figure 5.3A: Thin section showing crenulations in S1 in Quartz-muscovite-biotite schist exposed in Ukwa mine pit, Ukwa, T.S. no. 64C/05.

Figure 5.3B: Quartz-muscovite-biotite schist exposed in Jagantola mine pit, Ukwa, T.S. no. 64C/05.

5.2.1.3 Manganese ore horizon

Mn ore occurs as <1 cm to 30 cm thick bands interbanded with cherty manganiferous and ferruginous quartzite and impure calcareous rock as quartz-sericite/muscovite-calcite schist. Apart from manganiferous and ferruginous quartzite and calcareous rock, Mn ore is also interbanded occasionally with sericitic phyllite/schist, particularly at the lower portion of the horizon. The individual bands of quartzite and calcareous rock range in thickness from <1 cm to >1 m. The ore is either massive in nature or is banded with mm thin quartzite and carbonate partings. The ore is dark grey coloured and fine grained. The Mn ore horizon is continuous band (Fig 5.4) showing culmination and depression nature and due to deformation in the area. It occurs as massive, high grade ore as observed in UKBH-1, UKBH-2, UKBH-5 as well as interbanded with quartzite, low grade ore as observed in UKBH-3, UKBH-4, UKBH-6. A total of 10.01 m of thick ore horizon was intersected in 6 different boreholes (UKBH-1, 2,3,4,5 & 6) with maximum thickness of ore horizon 3.61m (UKBH-3) and minimum thickness of ore horizon 0.15 m (UKBH-1).

The ore horizon ranges in thickness from 6 cm to about 10 m (Khan and Fulmari, 1996) and from 0.15cm to 3.61m (present study). The cumulative thickness of the ore in the ore horizon

ranges from a few cm to 4.85 m (Khan and Fulmari, 1996) and 3.61m (present study) as revealed by drilling data. The ore horizon at places shows fractured to brecciated nature. The interstices in brecciated and fractured zones are occupied by quartz and carbonate veinlets and encrustations. The fracturing and brecciation deteriorates the quality of the ore.

The ore horizon perhaps transgresses from Lohangi to Mansar as has been evidenced by association of calcareous facies rocks with the ore and foot wall rock. Wherever, calcareous chert has been invaded by vein quartz, it has produced a quartz garnet rock with tiny pink garnets. At places honey yellow garnets are noticed at the contact wherever the carbonate bands have been traversed by vein quartz.

According to Khan and Fulmari, 1996, siltstone and clay bed are perhaps derived from calcareous phyllite in the area. The calcareous phyllite due to leaching may give rise to soft loose silty or clayey horizon depending on the mineral constitution and particle size of the original rock after removal of calcareous material by solution. This view is further supported by the presence of calcareous phyllite/schist layers associated with the ore horizon. These layers occur invariably at the bottom of ore horizon but at places they are also seen interbanded with it. Leaching following physico-chemical factors must be responsible for (1) presence of fractured, porous and permeable competent layers in the form of Mn ore and manganiferous quartzite within comparatively less permeable pelites which facilitated circulation of groundwater, (2) acidic nature of circulating water because of solution of sulphides (mostly pyrite) present in the ore horizon and adjoining rocks helping in leaching of calcareous constituents of the rock.



Figure 5.4A: S-0 as observed in Manganese ore horizon in MOIL mine pit, Ukwa (B), T.S. no. 64C/05.



Figure 5.4B: Undulations in Manganese ore horizon due to deformation in MOIL mine pit, Ukwa (B), T.S. no. 64C/05.

5.2.3 Structures

5.2.3.1 Primary Structures

In metasedimentaries the most important non diastrophic structure is bedding plane (So) (Fig 5.4)., which is very well preserved in ore horizon in the form of compositional layering represented by alternations of Mn ore and manganiferous quartzite, with occasional calcareous and rare argillaceous (sericitic schist) bands. The colour/compositional layering in footwall phyllite also represent the bedding plane of the rock. The banded nature of quartz-mica schist and calcareous mica schist is also because of original compositional layering. The bedding in ore horizon and associated footwall and hanging wall rocks strikes NE-SW to ENE-WSW and dips at low to moderate angles towards NW. Pencontemporaneous fracturing and folding are also recorded in ore horizons, as observed in Ukwa mine and also in drill cores.



Figure 5.5A: Showing relict of equigranular and porphyritic textures of the original granitic rock, exposed around Gangulpara, T.S. no. 64C/05.

5.2.3.2 Secondary Structures

According to Khan and Fulmari, 1996, the secondary structures in the gneiss are in the form of gneissic (shear) foliation and mineral lineations. The secondary structures in the metasedimentaries are in the form of axial plane schistosity (S_1) of regional (F_1) folds with which is associated intersection lineation (L_1) formed by the intersection of bedding with schistosity. The axial plane cleavage (S_2) of second-generation folds (F_2) is associated with minor folds and intersection lineation (L_2) formed by the intersection of axial plane with S_0 or S_1 . In the similar fashion the evidences of F_3 and F_4 and related planar and linear structures i.e. S_3 , L_3 and S_4 and L_4 are also recorded in the area which are better observed and understood in metasedimentaries. (After Khan et al. 1993-96)

The synthesis of structural data reveals that the area has undergone at least four phases of folding deformation (D_1 to D_4/F_1 to F_4). The first phase (D_1) was the major one which gave rise to ENE-WSW major (F_1) overturned folds. The Sausars present in Ukwa area are considered to be the southeastern limb of an overturned synform and regional foliation (S_1) as axial planar to F_1 folds. This phase has given rise to subhorizontal intersection lineations (L_1) formed by intersection of S_1 on S_0 plane. The second phase (D_2) has been represented by low west to WSW plunging

minor folds (F_2) which has developed subvertical crenulations (S_2) on S_0 and S_1 planes and L_2 intersection lineations and fold axes. (After Khan et al; 1993-1996.)

The third phase (D_3) is not very well developed through out but wherever recognisable, it is seen in the form of NNE-SSW trending warps (F_3) and associated axial planar cleavage (S_3) and intersections lineation (L_3) and fold axes. The S_3 strikes in NNE-SSW direction and dips sub vertically. L_3 is low NNE plunging. (After Khan et al; 1993-1996.)

The fourth phase of deformation (D_4) has given rise to NW-SE trending open warps on earlier formed planar structures, producing depressions and culminations. The F_4 are low north-westerly plunging with subvertical axial plane giving rise to NW-SE striking and sub vertical dipping axial plane cleavage (S_4) and low to moderate NW plunging intersection lineations (L_4) and fold axes. (After Khan et al; 1993-1996.)

The culminations and depressions in the levels of intersection of ore horizon is perhaps because of the combined effects of the different phases of deformations (Fig. 5.6).

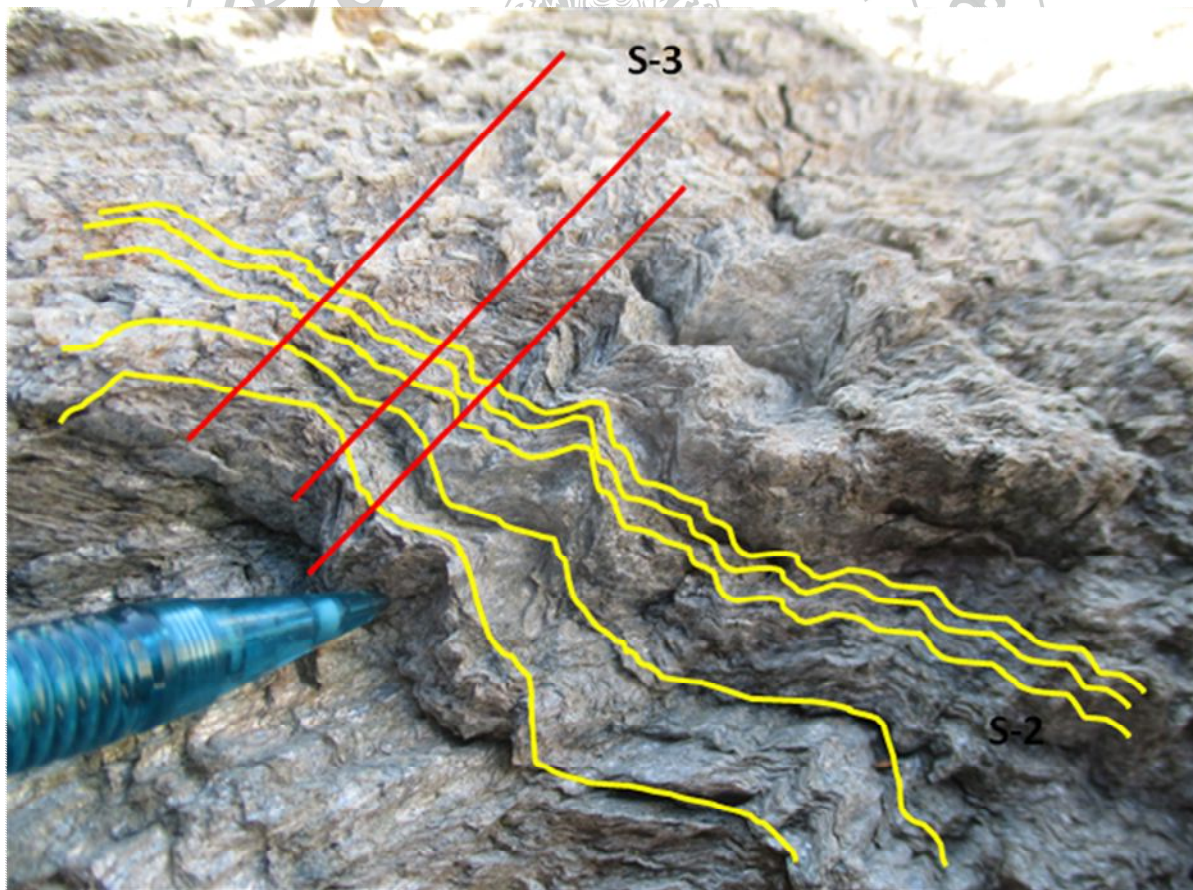


Figure 5.6: Sericite schist undergone different phases of deformation

The contact between older metamorphics and Sausars is tectonic which is also evidenced by mylonitization of gneisses along the contact. This feature is perhaps related to the first phase of deformation (D_1). It indicates that D_1 is represented by F_1 folding and probably thrusting also. The event has imparted shear foliation on gneisses showing parallelism with the S_1 of Sausars. Perhaps the original planar structure (gneissosity) of older metamorphics has been obliterated by D_1 along the contact with Sausars. (After Khan et al. 1993-1996)

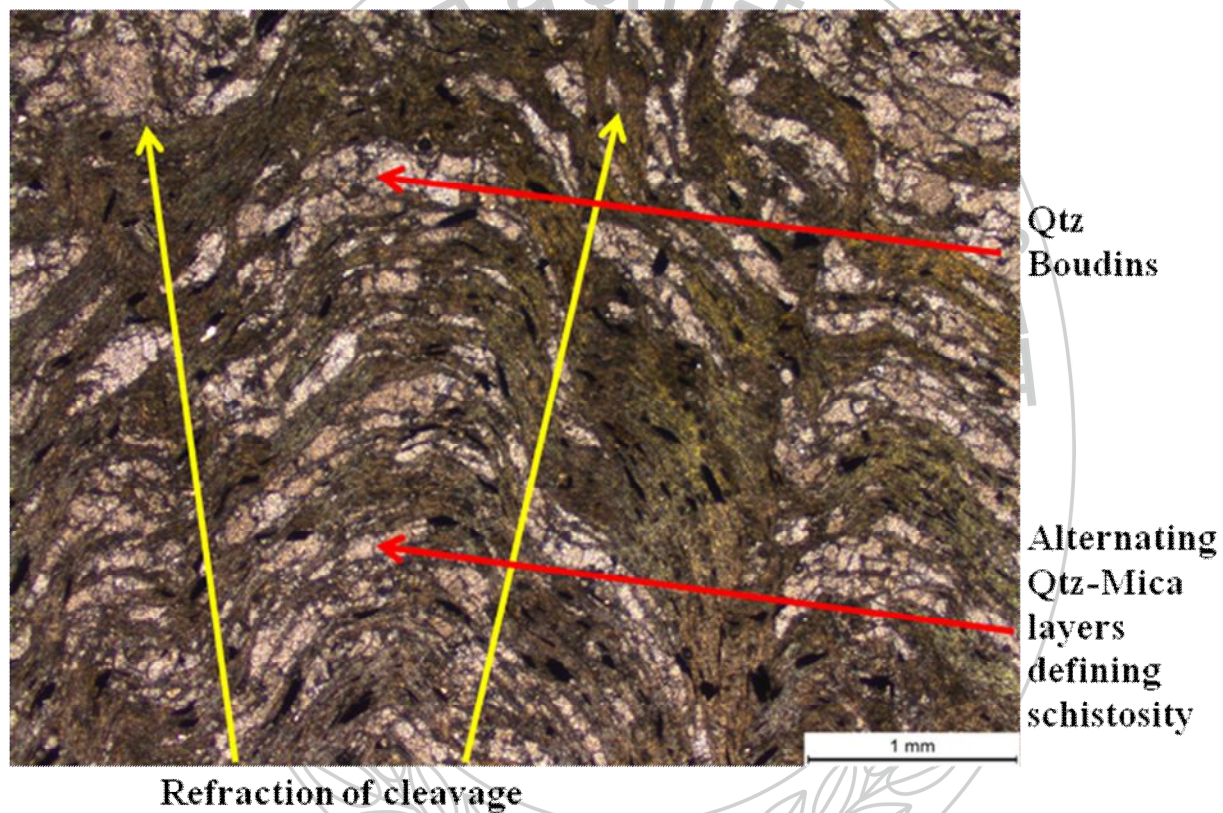


Fig 5.7: Thin section Of Quartz-Muscovite Schist indicating schistosity and cleavage refraction .(PS-I)

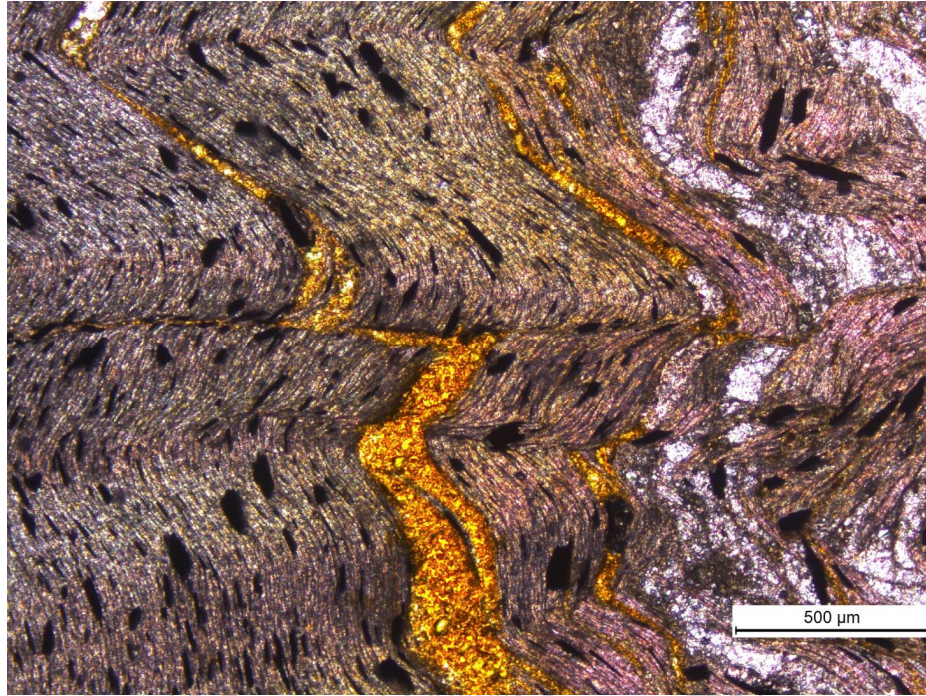


Fig 5.8: Thin section of Quartz-Muscovite Schist showing crenulations (PS-III)

5.2.4 Metamorphism

According to Devarajan and Hakim, 1992, the older metamorphic in the area have undergone metamorphism of lower granulite facies in the Bharweli Ukwa area. The Sausar have undergone amphibolite facies of metamorphism with the successively higher zones being encountered from south to north. The low grade phyllites of the Mansar Formation partly belong to the green schist facies. The chief constituent minerals are muscovite, chlorite and quartz. The associated pelitic schists of Mansars have undergone metamorphism of chlorite-biotite zone. The major constituent minerals are quartz and muscovite with subordinate amount of biotite and chlorite.

5.2.5 Mineralogy of the ore zone and ore texture

As per study by Khan and Fulmari 1996, the ore microscopic study reveals that the Mn ore comprises braunite, hollandite, bixbyite, hausmannite, haematite, pyrolusite and cryptomelane in varying proportion. Braunite is the most abundant mineral in the ore, followed by hollandite. Texturally the ore is either massive or banded. The massive type comprises mostly braunite and hollandite with granoblastic fabric while the banded ore has thin but distinct bands rich in braunite or hollandite or bixbyite.

5.2.6 Pitting and trenching

No pitting and trenching was carried because in work area ore horizon lies very deep.

5.2.7 Sampling

5.2.7.1 Core Samples

Core samples of -200 mesh size were prepared and submitted after drilling was completed. A total number of 51 Core samples have been prepared. Mn values determination from MnO is explained in Annexure-II. Details of samples collected from individual boreholes are described below table 5.3:

Borehole Number and Run	Ore horizon recovered (Run along Borehole)		Sample Number;	Sampled rocks	Maganese %
	From	To			
UKBH-01.	Run 83 to 86 m.				
	83.00	83.20	UKBH-01/CS/1/16-17	Ore horizon and Phyllite	5.878
UKBH-02.	a) Run 98 to 101 m.				
	100.45	100.65	UKBH-02/CS/1/16-17	Ore Horizon with fine interbanding of quartz.	12.600
	100.65	100.86	UKBH-02/CS/2/16-17	Ore Horizon with fine interbanding of quartz.	13.793
	100.86	101.00	UKBH-02/CS/3/16-17	Ore Horizon with fine interbanding of quartz.	32.49
	b)New Run from 101 to 104 m.				
	101.00	101.27	UKBH-02/CS/4/16-17	Ore Horizon with fine interbanding of quartz.	39.729
	101.27	101.47	UKBH-02/CS/5/16-17	Ore Horizon with fine interbanding of quartz.	37.646
	101.47	101.68	UKBH-02/CS/6/16-17	Ore Horizon with fine interbanding of quartz.	29.468
	101.68	101.88	UKBH-02/CS/7/16-17	Ore Horizon with fine interbanding of quartz.	37.39

	101.88	102.09	UKBH-02/CS/8/16-17	Ore Horizon with fine interbanding of quartz.	25.278
	102.09	102.29	UKBH-02/CS/9/16-17	Ore Horizon with fine interbanding of quartz.	6.522
	102.29	102.5	UKBH-02/CS/10/16-17	Ore Horizon with fine interbanding of quartz.	32.116
UKBH-03	a) Run from 122 to 125 m.				
	123.78	123.98	UKBH-03/CS/1/16-17	Ore Horizon interbanded with quartzite and carbonate material.	10.2616
	123.98	124.18	UKBH-03/CS/2/16-17	Ore Horizon interbanded with quartzite and carbonate material.	13.4601
	124.18	124.38	UKBH-03/CS/3/16-17	Ore Horizon interbanded with quartzite and carbonate material.	21.2589
	124.38	124.58	UKBH-03/CS/4/16-17	Ore Horizon interbanded with quartzite and carbonate material.	14.9935
	124.58	124.79	UKBH-03/CS/5/16-17	Ore Horizon interbanded with quartzite and carbonate material.	31.0171
	124.79	125.00	UKBH-03/CS/6/16-17	Ore Horizon interbanded with quartzite and carbonate material.	22.653
	b) Run from 125 to 128 m				
	125.00	125.19	UKBH-03/CS/7/16-17	Ore Horizon interbanded with quartzite and carbonate material.	21.9559
	125.19	125.39	UKBH-03/CS/8/16-17	Ore Horizon interbanded with quartzite and carbonate material.	30.5679

	125.39	125.59	UKBH-03/CS/9/16-17	Ore Horizon interbanded with quartzite and carbonate material.	33.5186
	125.59	125.8	UKBH-03/CS/10/16-17	Ore Horizon interbanded with quartzite and carbonate material.	28.8022
	125.8	126.00	UKBH-03/CS/11/16-17	Ore Horizon interbanded with quartzite and carbonate material.	38.5991
	126.00	126.20	UKBH-03/CS/12/16-17	Ore Horizon interbanded with quartzite and carbonate material.	21.0111
	126.20	126.40	UKBH-03/CS/13/16-17	Ore Horizon interbanded with quartzite and carbonate material.	34.7423
	126.40	126.60	UKBH-03/CS/14/16-17	Ore Horizon interbanded with quartzite and carbonate material.	23.2183
	126.60	126.81	UKBH-03/CS/15/16-17	Ore Horizon interbanded with quartzite and carbonate material.	23.7062
	126.81	127.01	UKBH-03/CS/16/16-17	Ore Horizon interbanded with quartzite and carbonate material.	14.5521
	127.01	127.21	UKBH-03/CS/17/16-17	Ore Horizon interbanded with quartzite and carbonate material.	24.2793
	127.21	127.425	UKBH-03/CS/18/16-17	Ore horizon and Phyllite	10.3081
Run 251 to 254 m.					
	251.7	251.9	UKBH-04/CS/1/16-17	Ore Horizon interbanded with quartzite and carbonate material.	18.3082
	251.9	252.1	UKBH-04/CS/2/16-17	Ore Horizon interbanded	24.1089

UKBH-04				with quartzite and carbonate material.	
	252.1	252.3	UKBH-04/CS/3/16-17	Ore Horizon interbanded with quartzite and carbonate material.	26.1303
	252.3	252.5	UKBH-04/CS/4/16-17	Ore Horizon interbanded with quartzite and carbonate material.	33.5961
	252.5	252.7	UKBH-04/CS/5/16-17	Ore Horizon interbanded with quartzite and carbonate material.	31.4431
	252.7	252.9	UKBH-04/CS/6/16-17	Ore Horizon interbanded with quartzite and carbonate material.	33.6503
	252.9	253.1	UKBH-04/CS/7/16-17	Ore Horizon interbanded with quartzite and carbonate material.	29.5999
	253.1	253.3	UKBH-04/CS/8/16-17	Ore Horizon interbanded with quartzite and carbonate material.	17.7661
	253.3	253.5	UKBH-04/CS/9/16-17	Ore Horizon and Gneissic material	12.1823
UKBH-05	a) Run 194 to 197.10 m.				
	196.92	197.10	UKBH-05/CS/1/16-17	Ore Horizon interbanded with quartzite and carbonate material.	16.7516
	b) Run 197.10 to 200 m.				
	197.12	197.93	UKBH-05/CS/2/16-17	Ore Horizon interbanded with quartzite and carbonate material.	10.2229
	197.93	197.52	UKBH-05/CS/3/16-17	Ore Horizon interbanded with quartzite and	21.4758

				carbonate material.	
	197.52	197.72	UKBH-05/CS/4/16-17	Ore Horizon interbanded with quartzite and carbonate material.	16.8677
	197.72	197.92	UKBH-05/CS/5/16-17	Ore Horizon interbanded with quartzite and carbonate material.	24.318
	197.92	198.12	UKBH-05/CS/6/16-17	Ore Horizon interbanded with quartzite and carbonate material.	11.2219
	198.12	198.32	UKBH-05/CS/7/16-17	Ore Horizon interbanded with quartzite and carbonate material.	5.04948
	198.32	198.52	UKBH-05/CS/8/16-17	Ore Horizon interbanded with quartzite and carbonate material.	10.5327
	198.52	198.79	UKBH-05/CS/9/16-17	Ore Horizon interbanded with quartzite and carbonate material.	10.9818
Run 251 to 254 m.					
UKBH-06	253.25	253.45	UKBH-06/CS/1/16-17	Ore Horizon	18.3082
	253.45	253.65	UKBH-06/CS/2/16-17	Ore Horizon	23.3732
	253.65	253.85	UKBH-06/CS/3/16-17	Ore Horizon	17.0614
	253.85	254.05	UKBH-06/CS/4/16-17	Ore Horizon with phyllite	1.13846

80 Kg of Manganese ore samples has been collected from core samples and have been submitted to IBM, Nagpur for beneficiation.

5.2.8 Discussion of results of chemical analysis of samples

To study the dispersion of elements in current block, samples were collected from mine sections as well as from road sections to study their chemical composition of all lithologies available in area. Elemental analysis of Mn, P, Si, Fe, Cu, Pb, Zn, Co, Ni and Cr were analysed in BRS and whole rock analysis was carried out in PCS to understand about protoliths and environmental conditions of metamorphism. A total of 51 nos. core samples, 20 nos. BRS, 10 nos. REE, 10 nos. PCS and 5 nos. of XRF samples were collected and analysed besides this 5 nos. of water samples were also collected and analysed. The results of core samples, BRS, PCS, REE and PS have been received. Results of XRF are yet to be received.

BRS samples indicated Mn value range from 3.11% to 56.241%, with P value ranging in from 0.04 to 1.67 %. Values of Fe vary from 1.31% to 14.196%. Due to high amount of quartz Si value is also high and varies from 2.832% to 41.03%. The elements like Cu, Pb, Zn, Ni, Co, Cr show sympathetic relations due to similar ionic radius. The Cu values range from 10 ppm to 850 ppm with one sample showing value upto 0.1%. Similar values of Pb, Zn, Ni, Co, Cr is observed. (Annexure-I)

Core samples were collected from whole ore horizon (ore plus waste rock). Individual details of samples are mentioned in Annexure-II. Here value of Mn found to varying from 5.0494 to 39.7298%. A summary of ore along with their weighted average is mentioned below in tabular form. Determination of weighted average of Mn is mentioned in Annexure-II. Here dilution is observed in UKBH-1 due to core loss. Therefore value of UKBH-1 will not be used in reserve estimation.

Total ore horizon (ore + waste) intersected in Western Ukwa Block					
Section Line	BH	Total Drill Depth in meter	Ore horizon in meter	Only Ore	Weighted average
AB	UKBH-1	113.35 m	83.0 -83.15m	0.15m	5.878%
	UKBH-5	200 m	196.92-198.79m	0.40m	23.0114%
CD	UKBH-2	116 m	100.45-102.50	1.40m	33.6151%
	UKBH-4	254 m	251.70-253.5 m	1.20m	29.9035%
EF	UKBH-3	128 m	123.75-127.425	2.40m	27.9785%

	UKBH-6	260 m	253.25-253.95	0.20m	23.3732%
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The Average values of Si for ore in six boreholes were found to be varying from 12.34% to 26.04%. Similarly for P it was found to be ranging from 0.07% to 0.56%. In iron and steel industry, the BIS: 1281-2005 (Reaffirmed 2008) specification is laid down for manganese ore. The specifications based on the user industry indicate that normally manganese ore containing 28 to 35% Mn is used. Ore size generally varies from 10 to 40 mm. For other constituents general stipulations are Fe: 16 to 22%, SiO₂: 2 to 8%, Al₂O₃: 5 to 8% and P: 0.3% maximum. However as per IBM mineral book 2014 there are industries like Nav Bharat Ferro-Alloys Ltd, Paloncha, Khammam(Specifications of ore consumed Mn: 28-50%, P: 0.1-0.3%, Si: 3.73328-13.9998% , Fe :5-8%) and Monnet Ispat & Energy Ltd, Raipur (Specifications of ore consumed Mn: 28 - 46%, Si: 2.799 – 15.866%, P 0.28%) which uses High Si ore. Based on these values ore horizon of three boreholes (UKBH-2, 3 and 4) of current FSP can be used for resource estimation. Though in case of UKBH-3 and UKBH-4, P value is slightly high but it can be compensated by mixing with high grade ore.

The analytical results of PCS samples (Annexure-V) are showing encouraging value of MnO ranging from 6.73% to 39.76%. Out of 10 nos. of PCS samples, 4 nos. of samples which were collected from Mn horizon of UKBH-2, 3, 4 and 8 were showing anomalous value of MnO ranging from 29.06% to 39.76%. Concentration of trace element viz. Ba is very high in these samples and ranging from 5497ppm to 39000 ppm and value of Sr is also high in 2 nos. of samples viz. PCS-6 (1120 ppm) and PCS-10 (1012 ppm). Higher values of Ba and Sr are observed which may be due to compatible relationship with Mn. Ba and Sr could have its origin from mica, feldspar, granites and clay.(another possibility is vein bearing quartz and calcite found in area) Other trace elements in PCS samples like Nb, Rb, Y, Zr are present in below detection limit (non compatible relation with Mn)

The concentration of REE elements (Annexure-VI) in 10 samples collected from granite gneiss, soil and Mn horizon are not showing encouraging value. Though some of the REE elements viz. La and Ce are showing higher values 318.92 ppm and 503.68 ppm respectively in sample no. REE-5. However some conclusions can be drawn for protolith from which material was eroded and deposited (later metamorphosed) and of gneiss; higher values of LREE indicates partial melting of LREE enriched source and negative Eu anomaly indicates fractionally

crystallized plagioclase or equilibrium of source with plagioclase bearing mantle from which earlier protolith was derived. (Fig 5.9)

Spider plot – REE chondrite (Nakamura 1974)

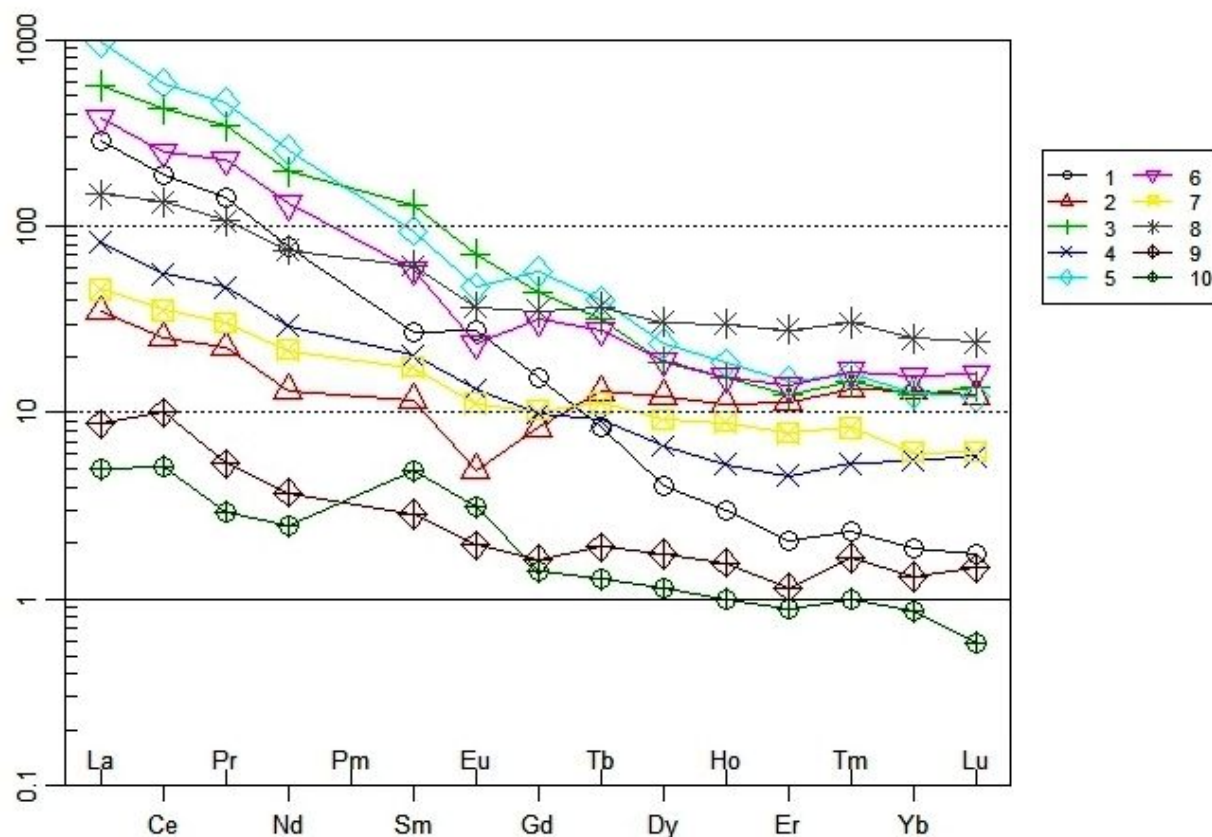


Fig 5.9: Plot of Chondrite Normalized value of REE samples (After Nakamura 1974).

5.2.9 Details of interpreted ore zone on the basis of geological investigation

During the course of G-2 stage exploration in parts of Toposheet nos. 64C/5 in Western Ukwa block, a total of 1071.35 m of drilling was done and 6 nos. of boreholes were drilled at second and third level to know strike and depth continuity of manganese horizon. The composite thickness of 10.01 m ore horizon was intersected in 6 different boreholes (UKBH-1, 2,3,4,5 & 6). The Maximum thickness of ore horizon was observed in UKBH-3 (3.61 m) and minimum

thickness of ore horizon was observed in UKBH-01 (0.15 m). Details of ore zone are provided in chapter 8.

5.3 Geophysical Explorations

As per earlier geophysical exploration (Gaonkar & Das, 1995), geophysical surveys like gravity, magnetic (V.F.) and self potential methods were taken up in Ukwa (21°58'30", 80°28'W) area, Balaghat district, M.P. for finding out extension along strike for the Ukwa manganese deposit in the soil covered gap area between Ukwa deposit and Jagantola trench in the Western Block and to the east of the Eastern Block (Fig. 5.10, 5.11, 5.12, 5.13).

The strata bound manganese bodies are located between the mica schist and the phyllites metasedimentaries. The metasedimentaries are overlying the granite gneiss basement. The bedding and the foliation trends are along ENE-WSW direction with moderate dips of about 25° to NW. The manganese horizon is exposed over a strike length of about 5 km and the thickness of the horizon ranges from 0.27 m to 6.1 m.

Two major magnetic anomalies, in the form of linear bands with strike length of about 3.5 km each and magnitude in the range of 100 to 400 gammas were picked up in the western block of this area (Fig. 5.10). The width is about 25 and 100 m for the southern and the northern anomalies respectively. The southern magnetic high was recorded over the main Ukwa mine trench, where the ore body with average width of 3 m is exposed at the surface. The anomaly shows a dominant positive high with absence of negative low to the north. This shape is attributed to the northward dip of the ore body. The gravity surveys (Fig. 5.11 & 5.12) closely corroborate the results of magnetic surveys in this area. Two linear gravity anomalies corroborate with the observed linear magnetic anomalies. The geophysical surveys indicate strike continuation of the exposed manganese mineralisation under soil cover, from Ukwa trench in the east to Jagantola trench in the west, for over 1.6 km. Presence of another gravity-magnetic linear anomaly in the soil covered area to the north of the ore body exposed in the main trench indicates interesting possibilities for the presence of manganese mineralisation. S.P. surveys were also carried out in this area (Fig. 5.11). The surveys have brought out feeble but definite low values of the order of 20 to 80 mv and corroborate with the gravity magnetic response over the manganese ore body exposed in trench and also support its extension to the west over a strike length of 1.6

km. No S.P. response was recorded over the northern gravity magnetic anomaly band. This could be due to the thick soil cover and the causative body being at large depth.

A magnetic peak with narrow width was picked up over manganese pit on Tr. TU4 in the eastern block. However this anomaly was not found to continue over the next traverse located at an interval of 300 m to east. From the appearance of rapid fluctuations in magnetic values on traverses laid further to the east in this block, it is inferred that Sausar formations have gone under Deccan Trap rubble.

5.4 Geochemical Explorations

Geochemical exploration have not been carried out in TS no 64C/05 so far due to inaccessibility in area (hilly terrain, dense jungle) and Naxals activities reported around.

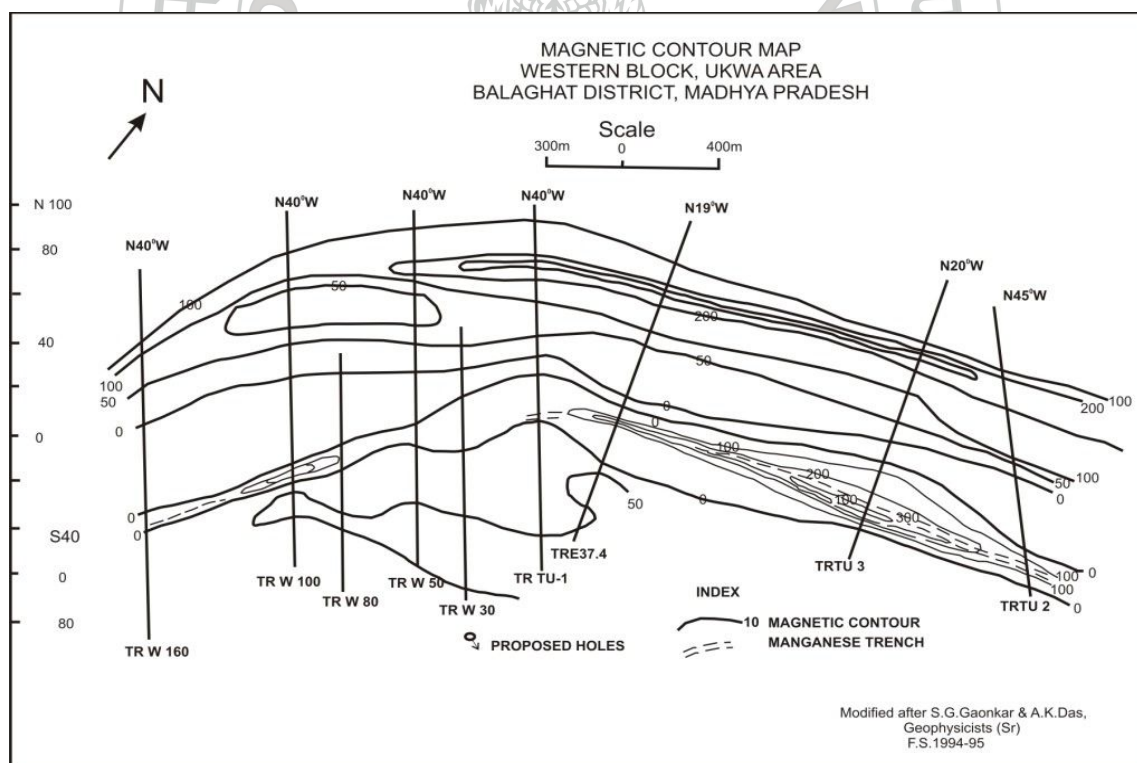


Figure 5.10 : Magnetic contour map of Western Ukwa Block (Reproduced from Gaonkar & Das, 1995).

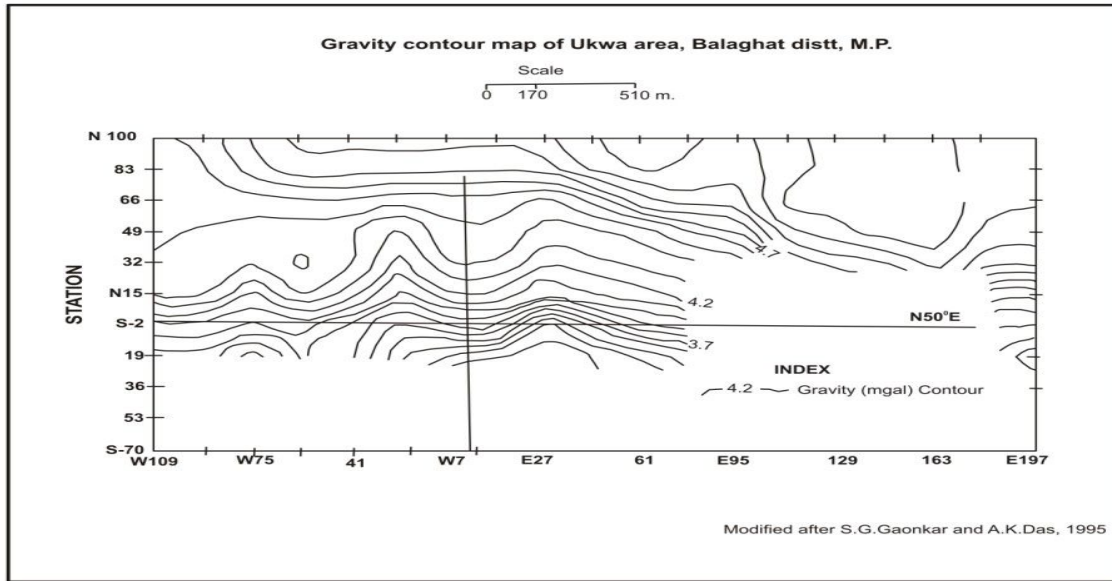


Figure 5.11 : Gravity contour map of Western Ukwa Block (Reproduced from Gaonkar & Das, 1995).

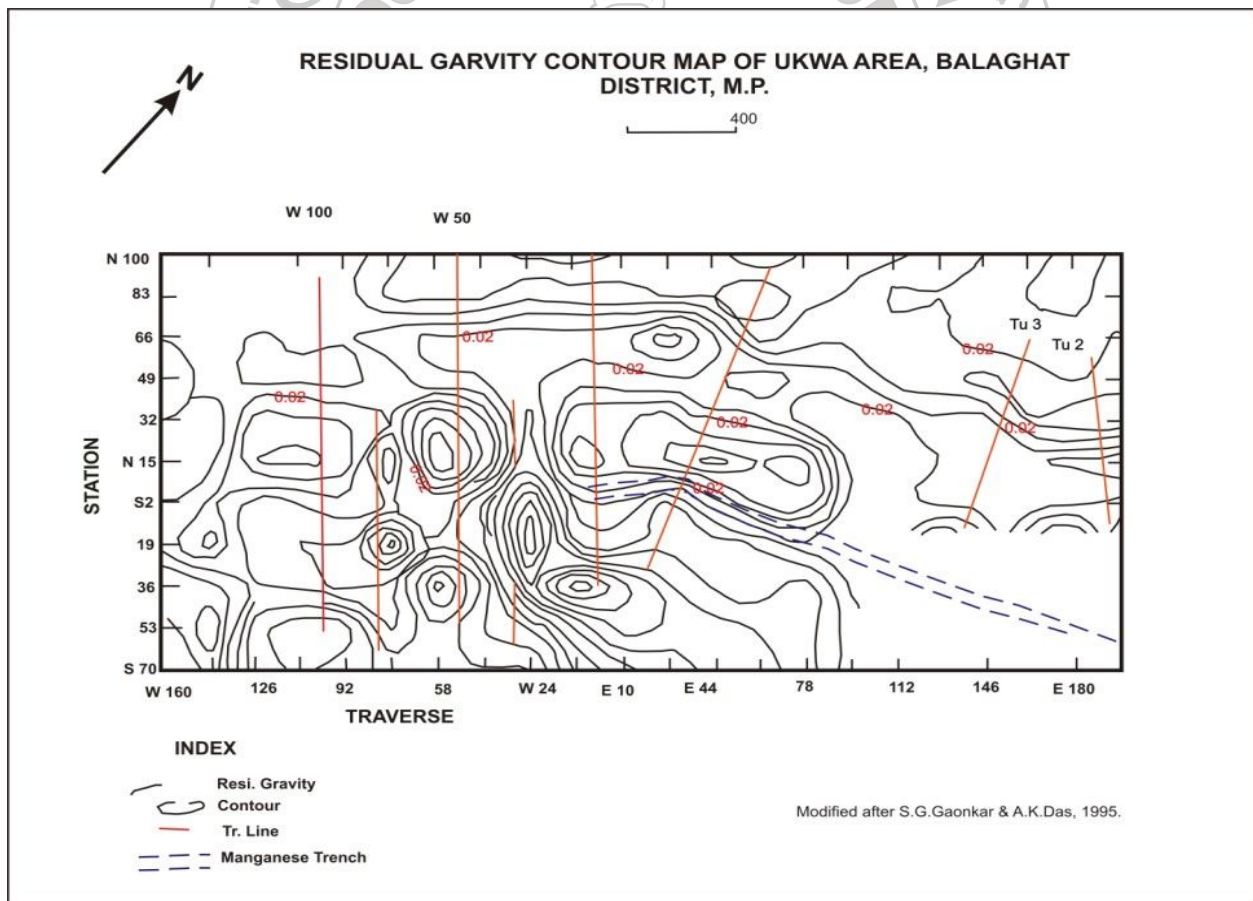


Figure 5.12 : Residual Gravity contour map of Western Ukwa Block (Reproduced from Gaonkar & Das, 1995).

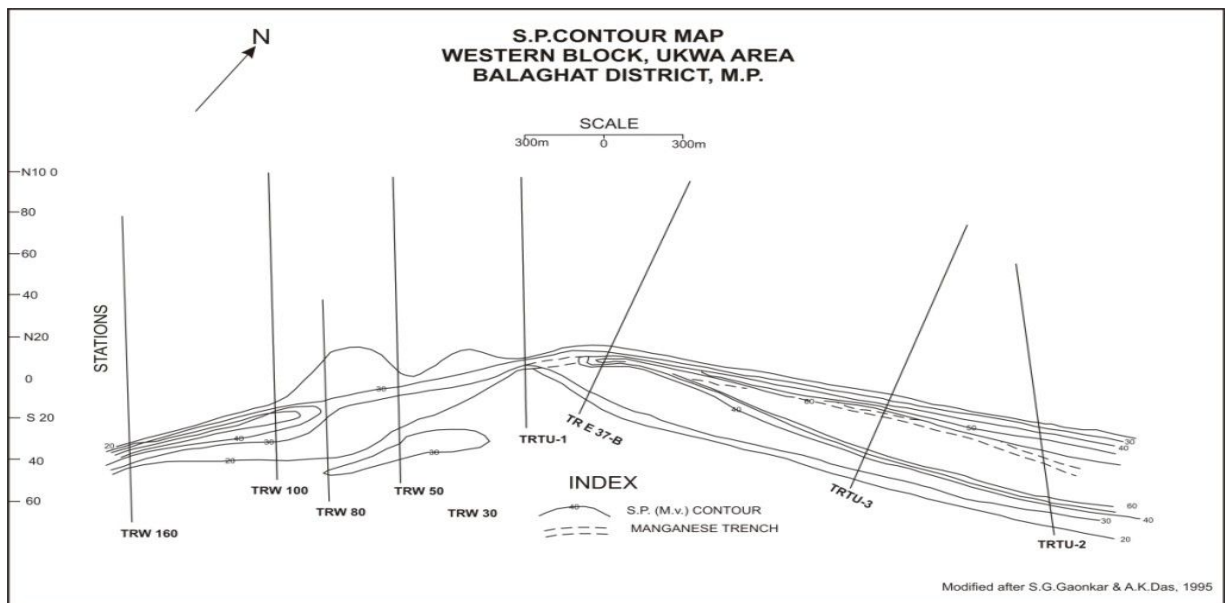
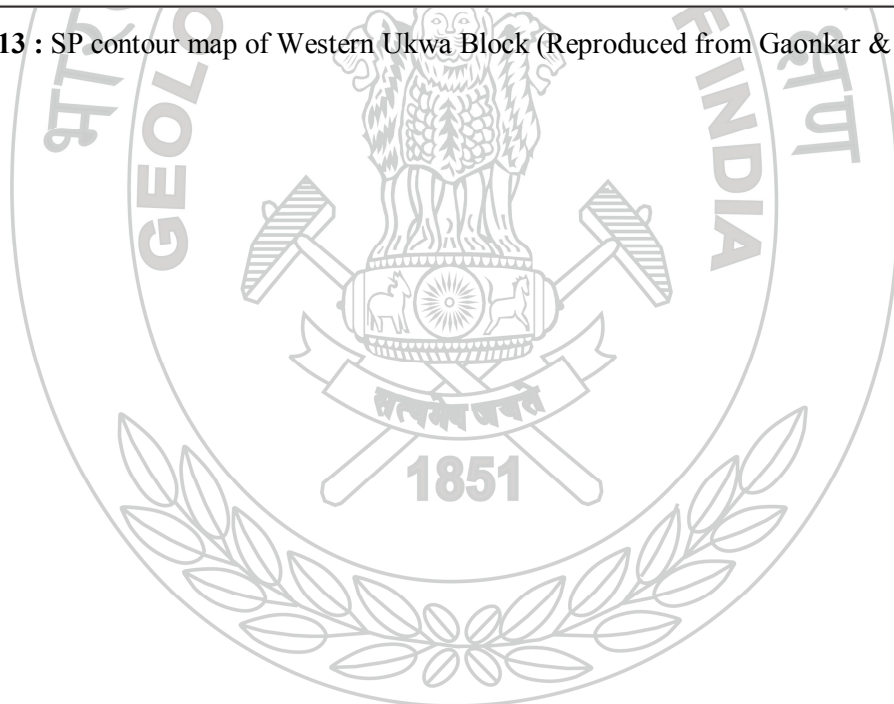


Figure 5.13 : SP contour map of Western Ukwa Block (Reproduced from Gaonkar & Das, 1995).



INTEGRATION OF GEOLOGICAL, GEOCHEMICAL AND GEOPHYSICAL EXPLORATION DATA AND INTERPRETATION THERE OFF

6.1 Creation of Geophysical, Geochemical, Geological (Lithological & Structural) and Outcrop maps on true scales

Only drilling components along with collection of samples were involved during current FSP that's why no geophysical, geochemical, geological (lithological & structural) and outcrop maps on true scales were prepared. However, earlier workers geophysical and geological (lithological & structural) and outcrop maps were provided along with interpretation in chapter 5 and Plate-III (Fig 5.10, 5.11, 5.12 and 5.13, references therein).

6.2 Overlay studies in GIS (Geological, Geochemical and Geophysical maps)

No overlay studies were carried out due to unavailability of maps on true scale. During current study detailed geological map of Khan and Fulmari, 1996 were used and digitized along with borehole locations of present as well as previous work in GIS for better representation. However, geophysical maps were digitized in Corel-12 software for better representation.

6.3 Semitransparent overlay maps

Semitransparent overlay map could not be prepared because of unavailability of map on true to the scale. However, soft copies of shape files are submitted for detailed geological map along with hard copy in CD/DVD.

ABIOTIC PARAMETERS

7.1 Soil, Surface water (Pre-monsoon and post monsoon), ground water and air sampling

The present area comprises of Mansar Group of rocks which are result of low grade metamorphism of pelitic rocks in the area. The lithounit consist of phyllite, quartz-muscovite-sericite-schist, Mn ore horizon and basement biotite gneiss mainly. Residual soils are common in the area and soil profile is well developed. Gully and sheet erosion is the major kind of soil erosion in this area. On the basis of texture, soil in the area are widely classified into four classes-

- A). Coarse soil:** It is mainly stony and gravelly soils characterized by low-grade soils with admixture of cobbles, pebbles and gravels generally found at the foot hills and along nalas.
- B). Medium soil:** It is sandy in nature and generally found near the river and stream beds with 60-70 percent sand and poor plant nutrients.
- C). Fine soil:** It loamy soils composed mostly of sand and silt with smaller amount of clay (40%-20% concentration respectively). This is the most common type of soil in the area.
- D). Very fine soil:** It is mainly clayey soils characterized by sticky when wet and very hard when dry.

5 nos. of water samples (post monsoon) were collected from dug well, hand pump and open pit to study about abiotic parameter. The details of water samples are given in below table no. 7.1.

Sl no.	Sample no.	Lat: Long	Remarks
1	64C05/SK/UKW1/ FS16-17	21°58'26":80°26'55"	pH 7.8, conductivity 393 μ s/cm, TDS 236ppm, Temp. 20.39°C
2	64C05/SK/UKW2/ FS16-17	21°58'24":80°26'52"	pH 6.76, conductivity 384 μ s/cm, TDS 230ppm, Temp. 25.35°C
3	64C05/SK/UKW3/ FS16-17	21°58'22":80°26'49"	pH 6.2, conductivity 228 μ s/cm, TDS 137ppm, Temp. 25.19°C

4	64C05/SK/UKW4/ FS16-17	21°58'26":80°26'41"	pH 6.12, conductivity 425µs/cm, TDS 255ppm, Temp. 26.22°C
5	64C05/SK/UKW5/ FS16-17	21°58'19":80°27'26"	pH 7.7, conductivity 83µs/cm, TDS 50ppm, Temp. 24.09°C

On the spot analysis of surface water indicates that manganese rich water shows low conductivity and TDS. In general the water condition in the area is good, less contaminated and drinkable. Chemical analysis of water sample indicates high value of Mn and Fe which is related to contamination of ground water by leaching from mine dump and geological formations. Rest of the values of elements is within crustal abundance limit.

Ground water condition in the area is quite good. The use of ground water is almost negligible in the area. Mica schist being the major litho unit in the country acts as good aquifer. Luxuriant growths of natural vegetation like palm and jamun trees indicate the shallow water table condition.

No air sampling was carried out during the course of drilling in Western Ukwa block.

MINERAL DEPOSIT

8.1 Surface indication of mineralization

A NE-SW trending ridge exposes lithounits of Mansar Group of rock viz. phyllite, quartz-muscovite-schist and Mn ore horizon (at places) in western Ukwa block. Earlier workers collected bed rock samples from manganiferous zone and analyses. The encouraging value of Mn leads to drilling in Ukwa area. At present there are no surficial outcrops present in the area. All the outcrops have been mined out in open cast mines by MOIL. In present scenario, drilling is performed here on the basis of previous work. To have idea about nature of mineralization underground mines and open cast mines were visited during the regional traverses. No surficial indication such as oxidised zone was encountered.

8.2 Mode of occurrence

Manganese ore horizon is part of Mansar horizon. However there is transgressive relationship between lower phyllitic (calcareous) and upper ore bearing Mansar formation as calcite is found in lower part of ore horizon in the form of thin veins. Mn ore occurs as <1 cm to 16 cm thick bands interbanded with cherty to jaspery manganiferous and ferruginous quartzite and impure calcareous rock as quartz-sericite/muscovite-calcite schist. The ore is either massive in nature or is banded with mm thin quartzite and carbonate partings. Very often biotite occurs as thin layers along bedding plane. The ore is dark grey coloured and fine grained and the ore horizon at places shows fractured to brecciated nature. The interstices in brecciated and fractured zones are occupied by quartz and carbonate veinlets and encrustations. The fracturing and brecciation deteriorates the quality of the ore.

8.3 Nature of Mineralisation

The ore horizon is shallow, thin and pockety at the eastern end of Ukwa deposit, whereas it is thin but continuous at the western end of the deposit. The surface and subsurface data have clearly indicated the syn-sedimentary stratiform origin of the ore, as evidenced by lithological association and conformable nature of ore bands with them. It shows pinch and swell structures, which may be due to combined effect of depositional feature with different phases of deformation. The culmination and depression in the levels of intersection of ore horizon along the strike and dip direction is perhaps because of combined effects of multiple deformation (Fig. 8.1).

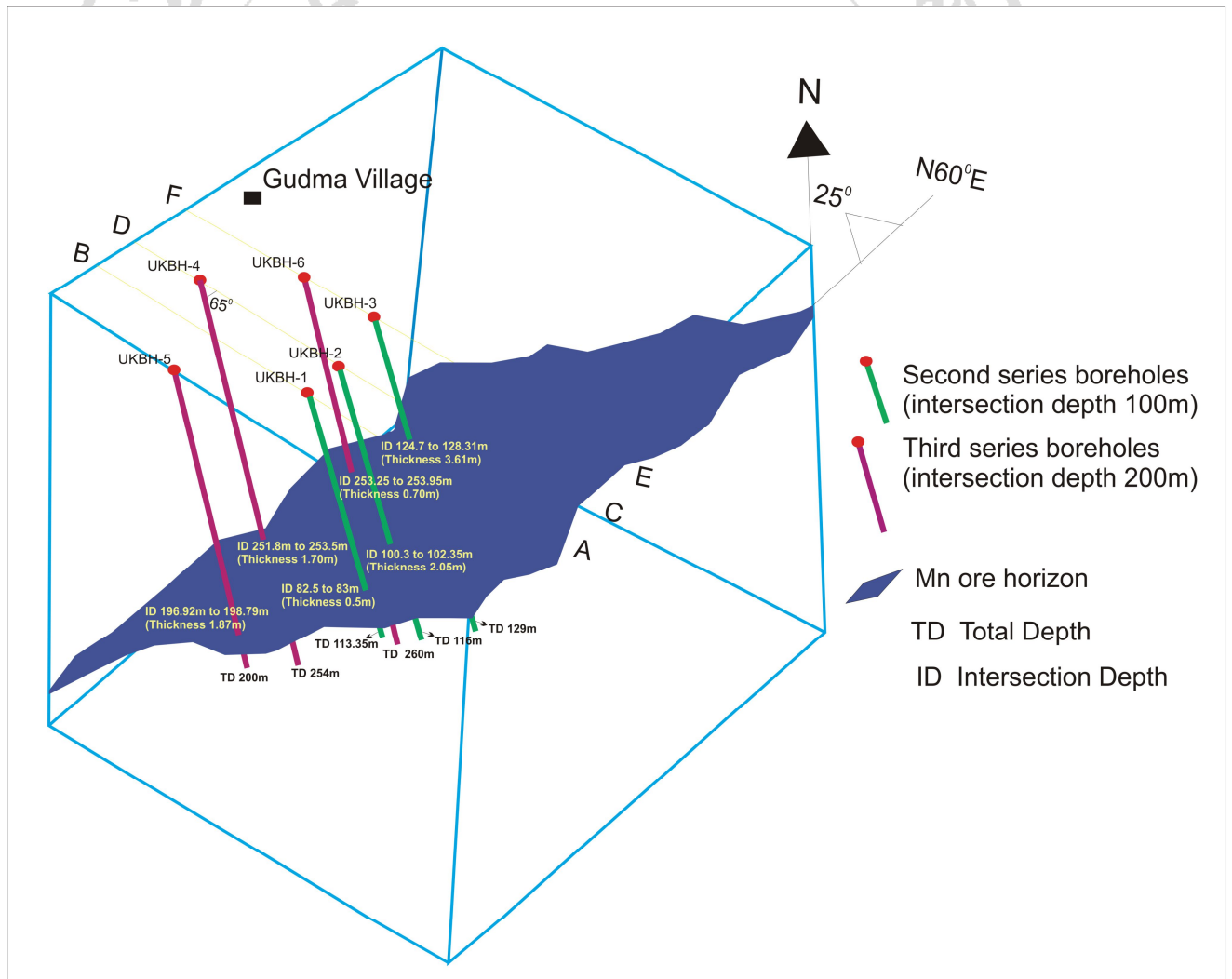


Figure 8.1 3D model of manganese mineralization in western Ukwa block.

8.4 Details of Mineralised Zone (ore plus waste rock within ore horizon)

Ukwa deposit is having a strike length of 5.4 km in total. The present western Ukwa block extends over a strike length of 1860 m and forms the western continuation of the Central Block. Width of mineralised zone found in this block varies from 0.15 m to 3.65 m (thickness of ore horizon as found during drilling). No geophysical or geochemical exploration is carried out during current work as objective of work was to augment the resource and increase confidence level. Details of mineralized zone are given in table no. 8.1 below as intersected during drilling.

Section Line	BH	Total Drill Depth in meter	Mineralized Zone (MZ) in meter	Thickness of MZ in meter
AB	UKBH-1	113.35 m	83.0 -83.15 m	0.15 m
	UKBH-5	200 m	196.92-198.79 m	1.87 m
CD	UKBH-2	116 m	100.45-102.50 m	2.05 m
	UKBH-4	254 m	251.70-253.5 m	1.80 m
EF	UKBH-3	128 m	123.75-127.425	3.645 m
	UKBH-6	260 m	253.25-253.95	0.70 m

Table 8.1 Depth of intersection of mineralization in Western Ukwa Block

8.5 Alteration zones and its relevance with mineralization

As there is no role of metamorphic fluids in deposition of ore so no alteration zones are expected in the area. As well as post depositional deformations have resulted only in change of disposition and structure of ore horizon (minor changes). Thus no alteration zones are observed in the area.

8.6 Genesis of Mineralisation/ Genetic model for mineralisation

The composition of footwall and hanging wall rocks and association of mangiferous cherry quartzite, carbonate rock with phyllitic intercalation in ore horizon suggests that the total association is characteristic of chemopelitic environment of deposition. The contacts of ore horizon with footwall and hanging wall rocks are quite sharp. Even within the ore horizon

different litho units have sharp contact with each other. The sequence could be represented as below:

Hanging Wall: Quartz sericite schist

Ore horizon

Foot Wall: Sericitic phyllite and schist with calcareous intercalations

The ore horizon has conformable relationship with hanging wall and footwall rocks. It has quite distinct lithological characters mostly representing chemogenic association within chemopelitic to pelitic sequence forming footwall and hanging wall rocks (Khan and Fulmari, 1996). No genetic model is being suggested by present authors.

The Proterozoic (c. 2.0Ga) Sausar Group, India, including the manganese deposits, have been complexly deformed and metamorphosed to grades ranging from lower greenschist facies to upper amphibolite facies. This sequence consists of metamorphosed equivalents of a limestone-shale-orthoquartzite assemblage (Roy 1966, 1981) indicating a shelf environment. The presence of dolomite at the top (Bichua Fm) indicates further shallowing of the basin. Interbanded Mn oxide orebodies (braunite, bixbyite, hollandite, jacobsonite, hausmannite) and Mn silicate-oxide rocks are interstratified with metapelites and orthoquartzites (Mansar Fm) and less commonly occur as conformable lenses in carbonate rocks of the older Lohangi Fm. The protoliths of these Mn rich rocks were Mn oxide/hydroxide sediments admixed with variable amounts of Fe, Si, and Al. The Mn silicate-carbonate rocks, derived from Mn carbonate progenitor with admixed impurities, also occur as isolated pockets in the Mn oxide ore horizon of the Mansar Fm. The Mn oxide deposits are inferred to have formed on the continental shelf during a sea-level highstand, when detrital supply was minimal. Deposition above the redox interface produced Mn oxides even on the carbonate substrate (Lohangi Fm). Mn carbonate sediments in the Mansar Fm were diagenetically derived from Mn oxides by reaction with calcareous partings in isolated pools where an evaporative condition developed. (Genetic diversity of manganese deposition in the terrestrial geological record - After S. Roy)

EXPLORATION BY DRILLING

9.1 Stage of exploration by drilling

The UNFC consists of a three dimensional system with the following three axes: Geological Assessment, Feasibility Assessment and Economic viability. The process of geological assessment is generally conducted in stages of increasing details. The typical successive stages of geological investigation i.e. reconnaissance, prospecting, general exploration and detailed exploration, generate resource data with a clearly defined degrees of geological assurance. These four stages are therefore used as geological assessment categories in the classification. The four stages of geological assessment are represented by 4 codes viz. (G-1) detailed exploration, (G-2) general exploration, (G-3) prospecting and (G-4) reconnaissance. As per UNFC classification, this exploration can be classified as G-2 stage as drilling is being carried out for augmentation of resource in between previous boreholes.

9.1.1 Field guidelines for adoption of United Nations Framework Classification

Manganese horizons in sedimentary and metasedimentary sequences of stratiform and stratabound nature are classified for UNFC norms on the following basis:

1. Geological survey:

- (i) Mapping on 1:25,000 to 1:5,000 or larger scale with triangulation points, benchmarks, if any shown.
- (ii) Linking of maps so prepared with topogrid.
- (iii) Assessment of lithology, structure, surface mineralisation, analysis of old history of mining.

2. Geochemical survey:

- (i) Detailed litho geochemical channel sampling from fresh rock exposures, trenches, pits.
- (ii) Recording of deleterious elements, likely by-product elements (e.g. Ga in bauxite, Ni, PGE etc. in chromite, Au in Fe ore, etc).

3. Geophysical survey:

- (i) Borehole geophysical survey.
- (ii) Special geophysical traverses for problem solving, if required.

4. Technological:

- (a) Pitting/trenching: systematic pitting/trenching for deciphering extent of mineralization at surface.
- (b) Drilling: grid reduction needed: spacing Iron and manganese ore-100 to 200m.
- (c) Sampling:
 - (i) Systematic pit and trench sampling, deep pitting if necessary;
 - (ii) Core sampling: lithology and strength of mineralisation wise (check sampling -10%).
- (d) Laboratory scale scanning/chemical analysis,
- (e) Bulk sampling if necessary for testing processing technology.
- (f) Collection of abiotic geo-environmental parameters.

5. Petrographic: Study of petrographic character of rocks including mineral identification, grain size, texture etc.

9.2 Methodology of drilling with details of type of drilling

Diamond core drilling was used for drilling in Ukwa western block. The drill bit used here is composed of group of small, industrial grade diamonds set into a metallic, soft matrix. As the ground is drilled, the matrix will wear away and expose more diamonds. This bit is attached to a drill rod, which is around 10 feet in length, and then more members of pipe can be attached to the top of this so a greater depth can be drilled. Inside the drill rod, a core tube is attached to a cable via a latching mechanism. The core tube is lifted to the surface using the cable, so the solid core can be removed. There are two primary types of diamond drilling-rotary drilling and wireline drilling. Drilling work in Ukwa western block was outsourced to M/s Gemko-Kati. Core drilling (NX size) was carried out with KDR-750 machine using wireline drilling method.

9.3 Borehole planning

As the objective was to augment the resource so all the previous boreholes along with details listed in Annexure-III are taken into consideration for borehole planning even though all the first series of boreholes of previous work lies within the MOIL leasehold boundary. After survey work was carried out, it was found that no first series of boreholes could be drilled so it was proposed to intersect the ore horizon at 100 m and 200 m vertical depths as per FSP (Two deeper level intersections and grid spacing is reduced in between 100-200 m as per G2 rule). Earlier workers had drilled at spacing of 190 to 260 m. As the objective of work was to augment the reserve, spacing between section lines was proposed at 130 m interval. At ground conditions section lines were placed in range from 100 to 141 m. Graphical lithologs of current boreholes have been provided in Annexure-X.

9.4 Borehole Logging

Borehole logging was carried out and results are described as summarized litholog in tabular form. UKBH-1, UKBH-2 and UKBH-3 are second level boreholes while UKBH-4, UKBH-5 and UKBH-6 are third level boreholes. All the UKBH series boreholes are 65° South easterly inclined.

BOREHOLE NO. UKBH-1:

(A) Location: Western Block, Section Line No. A-B. Lat/Long: 21°58'13":80°26'54".

(B) R.L at Collar: 600.060 m.

(C) Date of commencement: 09/01/2017.

(D) Date of completion: 17/01/2017.

(E) Total Depth: 113.35 m.

(F) Vertical Depth of intersection: 75.22 m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	8	Yellow silty soil with quartz fragments.	

8	17	Lateritic soil with quartz fragments.	
17	26	Weathered garnetiferous muscovite schist.	
26	35	Garnetiferous quartz muscovite schist.	
35	68	Interbanded Garnetiferous quartz muscovite schist and quartzite with rare garnets.	
68	74	Quartz muscovite schist with rare garnets.	
74	83	Sericite schist(With Chlorite increasing downward)	
83	86	Sericite schist (with Mn ore). Mn ore is brecciated and interbanded with quartzite.	Core recovery = 0% during initial drilling. Mn ore of .15m was recovered during the re drilling at 83 m, core recovery =70% during re drilling.(Re drilling was started on 24/03/17; completed on 27/03/17) . Ore is 0.15m thick.
86	92	Phyllite	
92	113.35	Biotite Gneiss with veins of calcite and quartz and angular to rounded fragments of pegmatite	Initially very fine grained due to tectonic contact. Specks of magnetite and pyrite are observed within gneiss in disseminated form.

BOREHOLE NO. UKBH-2 :

(A) Location: Western Block, Section Line No. C-D.Lat/Long:21°58'19":80°27'00"

(B) R.L at Collar: 603.066 m.

(C) Date of commencement: 20/01/2017.

(D) Date of completion: 27/01/2017.

(E) Total Depth: 116 m.

(F) Vertical Depth of intersection: 91.03 m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	5	Yellow clayey soil	
5	14	Lateritic soil with quartz fragments	
14	26	Weathered garnetiferous muscovite schist	
26	35	Garnetiferous muscovite schist	
35	47	Quartz muscovite Schist	
47	62	Garnetiferous quartz muscovite schist	
62	85.95	Quartz muscovite sericite schist with rare garnets.	
85.95	95	Quartz Muscovite Schist	
95	100.45	Sericite Schist with chlorite.	
100.45	102.5	Mn ore of massive nature was observed with fine interbanding of quartz.(ore horizon)	Mn ore horizon of 1.43m was recovered, core recovery = 69.756%, core loss is .62 m. Brecciation is observed. Actual thickness of ore is 1.40m .

102.5	106	Phyllite.	
106	116	Biotite Gneiss with veins of quartz and calcite with angular to rounded fragments of pegmatite.	Initially very fine grained due to tectonic contact. Specks of magnetite and pyrite are observed within gneiss in disseminated form

BOREHOLE NO. UKBH-3:

(A) Location: Western Block, Section Line No. E-F.Lat:long.21°58'22":80°27'10".

(B) R.L at Collar: 601.208 m.

(C) Date of commencement: 29/01/2017.

(D) Date of completion: 05/02/2017.

(E) Total Depth: 128 m.

(F) Vertical Depth of intersection: 112.18 m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	14	Red lateritic soil with quartz fragments.	
14	23	Yellow silty soil with fragments of quartz, laterite and schist.	
23	32	Weathered garnetiferous muscovite schist.	
32	56	Garnetiferous quartz muscovite schist.	Stringers of hematite are found to be intruding along weak planes.
56	77	Quartz muscovite schist.	
77	97.95	Quartz muscovite schist with	

		chlorite.	
97.95	113	Sericite schist with chlorite.	
113	123.78	Sericite schist.	Specks of pyrite and magnetite are observed in it.
123.78	127.425	Mn ore horizon (Mn ore inter banded with Mn Quartzite) is traversed by carbonate and quartz veinlets. (Pink coloured carbonate material).	Core recovery is 100% with Mn bands varying from <1 mm to 11cm thick bands. Evidence of faulting (small scale and oblique slip) and brecciation is observed. Actual thickness of Ore is 2.40m.
127.425	128	Fine grained gneissic material.	Due to tectonic nature of contact.

BOREHOLE NO. UKBH-4:

(A) Location: Western Block, Section Line No. C-D.Lat:Long.21°58'25":80°26'55".

(B) R.L at Collar: 599.313 m.

(C) Date of commencement: 09/02/2017.

(D) Date of completion: 22/02/2017.

(E) Total Depth: 254 m.

(F) Vertical Depth of intersection: 228.11 m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	8	Reddish brown sandy soil with quartz fragments.	
8	26	Lateritic soil.	
26	44	Yellowish and weathered	

		quartz muscovite schist.	
44	59	Weathered quartz muscovite schist.	
59	89	Quartz muscovite schist.	
89	98	Quartz muscovite schist with rare garnet.	Specks of Magnetite are observed in disseminated form.
98	146	Quartz muscovite schist with chlorite and rare garnet.	Specks of Magnetite and Pyrite are observed in disseminated form.
146	218	Garnetiferous quartz muscovite schist.	
218	236	Sericite schist.(with increasing chlorite with depth)	
236	251.7	Quartz muscovite schist.	
251.7	253.5	Mn ore horizon (Mn ore inter banded with Mn Quartzite) is traversed by carbonate and quartz veinlets. (Pink coloured carbonate material).	Core recovery is 100% with Mn bands varying from <1 mm to 16 cm thick bands. Brecciation is observed. Actual thickness of ore is 1.20m.
253.5	254	Fine grained gneissic material.	Due to tectonic nature of contact.

BOREHOLE NO. UKBH-5:

(A) Location: Western Block, Section Line No. A-B.Lat:Long.21°58'19'':80°26'49'.

(B) R.L at Collar: 602.066 m.

(C) Date of commencement: 25/02/2017.

(D) Date of completion: 01/03/2017.

(E) Total Depth: 200 m.

(F) Vertical Depth of intersection: 178.47 m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	9	Yellow silty soil with rock fragments.	
9	17	Red lateritic soil.	
17	32	Yellow clayey soil with rock fragments.	
32	56	Yellow clayey soil along with weathered garnetiferous muscovite schist.	
56	71	Weathered garnetiferous muscovite schist.	
71	131	Garnetiferous quartz muscovite schist.	Stringers of Fe along weak planes.
131	170	Quartz muscovite schist.	
170	196.92	Sericite schist with chlorite.	
196.92	198.79	Mn ore horizon(Mn ore inter banded with Mn Quartzite) is traversed by carbonate and quartz veinlets. (Pink coloured carbonate material).	Mn ore horizon of 1.42 m was recovered, core recovery = 75.93%, 0.45 m was collected in powdery form. Brecciation is observed. Actual thickness of ore is 0.40m.
198.79	200	Fine grained gneissic material.	Due to tectonic nature of contact.

BOREHOLE NO. UKBH-6:

(A) Location: Western Block, Section Line No. E-F. Lat:Long.21°58'29":80°27'5".

(B) R.L at Collar: 599.837 m.

(C) Date of commencement: 05/03/2017.

(D) Date of completion: 16/03/2017.

(E) Total Depth: 260 m.

(F) Vertical Depth of intersection: 229.52 m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	11	Yellow clayey soil.	
11	20	Red lateritic soil with rock fragments.	
20	44	Yellow clayey soil.	
44	50	Yellow clayey soil with weathered quartz muscovite schist.	
50	98	Quartz muscovite schist with chlorite traversed by calcite vein along weak planes.	
98	116	Quartz muscovite schist with garnet.	Specks of magnetite is found in disseminated form
116	140	Quartz muscovite-sericite schist.	
140	158	Garnetiferous quartz muscovite schist.	
158	182	Quartz muscovite schist with chlorite.	
182	203	Garnetiferous quartz muscovite sericite schist.	

203	253.25	Quartz muscovite sericite schist.	
253.25	253.95	Mn ore horizon (Mn ore inter banded with Mn Quartzite) is traversed by carbonate and quartz veinlets. (Pink coloured carbonate material).	Core recovery is 100% with Mn bands varying from <1 mm to 5 cm thick bands. Brecciation is observed. Actual thickness of ore is 0.20m.
253.95	257	Phyllite	
257	260	Fine grained gneissic material.	Due to tectonic nature of contact.

9.5 Core Recovery Percentage

Core recovery percentage of individual boreholes for whole run was calculated and presented here in tabular form.

UKBH-1

From	To	Core Recovery
0	2	0
2	5	0
5	8	0
8	11	0
11	14	0
14	17	18.33
17	20	63
20	23	0
23	26	86
26	29	97.33
29	32	98
32	35	97.33
35	38	98.66
38	41	97.833
41	44	96.666
44	47	99.166
47	50	95.666
50	53	99.66

53	56	97
56	59	98.66
59	62	100
62	65	100
65	68	98
68	71	100
71	74	100
74	77	99.333
77	80	99
80	83	100
83	86	15.666
86	89	26.66
89	92	88.66
92	95	100
95	98	99.33
98	101	100
101	103.95	100
103.95	107	100
107	109.95	98.983
109.95	112.95	100
112.95	113.95	100

UKBH-2

From	To	Core Recovery
0	2	0
2	5	0
5	8	0
8	11	0
11	14	0
14	17	0
17	20	0
20	23	3.33
23	26	62.33
26	29	99.33
29	32	99
32	35	97.33
35	38	97.33
38	41	97.66
41	44	99
44	47	100
47	50	100
50	53	100
53	56	100

UKBH-3

From	To	Core Recovery
0	2	0
2	5	0
5	8	0
8	11	0
11	14	0
14	17	0
17	20	0
20	23	0
23	26	32.666
26	29	98.66
29	32	100
32	35	99.66
35	38	100
38	41	100
41	44	97.33
44	47	100
47	50	99.33
50	53	100
53	56	98
56	58.9	100

56	59	99.33
59	62	100
62	65	100
65	68	98
68	71	100
71	73.95	100
73.95	76.95	100
76.95	79.95	100
76.95	82.95	97.33
82.95	85.95	100
85.95	89	100
89	92	100
92	95	100
95	98	100
98	101	95.66
101	104	79
104	107	100
107	110	99
110	113	100
113	116	100

58.932	62	99.0323
62	64.95	98.644
64.95	67.95	100
67.95	71	99.672
71	73.95	100
73.95	77	100
77	80	100
80	82.95	100
82.95	86	100
86	89	100
89	92	99.33
92	95	100
95	97.95	100
97.95	101	99.344
101	104	99.667
104	106.9	98.62
106.9	110	99.354
110	113	100
113	116	98
116	119	100
119	122	100
122	125	100
125	128	100

UKBH-4

From	To	Core Recovery
0	2	0
2	5	0
5	8	0
8	11	0
11	14	0
14	17	0
17	20	0
20	23	0
23	26	0
26	29	21.66
29	32	21
32	35	36.6
35	38	24
38	41	0
41	44	17.66
44	47	47
47	50	65.33
50	53	36.6
53	56	52.33
56	59	38.6
59	62	75.33
62	65	20
65	68	80
68	71	99
71	74	100
74	77	94.6
77	80	100
80	83	100
83	86	100
86	89	100
89	92	100
92	95	100
95	98	100
98	101	100
101	104	100
104	107	100
107	110	100
110	113	100
113	116	100
116	119	100
119	122	100
122	125	100

125	128	100
128	131	100
131	134	100
134	137	100
137	140	98
140	143	100
143	146	100
146	149	100
149	152	100
152	155	100
155	158	100
158	160.95	98.98
160.95	163.95	99.33
163.95	167	100
170	173	100
173	176	100
176	179	100
179	182	100
182	185	100
185	188	100
188	191	100
191	194	100
194	197	98.33
197	200	100
200	203	100
203	206	100
206	209	100
209	212	100
212	215	100
215	218	100
218	221	100
221	224	100
224	227	100
227	230	100
230	233	100
233	236	99.33
236	239	100
239	242	100
242	245	100
245	248	100
248	251	100
251	254	100

UKBH-5

From	To	Core Recovery
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0	2	0
2	5	0
5	8	0
8	11	0
11	14	0
14	17	0
17	20	0
20	23	0
23	26	0
26	29	0
29	32	0
32	35	10
35	38	0
38	41	23.66
41	44	39.66
44	47	19.33
47	50	51
50	53	91.33
53	56	95
56	59	88.66
59	62	85.66
62	65	86.66
65	68	100
68	71	92.33
71	74	99
74	77	100
77	80	100
80	83	100
83	86	99.33
86	89	99.66
89	92	98
92	95	95.66
95	98	100
98	101	100
101	104	100
104	107	100
107	110	100
110	113	99.33
113	116	100
116	119	100
119	122	99.33
122	125	100
125	128	100
128	131	100
131	134	98.66
134	137	100
137	140	100

140	143	99
143	146	100
146	149	100
149	152	99
152	155	100
155	158	100
158	160.9	100
160.9	164	100
164	167	100
167	170	100
170	173	100
173	176	97.6667
176	179	100
179	182	100
182	185	100
185	188	100
188	191	99
191	194	100
194	197.1	98.66
197.1	200	100

UKBH-6

From	To	Core Recovery
0	2	0
2	5	0
5	8	0
8	11	0
11	14	0
14	17	0
17	20	0
20	23	0
23	26	0
26	29	0
29	32	0
32	35	0
35	38	0
38	41	0
41	44	0
44	47	36.33
47	50	51.33
50	53	100
53	56	100
56	59	100
59	62	100
62	65	100

65	68	99
68	71	100
71	74	100
74	77	100
77	80	100
80	83	100
83	86	100
86	89	100
89	92	100
92	95	100
95	98	100
98	101	100
101	104	100
104	107	100
107	110	100
110	113	98.3334
113	116	100
116	118.9	100
118.9	121.95	100
121.95	125	100
125	128	99.33
128	131	100
131	134	100
134	137	100
137	140	100
140	143	99.33
143	146	100
146	149	100
149	152	100
152	155	100
155	158	100
158	161	100
161	164	100
164	167	100
167	170	100
170	173	100
173	176	100
176	179	100
179	182	100
182	185	98.33
185	188	100
188	191	100
191	194	100
194	197	100
197	200	100
200	203	99
203	206	100

206	209	100
209	212	100
212	215	100
215	218	100
218	221	99.33
221	224	100
224	227	100
227	230	100
230	233	99.33

233	236	98.3
236	239	99.33
239	242	100
242	245	100
245	248	100
248	251	100
251	253.9	100
253.9	256.1	100
256.1	257	100
257	260	100

9.6 Geophysical logging of borehole and correlation with borehole Geology

Geophysical borehole loggings have been carried out in boreholes UKBH-1, UKBH-2, UKBH-3, UKBH-5 and UKBH-6, (except one borehole UKBH-4) Ukwa block, Balaghat, Madhya Pradesh by deploying Microllogger unit and acquired the data of Self Potential, Single Point Resistance, Natural Gamma and Resistivity by the geophysicist, GSI, Nagpur. Resistance / Resistivity logs in conjunction with SP log prove useful in detection of mineralized zones. The details of geophysical borehole loggings for each borehole is listed below:

1) UKBH-1

Total drilled depth – 113.95 m; Casing- 19.50 m

Reported mineralization is: -

(i) 83.00-86.00 m	(ii) 92.00-107.00 m
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2) UKBH-2

Total drilled depth – 116.00 m; Casing- 27.00 m

Reported mineralization is: -

(i) 100.45-102.50 m	(ii) 106.00-116.00 m
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3) UKBH-3

Total drilled depth – 128.00 m; Casing- 27.00 m

Reported mineralization is: -

(i) 32.00-56.00 m	(ii) 113.00-123.78 m	(iii) 123.78-127.43 m
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4) UKBH-5

Total drilled depth – 200.00 m; Casing- 51.00 m

Reported mineralization is: -

(i) 71.00-131.00 m	(ii) 196.92-198.79 m
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5) UKBH-6

Total drilled depth – 260.00 m; Casing- 44.00 m

Reported mineralization is: -

(i) 98.00-116.00 m	(ii) 253.25-253.95 m
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The conductive zones are identified on the basis of Low Single Point Resistance, Low Short & Long Normal Resistivity, High Self Potential and Low Natural Gamma. The details are given below:

1) UKBH-1

This borehole was logged on 17-01-2017 up to depth of 106.13 m and as per analysis and interpretation of the data of Self Potential, Single Point Resistance, Natural Gamma and Resistivity, one conductive zone has been identified between 81.35 – 96.00 m (Fig.9.1), which is listed below:-

(i) 81.35 – 96.00 m

2) UKBH-2

This borehole was logged on 04-02-2017 up to depth of 114.25 m and as per analysis and interpretation of the data of Self Potential, Single Point Resistance, Natural Gamma and Resistivity, one conductive zone has been identified between 100.30 – 103.20 m (Fig.9.2), which is listed below:-

(i) 100.30 – 103.20 m

3) UKBH-3

This borehole was logged on 08-02-2017 up to depth of 128.21 m and as per analysis and interpretation of the data of Self Potential, Single Point Resistance, Natural Gamma and Resistivity, one conductive zone has been identified between 115.10 – 120.40 m (Fig.9.3), which is listed below:-

(i) 115.10 – 120.40 m

4) UKBH-5

This borehole was logged on 02-03-2017 up to depth of 119.14 m and as per analysis and interpretation of the data of Self Potential, Single Point Resistance, Natural Gamma and Resistivity, four conductive zone has been identified between 97.50 – 197.20 m (Fig.9.4), which are listed below:-

(i) 97.50-98.85 m	(ii) 112.15-116.30 m
(iii)187.50-188.70 m	(iv)190.50-197.20 m

5) UKBH-6

This borehole was logged on 17-03-2017 up to depth of 253.53 m and as per analysis and interpretation of the data of Self Potential, Single Point Resistance, Natural Gamma and Resistivity, nine conductive zone has been identified between 60.00 – 246.00 m (Fig.9.5), which are listed below:-

(i) 60.00-61.00 m	(ii) 100.10-101.00 m	(iii)113.05-114.00 m
(iv)129.00-130.00 m	(v) 131.00-132.00 m	(vi)191.75-192.75 m
(vii)196.70-198.00 m	(viii)239.90-241.00 m	(ix)243.00-246.00 m

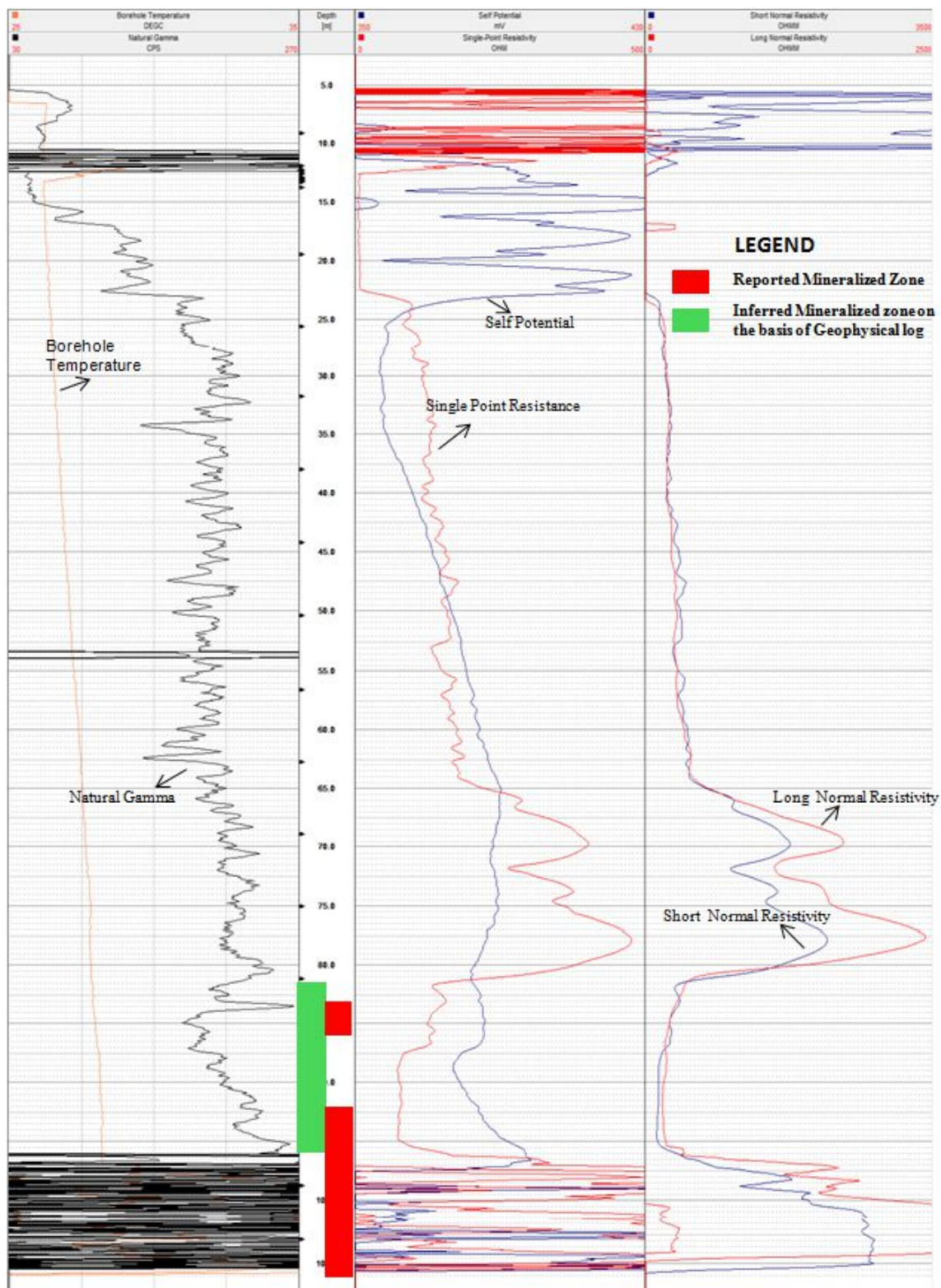


Figure 9.1 Geophysical log of borehole no. UKBH-1, Ukwa block, Balaghat, Madhya Pradesh.

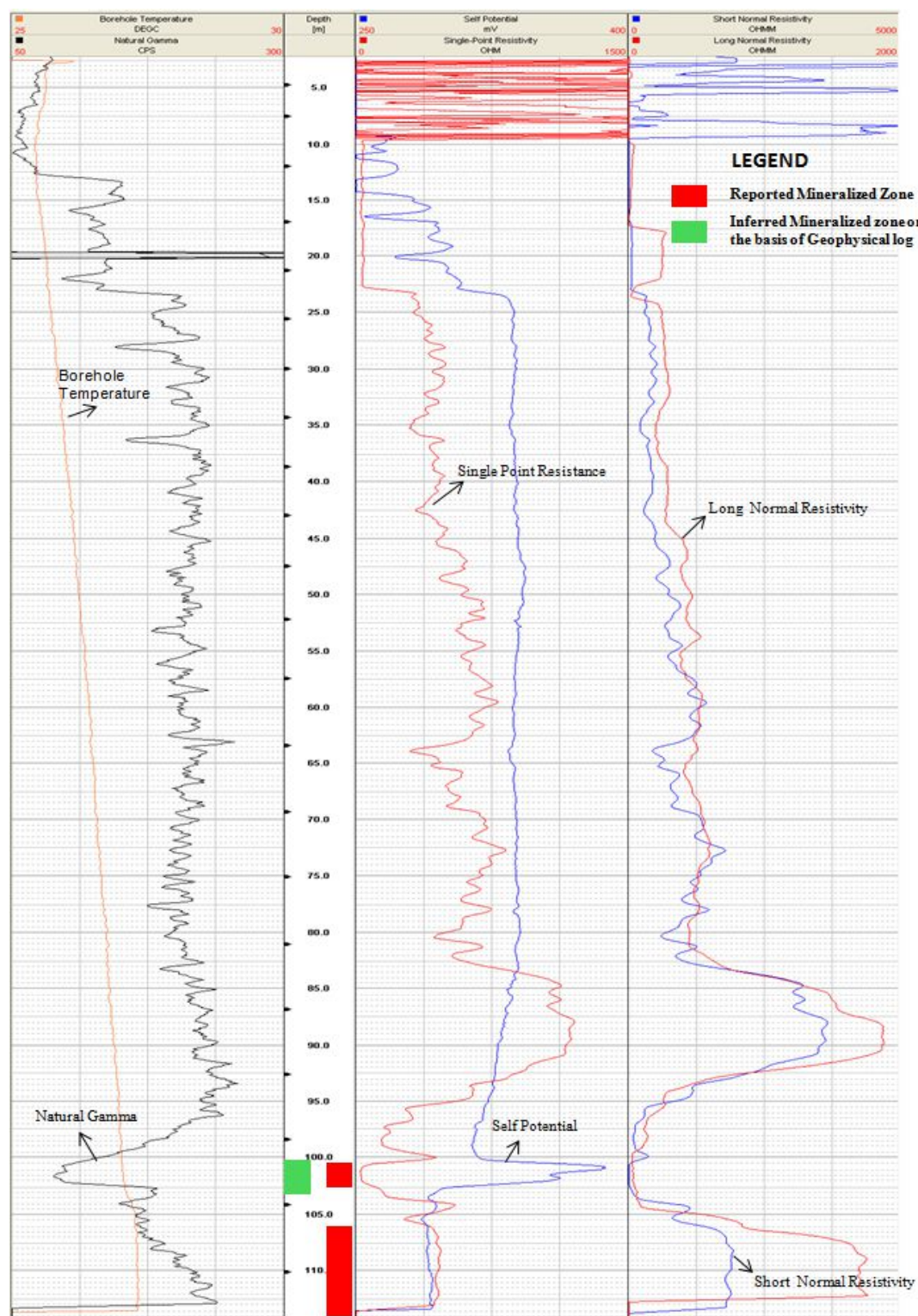


Figure 9.2 Geophysical log of borehole no. UKBH-2, U kwa block, Balaghat, Madhya Pradesh.

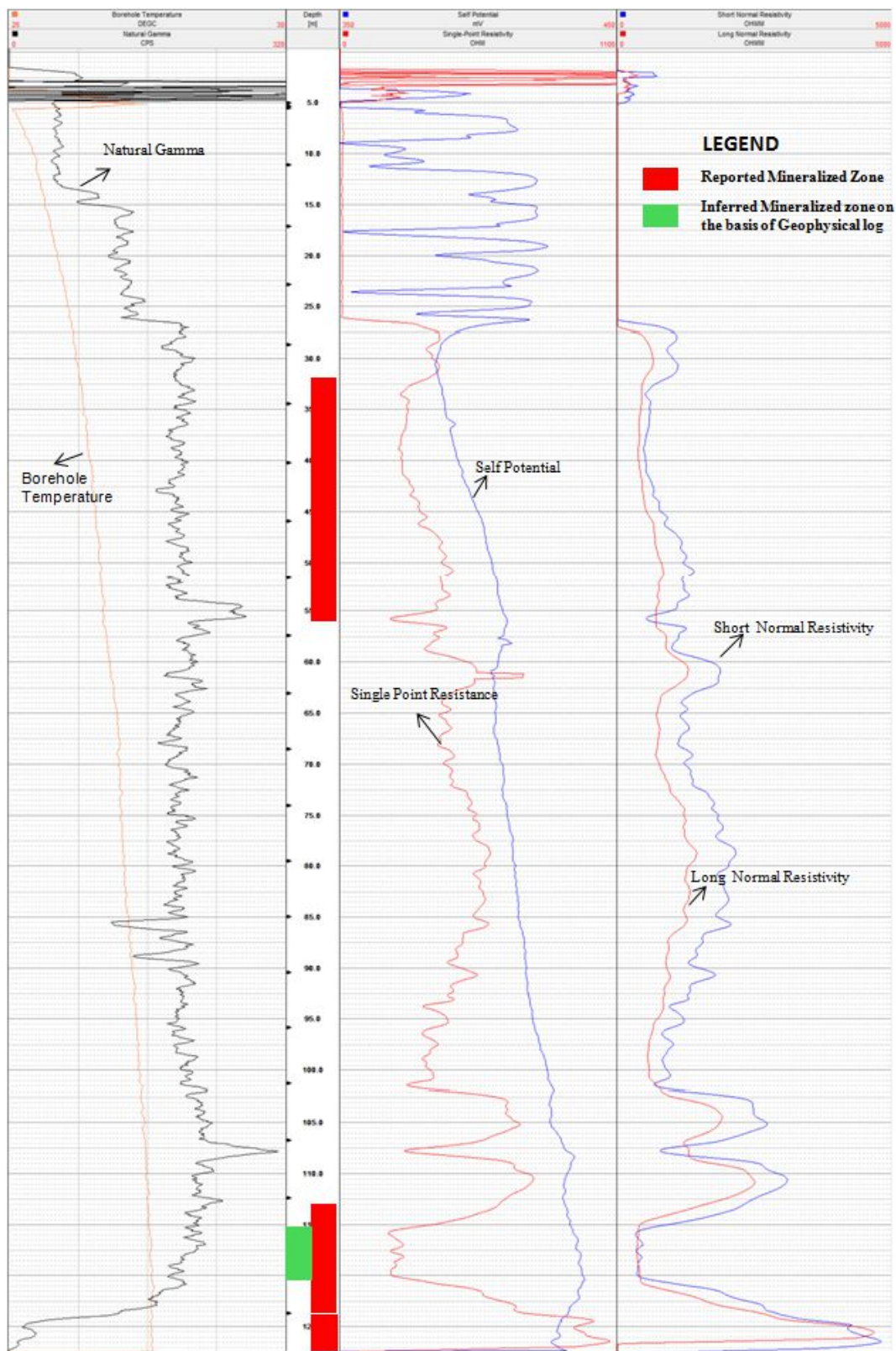


Figure 9.3 Geophysical log of borehole no. UKBH-3, Ukwa block, Balaghat, Madhya Pradesh.

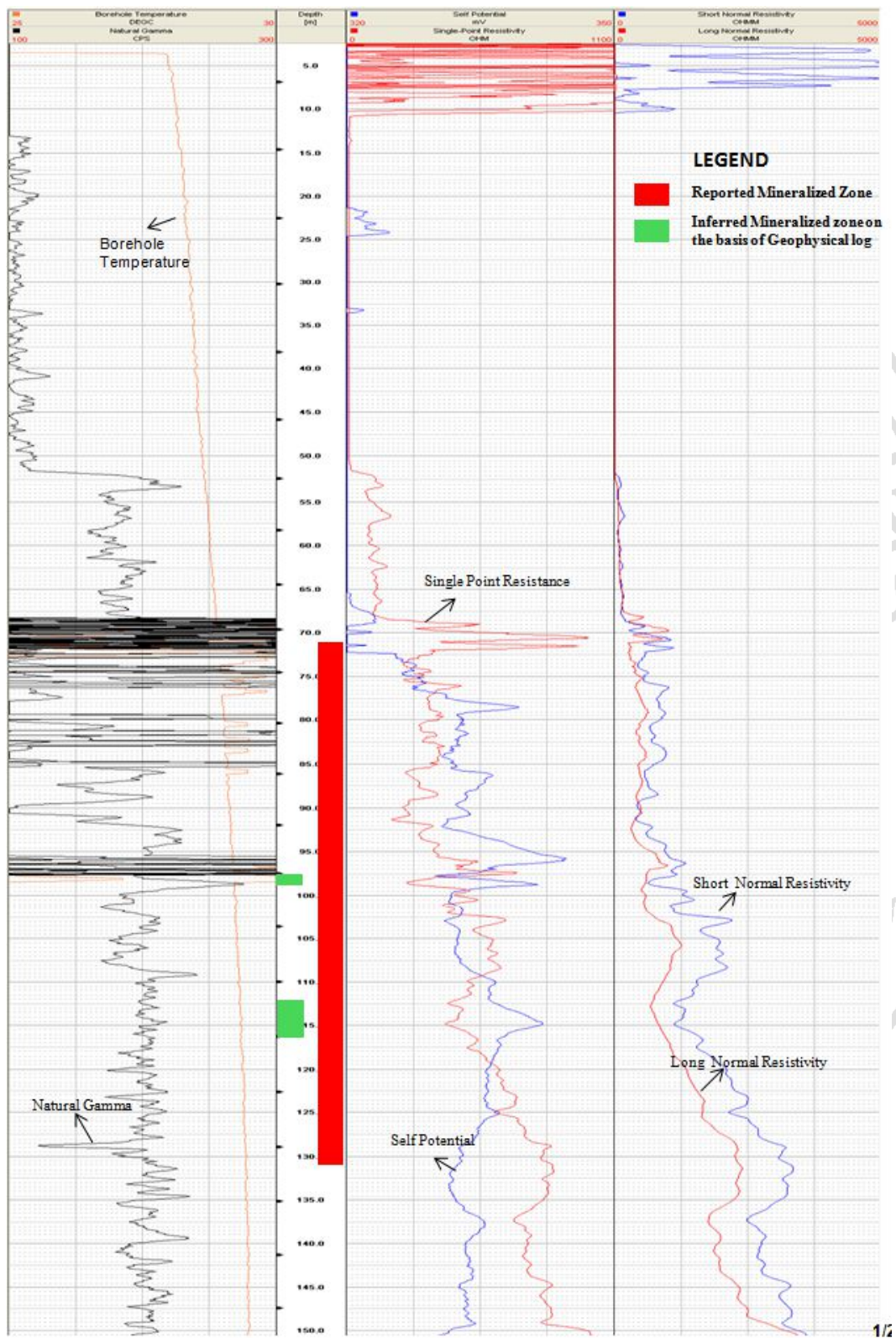


Figure 9.4 Geophysical log of borehole no. UKBH-5, Ukwa block, Balaghat, Madhya Pradesh.

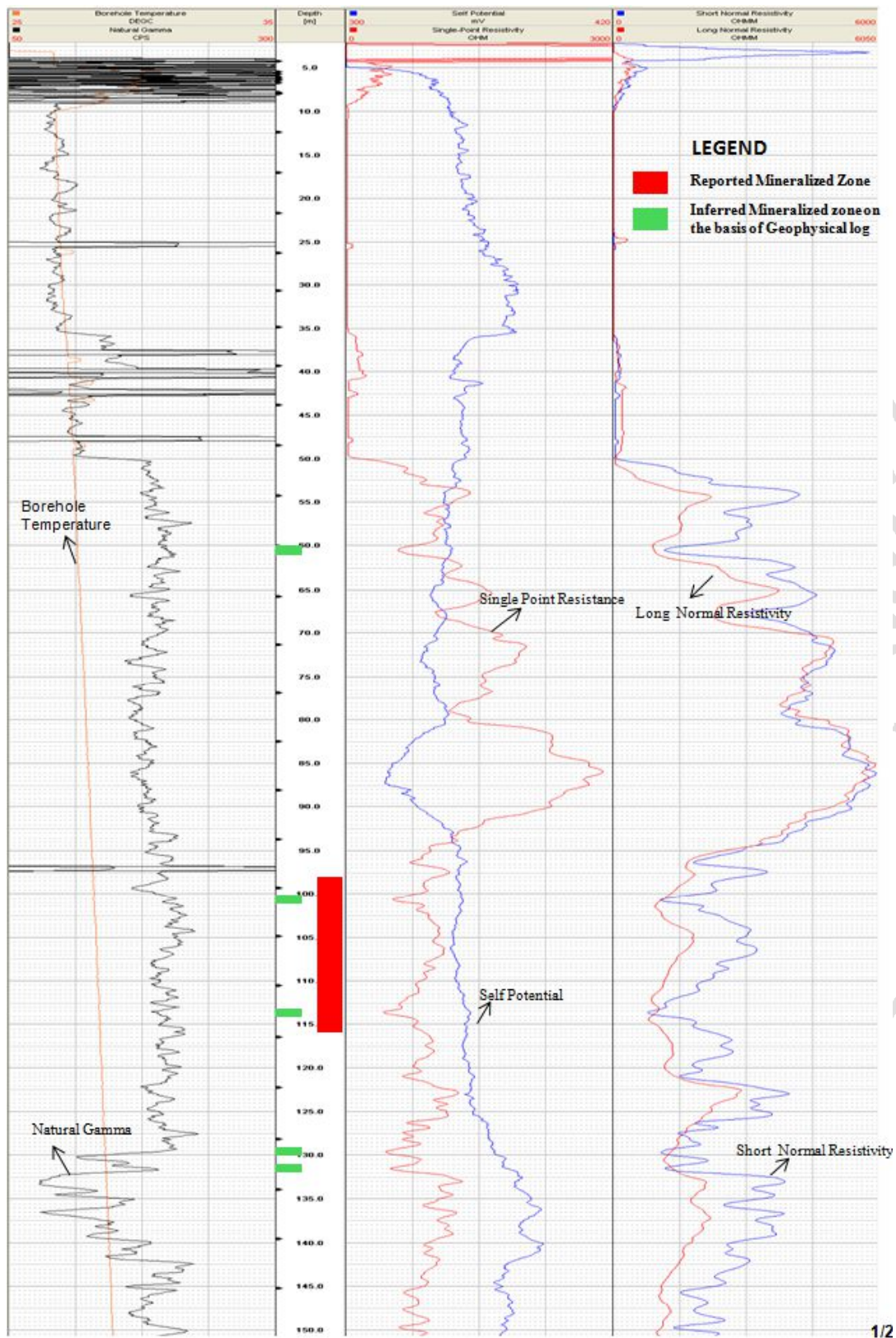


Figure 9.5 Geophysical log of borehole no. UKBH-6, U kwa block, Balaghat, Madhya Pradesh.

9.7. Mineralogy of the ore zone

Mn ore (psilomelane, braunite, rhodochrosite, hollandite, bixbyite, hausmannite) occurs as <1 cm to 16 cm thick bands interbanded with cherty to jasper rich manganiferous and ferruginous quartzite and impure calcareous rock as quartz sericite/muscovite-calcite schist (Fig 9.7). Apart from manganiferous and ferruginous quartzite and calcareous rock, Mn ore is also interbanded occasionally with sericitic phyllite/schist, particularly at the lower portion of the horizon. The ore is either massive in nature or is banded with mm thin quartzite and carbonate partings, at UKBH-1 highly friable nature of ore horizon was observed. Brecciation is also very common in ore horizon as small scale faults (Fig 9.6) are also observed in ore horizon. Very often biotite occurs as mm-thin layers along bedding plane. The ore is dark grey coloured and fine grained.

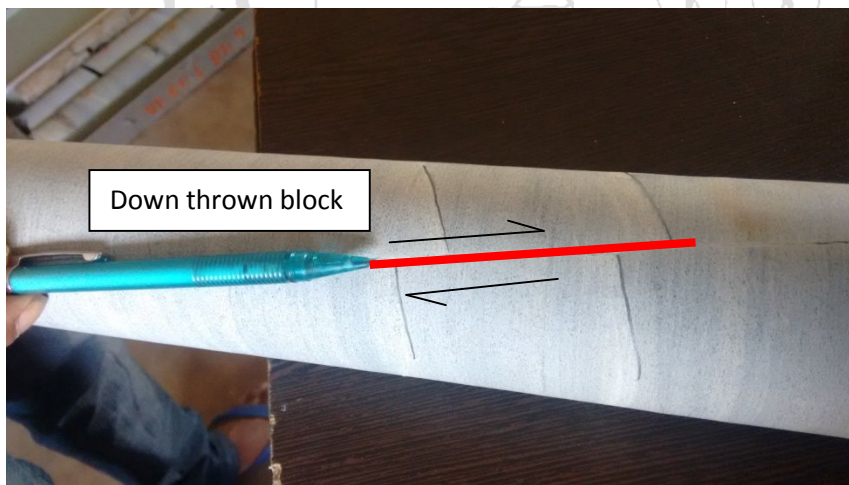


Fig: 9.6 Normal fault as observed in Sericite schist just above ore horizon with throw of 1cm.



Fig: 9.7 Thin partings of Calcite interbanded with ore (mm scale) along with schist.

9.8 Borehole deviation test and methodology

Borehole deviation test was carried out using etching method by hydrofluoric acid (HF). Test tubes were inserted at every -30 m RL for an hour with HF. Once test tubes were etched; angles were measured and deviation angle was determined (Fig. 9.8).

UKBH-1: No deviation was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	30	71°	65°
30	60	71°	65°
60	90	71°	65°
90	113.35	71°	65°

UKBH-2: Deviation of -1.5^0 to 3^0 was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	30	70°	63.5°
30	60	70°	63.5°
60	90	71°	65°
90	116	74°	68°

UKBH-3: Deviation of -2.5^0 was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	30	69°	62.5°
30	60	69°	62.5°
60	90	69°	62.5°
90	116	69°	62.5°

UKBH-4: Deviation of -3^0 to 3^0 was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	30	68°	62°
30	60	68°	62°
60	90	69°	62.5°
90	120	69°	62.5°
120	150	69°	62.5°
150	180	70°	63.5°
180	210	70°	63.5°
210	240	71°	65°
240	254	74°	68°

UKBH-5: Deviation of 1^0 to 3^0 was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	30	72°	66°
30	60	72°	66°
60	90	72°	66°
90	120	72°	66°
120	150	72°	66°
150	180	74°	68°
180	200	74°	68°

UKBH-6: Deviation of -2.5^0 to -3^0 was observed in this borehole.

FROM	TO	ANGLE MEASURED	ACTUAL ANGLE
0	30	68°	62°
30	60	68°	62°
60	90	68°	62°
90	120	69°	62.5°
120	150	70°	63.5°
150	180	70°	63.5°
180	210	70°	63.5°
210	240	71°	65°
240	254	71°	65°

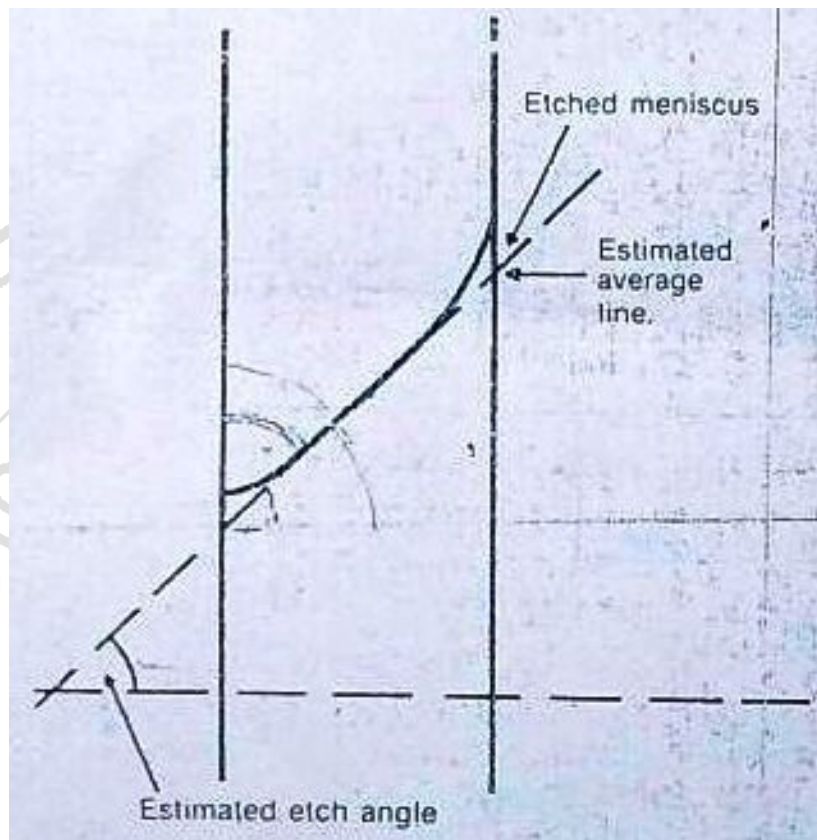


Figure 9.8: Diagram showing etch angle measurement from test tube (D.A. Berkman,)

9.9) Methodology of the ore zone sampling

Samples were collected from ore zone for analysis and submitted to laboratory in the form of core samples. Core samples of -200 mesh size were prepared and submitted for necessary chemical analysis. A total number of 51 Core samples have been collected out of 10.01 m of ore horizon recovered. Unbiased sampling has been carried out from ore horizon (ore plus waste rock). Samples were prepared as per run of drilling.

9.10 Sample preparation, Chemical analysis and laboratory procedures

Chemical results have been received and are described in chapter 5.

9.11 Check samples

No check samples were submitted for analysis. However comparison of chemical analysis with earlier worker chemical analysis have been done.

9.12 Details of Ore Zones of the boreholes

The details of the ore zone of each borehole is given below in table no. 9.1.

Depth of intersection of mineralization in Western Ukwa Block				
Section Line	BH	Total Drill Depth in meter	Mineralized Zone (MZ) in meter	Thickness of MZ in meter
AB	UKBH-1	113.35 m	83.0 -83.15m	0.15m
	UKBH-5	200 m	196.92-198.79m	1.87m
CD	UKBH-2	116 m	100.45-102.50	2.05m
	UKBH-4	254 m	251.70-253.5 m	1.80m
EF	UKBH-3	128 m	123.75-127.425	3.645m
	UKBH-6	260 m	253.25-253.95	0.70m

Here mineralization is describing thickness of ore horizon i.e. ore plus waste rock. Though mineralisation is proved upto -300m dip length as well along strike, change in intersection RL of ore horizon along borehole is attributed due to 4 stages of deformation area has undergone. It is difficult to correlate ore zone as culmination and depression along the strike as well as dip resulted in erratic behaviour of ore body even at same RL.

ORE BENEFICIATION STUDY

Ore beneficiation is a process for extractive metallurgy which improves the economic value of the ore by removing the gangue minerals, results in a higher grade product (concentrate) and a waste stream (tailings).

Manganese ore beneficiation study will be carried out as subject to the availability of 250 kg of sample. For this reason communication have been made to MOIL, Nagpur as well as MOIL, Bharveli, Balaghat. Due to no response from MOIL, sample for ore beneficiation was collected from core samples; and 80 kg sample is submitted to IBM for beneficiation studies. Details will be provided subsequently after the study.

Till the Bulk density data is received; bulk density of 3.55 will be used based on weighted average for block for gross tonnage calculation . Choudhary et al (1979) calculated average specific gravity for a particular grade of Mn ore and utilised that data for reserve estimation taking into consideration the 90% of that value as effective specific gravity and leaving 10% towards voids present in the ore. They indicated the values of Sp. gravity for particular Mn concentration is provided below in tabular form.

%Mn	Average Sp. Gravity	Effective Sp. gravity
25-30	3.86	3.47
30-35	3.72	3.34
35-40	3.95	3.55
40-45	4.36	3.92
45-50	4.37	3.93

Table 10.1: Showing values of specific gravity against concentration.

This table is used now for reserve estimation in chapter 12.

GEOTECHNICAL STUDIES ON BOREHOLE CORE SAMPLES OF MINERALISED ZONE, HANGING WALL AND FOOTWALL SIDE

No geotechnical studies were carried out during current FSP; however RQD analysis of core samples was done during borehole logging. Details of RQD analysis is given below:

11.1 RQD (Rock quality designation studies): RQD analysis of cores collected were analyzed. Methodology used is explained in figure below.

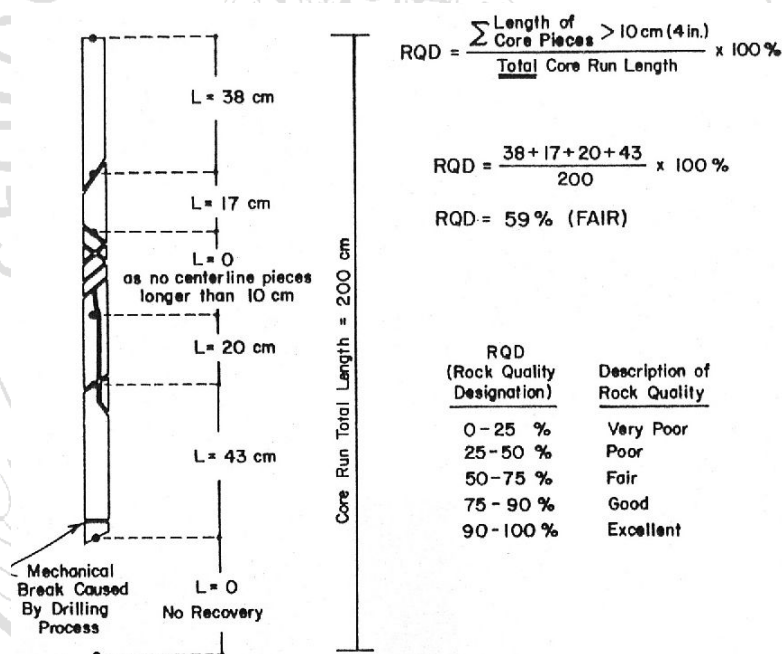


Figure 11.1: Diagram illustrating RQD calculation (D.U. Deere and D.W. Deere, 1988).

Data obtained after analysis is described in tabular form. RQD value of only core samples after weathered horizon is mentioned below (Run with ore horizon is marked in bold).

UKBH-1			UKBH-2			UKBH-3		
From	To	RQD	From	To	RQD	From	To	RQD
14	17	0	20	23	3.33	23	26	0
17	20	23	23	26	16	26	29	58.66
20	23	0	26	29	47	29	32	62
23	26	38	29	32	26.66	32	35	42.666
26	29	44.86	32	35	3.3	35	38	60.33
29	32	50	35	38	11	38	41	68
32	35	68.04	38	41	70.33	41	44	71
35	38	51.3851	41	44	64.33	44	47	75
38	41	47.7	44	47	76.33	47	50	82.33
41	44	80.33	47	50	86.33	50	53	95.333
44	47	78.668	50	53	93.66	53	56	82
47	50	65.331	53	56	95	56	58.9	90.344
50	53	73.07	56	59	91.66	58.932	62	95.1612
53	56	68.384	59	62	95	62	64.95	92.203
56	59	47.297	62	65	53.66	64.95	67.95	90.33
59	62	84.4333	65	68	91	67.95	71	96.7213
62	65	69.566	68	71	93	71	73.95	84.4067
65	68	85.714	71	73.95	97.96	73.95	77	95.57
68	71	75.233	73.95	76.95	94	77	80	95
71	74	90.166	76.95	79.95	82.33	80	82.95	89.491
74	77	81.543	76.95	82.95	77	82.95	86	100
77	80	93.93	82.95	85.95	90	86	89	98

80	83	92.33	85.95	89	81.63	89	92	95.66
83	86	7.66	89	92	95	92	95	90.666
86	89	0	92	95	64	95	97.95	97.2881
89	92	0	95	98	71.66	97.95	101	96.0655
92	95	94	98	101	74	101	104	90
95	98	78	101	104	33.33	104	106.9	87.931
98	101	89	104	107	91	106.9	110	86.7741
101	103.95	100	107	110	95.66	110	113	96.667
103.95	107	89.836	110	113	87	113	116	72.666
107	109.95	95.593	113	116	100	116	119	32
109.95	112.95	97				119	122	38.666
112.95	113.95	100				122	125	92.666
						125	128	88.66

UKBH-4

UKBH-5

UKBH-6

From	To	RQD	From	To	RQD	From	To	RQD
41	44	0	50	53	17	50	53	12.66
44	47	0	53	56	45.33	53	56	51.33
47	50	0	56	59	32.33	56	59	97
50	53	0	59	62	25.66	59	62	96.67
53	56	0	62	65	10.66	62	65	88.66
56	59	0	65	68	50	65	68	93
59	62	0	68	71	43	68	71	100

62	65	7	71	74	47	71	74	70.33
65	68	11.66	74	77	34	74	77	93
68	71	36	77	80	34	77	80	94.333
71	74	74.33	80	83	42.33	80	83	91.6667
74	77	75.66	83	86	31	83	86	95.6667
77	80	97.66	86	89	62.333	86	89	98
80	83	95.33	89	92	29	89	92	97.3334
83	86	89	92	95	59	92	95	94.6667
86	89	57	95	98	83.66	95	98	95
89	92	86	98	101	85.66	98	101	91
92	95	92	101	104	86	101	104	82
95	98	93	104	107	96	104	107	92.333
98	101	59	107	110	88	107	110	98
101	104	87.66	110	113	94.33	110	113	80.667
104	107	98.66	113	116	96.33	113	116	91.66
107	110	91.33	116	119	91	116	118.9	87.93
110	113	92	119	122	98.66	118.9	121.95	94.09
113	116	97	122	125	95	121.95	125	88.85
116	119	90.66	125	128	98.33	125	128	60.2
119	122	97.33	128	131	92	128	131	79.66
122	125	96.33	131	134	100	131	134	80
125	128	94.66	134	137	98.6666	134	137	89.33
128	131	96	137	140	98.33	137	140	93.66
131	134	90	140	143	92.33	140	143	86.91

134	137	95.33	143	146	96.33	143	146	94.33
137	140	93	146	149	96.33	146	149	96.66
140	143	86	149	152	31	149	152	92
143	146	91.66	152	155	95.66	152	155	89.33
146	149	98.66	155	158	98	155	158	33
149	152	94.66	158	160.9	94.333	158	161	68.12
152	155	98	160.9	164	98.387	161	164	93
155	158	90.66	164	167	93.79	164	167	77.66
158	160.95	94.23	167	170	98	167	170	100
160.95	163.95	78	170	173	100	170	173	98.33
163.95	167	96.66	173	176	97	173	176	100
170	173	97.66	176	179	99	176	179	99
173	176	97	179	182	95.666	179	182	94.33
176	179	96	182	185	100	182	185	93.89
179	182	97.99	185	188	94.33	185	188	100
182	185	98.33	188	191	92.25	188	191	94.66
185	188	98.66	191	194	89.33	191	194	92
188	191	94	194	197.1	90.87	194	197	93.66
191	194	98.66	197.1	200	70	197	200	96
194	197	96.61				200	203	91.58
197	200	92				203	206	96.33
200	203	98.33				206	209	96.33
203	206	94.66				209	212	96

206	209	93.33	212	215	100
209	212	73	215	218	100
212	215	90	218	221	96.97
215	218	93.33	221	224	98
218	221	87	224	227	95.33
221	224	87	227	230	98.66
224	227	92	230	233	94.63
227	230	87	233	236	77.28
230	233	100	236	239	79.19
233	236	100	239	242	81
236	239	100	242	245	74
239	242	100	245	248	81
242	245	85	248	251	87.66
245	248	40.33	251	253.9	79.65
248	251	79.33	253.9	256.1	100
251	254	49	256.1	257	100
			257	260	100

RESOURCE ESTIMATION

12.1 Introduction

Resource estimation of ore resource is based on persistence of the ore horizon along its strike and dip and the thickness of the ore horizon and grade of the manganese ore. All factors have been determined in the exploration by drilling. Data from earlier exploration has also been utilised to the possible extent.

Previous workers (Khan and Fulmari, 1996) have estimated recoverable resource of ore as 3.01mt in Ukwa western block and gross tonnage (ore + waste) of the part in which augmentation work has been carried out as 3.597 mt at 40.34% of Mn (Annexure-IX). As current works is based on augmentation of ore resource, a part of western block is considered here for drilling with close intervals and resource estimation. LV section and cross sections for resource estimation has been prepared. As current area is part of whole block so tonnage estimated here is not of whole block but the part which has been assigned for augmentation along with first series of boreholes in MOIL boundary.

12.2 Detailed description of ore zones with core recovery

Depth of intersection of ore horizon is listed below in table no. 12.1 along with description of core recovery.

Table 12.1 Depth of intersection of mineralization in Western Ukwa Block

Depth of intersection of mineralization in Western Ukwa Block					
Section Line	BH	Total Depth in meter	Drill in (MZ) in meter	Actual thickness of ore	Description
AB	UKBH-1	113.35 m	83.0 -83.15m	0.15m	Core recovery = 0% during

					initial drilling. Mn ore of 0.15m was recovered during the re drilling at 83 m, core recovery =70% during re drilling. Ore horizon is highly brecciated.
	UKBH-5	200 m	196.92-198.79m	0.40m	Mn ore horizon of 1.42m was recovered, core recovery = 75.93%, 0.45 m was collected in powdery form. Brecciation is observed. Ore horizon here is represented by Mn ore interbanded with Mn quartzite and traversed by calcite and quartz vein.
CD	UKBH-2	116 m	100.45-102.50	1.40m	Mn ore horizon of 1.43m was recovered, core recovery = 69.756%, core loss is 0.62m. Lost core is collected in powdery form from sludge of drilling. Ore horizon is massive here interbanded with Mn quartzite.
	UKBH-4	254 m	251.70-253.5 m	1.20m	Core recovery is 100% with Mn bands varying from <1mm to 16cm thick

					bands. Brecciation is observed. Ore horizon here is represented by Mn ore interbanded with Mn quartzite and traversed by calcite and quartz vein.
EF	UKBH-3	128 m	123.75-127.425	2.4m	Core recovery is 100% with Mn bands varying from <1mm to 11cm thick bands. Evidence of faulting (small scale and oblique slip) and brecciation is observed. Ore horizon is traversed by silica veins. Yellowish and red coloured (Fe bearing) oxidised mineral is observed.
	UKBH-6	260 m	253.25-253.95	0.20m	Core recovery is 100% with Mn bands varying from <1mm to 5 cm thick bands. Brecciation is observed. Ore horizon is here represented by Mn quartzite with Mn ore. Ore horizon is traversed by silica veins. Yellowish and red coloured (Fe bearing) oxidised mineral is observed.

12.3 Cut off grade consideration

Cut off grade of previous work will be used here; as current works objective is to augment the reserve. As per IBM mineral book, 2015, the lowest Mn grade to be considered is 35%. However in iron and steel industry, the BIS: 1281-2005 (Reaffirmed 2008) specification is laid down for manganese ore. The specifications based on the user industry indicate that normally manganese ore containing 28 to 35% Mn is used. Ore size generally varies from 10 to 40 mm. For other constituents general stipulations are Fe: 16 to 22%, SiO₂: 2 to 8%, Al₂O₃: 5 to 8% and P: 0.3% maximum. However as per IBM mineral book 2014 there are industries like Nav Bharat Ferro-Alloys Ltd, Paloncha, Khammam(Specifications of ore consumed Mn: 28-50%, P: 0.1-0.3%, Si: 3.73328-13.9998% , Fe :5-8%) and Monnet Ispat & Energy Ltd, Raipur (Specifications of ore consumed Mn: 28 - 46%, Si: 2.799 – 15.866%, P 0.28%) which uses High Si ore. Based on these values ore horizon of three boreholes (UKBH-2, 3 and 4) of current FSP can be used for resource estimation. Though in case of UKBH-3 and UKBH-4, P value is slightly high but it can be compensated by mixing with high grade ore. Therefore here cut off value considered for resource estimation is 28%.

12.4 Minimum Stopping width consideration

The thickness of ore horizon varies from 0.15 m to 3.61 m as observed during drilling. In general the minimum stopping width is taken as 2.0 m for underground mining methods. However, for open cast mining the bench thickness is taken as stopping width, which varies from 2 m to 5 m. Here minimum stopping width should be considered as 2 m for underground mining as open cast mining of ore is already being carried out in adjacent area.

12.5 Correlation of ore lodes

The mineralised zone may be termed as the zone, which contains the ore mineral in traces or in visible quantity. However, the zones that contain the ore mineral or element equivalent to or

more than the cut off grade are defined as lode or ore zones. Here ore horizon is marked on the basis of chemical results of ore. The correlation has been constructed by joining the ore zone intersection points of all the boreholes falling in each section line (Fig. 12.1, 12.2 and 12.3). Strike continuity in whole block as well as dip continuity upto -250 RL can be confirmed in whole block. Although in reserve estimation there is break in strike continuity due to value of Mn falling below cut off grade. Bottoming of ore horizon is not observed yet. On the basis of correlation, the cross sectional area of the intersected ore zone has been calculated, which is the primary factor for estimation of the resources.

12.6 Description of lodes

Ore lodes have been described above in 12.2 sections.

12.7 Preparation of LV sections

Cross sections were prepared across the strike of the ore body for all the section lines (Fig. 12.1, 2 and 3), along which boreholes were drilled. L-V section along the strike of the ore body (Plate-II) is also prepared where all the six boreholes (viz. UKBH-1 to UKBH-6) have intersected ore horizon. Here only those current boreholes are considered which have Mn value above or equivalent to cutoff grade. Previous boreholes were also used in preparation of Cross section as well as in LV section. The manganese gross ore resource (ore + waste) has been calculated based on the mineralized area marked in each cross section after correlation of the boreholes. The 2-D subsurface geometry of the ore body across and along the strike can be reconstructed from the cross sections and L-V section respectively.

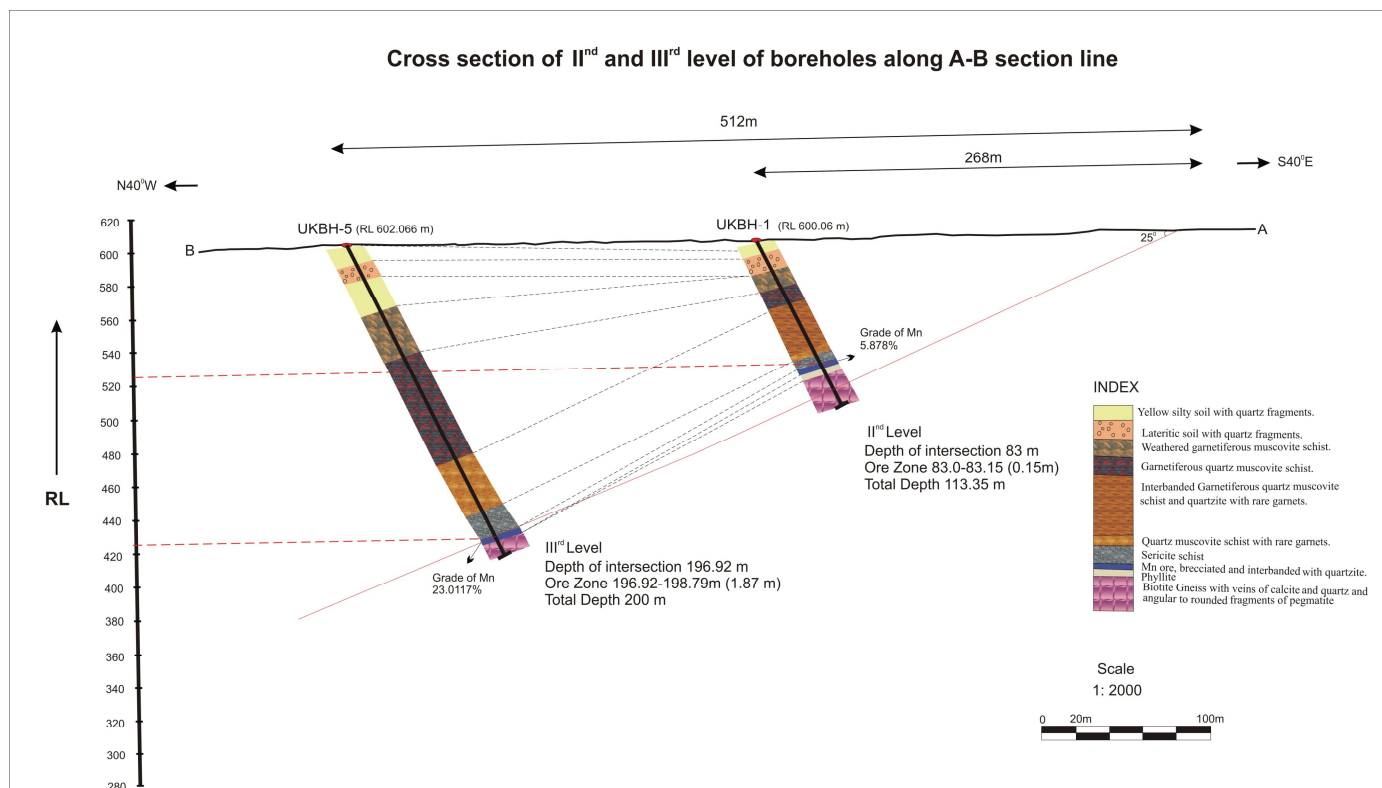


Figure 12.1 Cross section along AB section line.

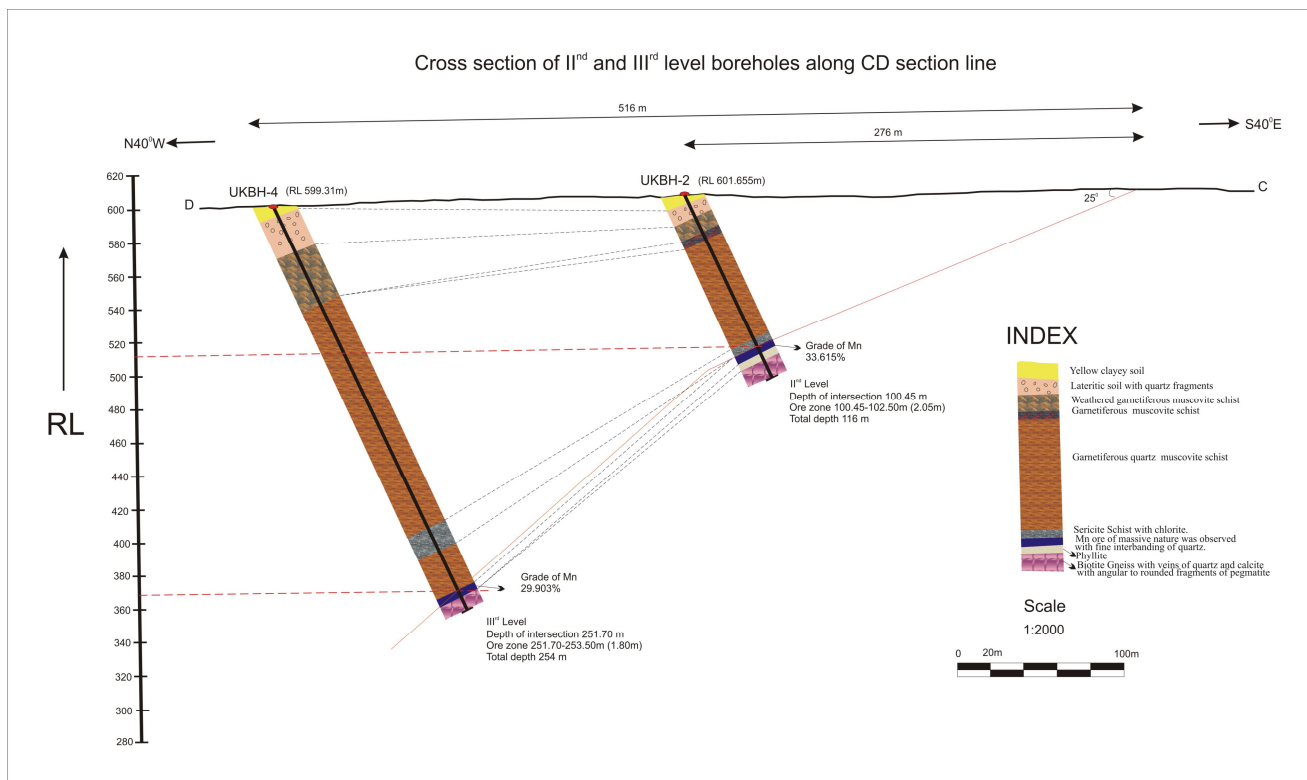


Figure 12.2 Cross section along CD section line.

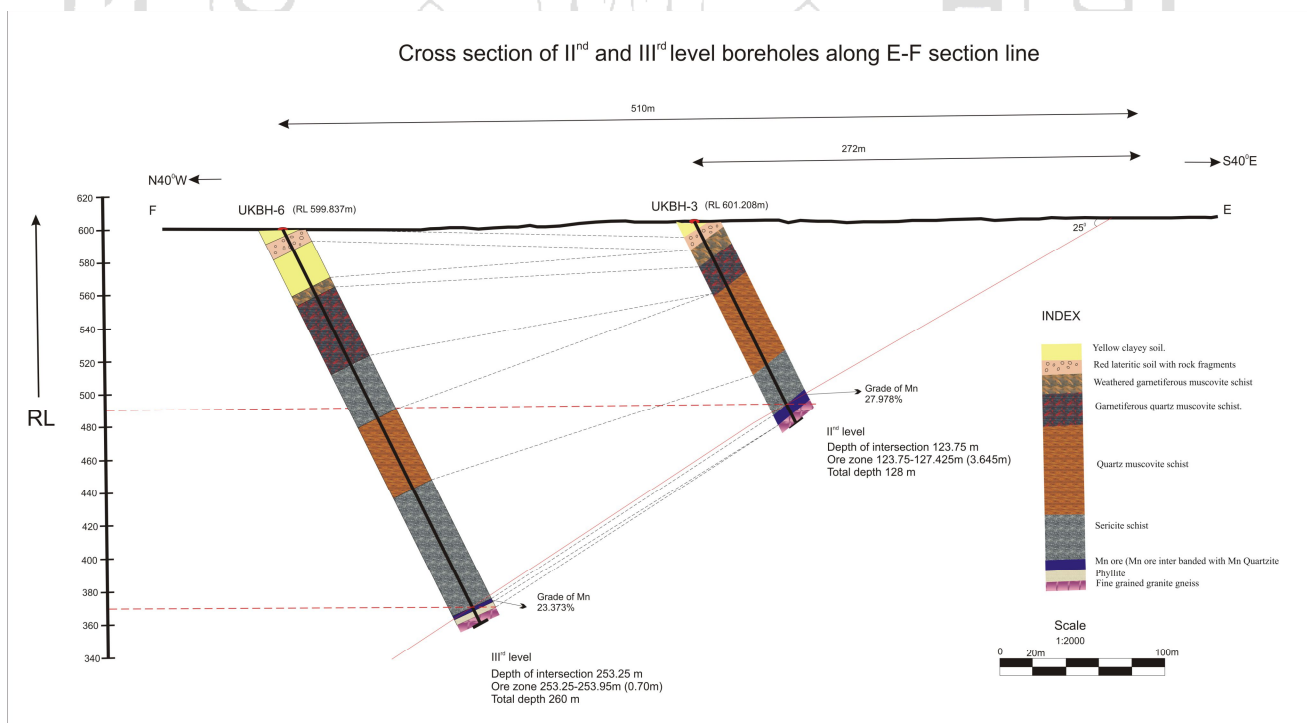


Figure 12.3 Cross section along EF section line.

12.8 Preparation of Level Plan

Level Plan has been prepared at -100m and it showed that area has multiple horizons which may be a wrong interpretation as area has only one stratiform ore horizon. The reason behind this is four phases of deformation which resulted in culmination and depression in the levels of intersection of ore horizon along the dip and strike. It is reflected in the cross sections and LV sections.

12.9 Assumptions for resource estimation

The following assumptions were followed while calculating the resource:

1. As the objective was to augment the resource so all the previous boreholes along with details listed in Annexure-III are taken into consideration for resource estimation even though four of the first series of boreholes of previous work lies within the MOIL leasehold boundary.
2. Data from a particular section line is considered effective, along the strike of the ore horizon, over half the distance to the section lines placed, immediately to its east and west. In cases where value is found to be less than cut off grade considered in adjacent section lines along same level; strike influence of section line having value of Mn is reduced to 25% for that level.
3. Data from the eastern most section line No. W-4 is considered effective - (a) towards west over half the distance up to E-F and (b) towards east over half the distance between W-4 and E-F. (For boreholes GSU-21, GSU-24 and GSU-34). For Borehole UKH-16 strike influence on either side is 25% of distance between section lines E-F and W-4. (as values are below cut off in borehole UKBH-6).
4. Data from the western most section line No. W-1 is considered effective - (a) towards west over 25% the distance up to A-B and (b) towards east over 25% the distance between W-1 and A-B. (as values are below cut off in boreholes of section line A-B)
5. Weathered horizon is considered upto 10 m from surface RL.
6. The dip length from the uppermost intersections is considered effective from the upper level as defined at (4) above, to a lower level placed half way between the intersection level under consideration and the next lower level of intersection.

7. For the next lower intersection the dip length is effective from the lower level assigned for the higher intersection as defined at (5) above, to a lower level placed half way between the intersection level under considerations and the next lower level of intersection, and so on.
8. For the lowermost intersection, the dip length is effective from the upper level assigned to it, as indicated at (5) and (6) above, to the actual level of intersection under consideration and the level placed below, equivalent to half the level difference between the upper level and the actual level of intersection and up to the lowermost level considered at further 20m along borehole.
9. Thickness of the manganese ore horizon and the manganese ore have already been worked out and are indicated in this chapter as well as in chapter 9.
10. Here previous boreholes have been utilised to determine weighted average of Mn for the whole block which is 35.138.
11. For estimates of gross tonnage of material the specific gravity used is 3.55 as of table specified in chapter 10.
12. Tonnage = volume x Sp. gravity (Tonnage is calculated using this formula).

12.10 Methodology of ore resource estimation

12.10.1 Longitudinal-Vertical Section method

The previous as well as present sections lines were used for preparation of LV section (Table no. 12.3) along the earlier prepared strike line. Each intersection has been placed within a panel which is defined by the following parameters.

- The intersection R.L. is marked on the appropriate level.
- The distance between the vertical lines on either side of the intersection point indicates the effective strike extent for the intersection.
- The horizontal lines placed on above the intersection point and the other below it, indicate the upper and the lower levels respectively between which the dip length is effective.
- The dip of the ore horizon is indicated at the top left hand corner of the panel.
- Borehole No. and the thickness, manganese and phosphorous contents of the ore revealed by the said borehole are mentioned in annexure pertaining to core samples (Annexure-II).

- Two categories of tonnage are calculated: indicated and inferred.
- Tonnage of gross resource(ore + waste) indicated by each intersection is indicated in table 12.3

12.10.2. Cross section method

There are two rules following which the resource is estimated by Cross Section method.

- Rule of nearest point.
- Rule of Gradual change.

The rule of nearest point of cross sectional area method for estimation of resource has been adopted with reference to the ore zone intersected in each borehole. The entire area along individual cross section is calculated for all the boreholes falling in the same cross section This is also known as rule of equal influence and according to this the value of any point between two stations is considered constant, equal to the value of the nearest station. In a general case of borehole A and B with thickness t_1 and t_2 , the value of each one extent to midpoint 'M' between holes (Fig 12.4). Any point on line AB or in the area of t_1 or t_2 , except M; is inside the "linear or area of influence" of a station A or B and near to it than to the adjoining one. Thus this property gives the rule its name of nearest point.

Here true thickness is determined by multiplying app thickness of ore horizon (ore + waste) with sin of angle measured in borehole for ore horizon with respect to core axis.

Illustration-1 (Calculation of resource of individual borehole lying in a cross section with variable strike length – e.g. borehole UKBH-04 lying in section line C-D):

The cross sectional area for Mn ore as intersected in UKBH-04 falling in C-D line is calculated by using graph sheet. In this method the total cross-sectional area with 217 m dip influence (108.5 m towards north-westerly and 108.5 m towards southeasterly) in UKBH-04 is multiplied with the strike influence of the borehole 141.5 m (along the strike 60 m westerly and 81.5 m easterly) along with true thickness as 1.6914m. The volume thus calculated is 51935.28 m^3 .

(Table no. 12.2). The volume thus calculated is multiplied by specific gravity of 3.55 for determination of tonnage of ore horizon (ore + waste) which is 184370.2536 tonne.

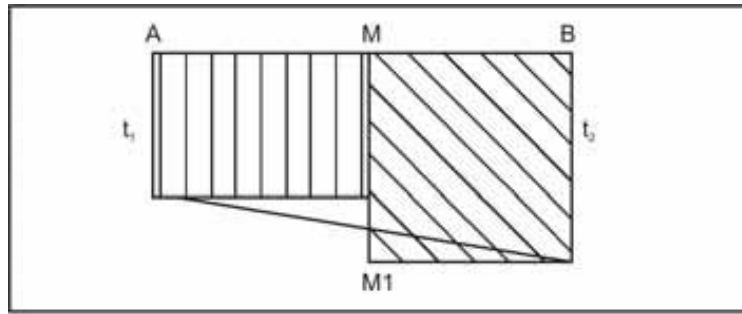


Fig 12.4: Interpretation of values between two boreholes A and B.



Table 12.2 Gross resource (ore + waste) of material estimated by cross section method.

Section Line	B.H. No.	Angle of lode with core axis	App. Thickness along borehole (in m)	True Thickness Calculated (in m)	Lode RL		Dip length	DIP RL (m)		Strike Length (in m)	Mn (in %)	Specific Gravity used based on weighted average of block	Volume(m ³)	Tonnage	Category
					Upper (m)	Lower (m)		Upper	Lower						
W-1	GSU-30	70°	1.55	1.4565	562	560.5	123	577	545	52.5	37.7	3.55	9405.349	33388.98806	Indicated
							44	590	577	52.5	37.7	3.55	3364.515	11944.02825	Inferred
W-1	GSU-31	75°	1.45	1.4005	529.5	528	196.5	545	467	52.5	34.13	3.55	14447.90813	51290.07384	Indicated
W-1	UKH-7	65°	0.75	0.67973	404.5	403	266	467	341.5	52.5	33.91	3.55	9492.42945	33698.12455	Indicated
							20	341.5	332	52.5	33.91	3.55	713.7165	2533.693575	Inferred
W-2	UKH-4	59°	2	1.71433	411.5	410	182	460	361	87.5	36.91	3.55	27300.70525	96917.50364	Indicated
							20	361	351	87.5	36.91	3.55	3000.0775	10650.27513	Inferred
W-2	GUG-2	70°	1.31	1.2309	510.5	509.5	182	541.5	460	87.5	40.63	3.55	19602.0825	69587.39288	Indicated
W-2	GSU-29	72°	0.9	0.8559	578.3	577.5	142	590	541.5	87.5	39.1	3.55	10634.5575	37752.67913	Indicated
C-D	UKBH-4	70°	1.8	1.6914	369	367.5	217	439	296.5	141.5	29.9035	3.55	51935.2827	184370.2536	Indicated
							20	296.5	284	141.5	29.9035	3.55	4786.662	16992.6501	Inferred
C-D	UKBH-2	85°	2.05	2.04219	511.5	509.5	217	580	439	141.5	33.6151	3.55	62706.46505	222607.9509	Indicated
							15	590	580	141.5	33.6151	3.55	4334.548275	15387.64638	Inferred
W-3	UKH-14	54°	5.4	4.36869	386.5	382.5	60	404	365	108.5	37.82	3.55	28440.1719	100962.6102	Indicated
							20	365	352	108.5	37.82	3.55	9480.0573	33654.20342	Inferred
W-3	UKH-3	54°	2.9	2.346149	425.5	423	60	441	404	108.5	43.63	3.55	15273.4299	54220.67646	Indicated

W-3	GSU-27	65°	3.05	2.76423	505	502	165.5	528	441	140.5	34.36	3.55	64275.94913	228179.6194	Indicated
W-3	GSU-26	65°	1.25	1.132884	554.5	553	111	577.5	528	140.5	45.85	3.55	17667.89242	62721.0181	Indicated
							32	590	577.5	140.5	45.85	3.55	5093.446464	18081.73495	Inferred
E-F	UKBH-3	75°	3.645	3.520799	488.5	485.5	238	546	428	134	27.9785	3.55	112285.3217	398612.8921	Indicated
							92	590	546	134	27.9785	3.55	43404.41007	154085.6558	Inferred
W-4	UKH-16	60°	1.2	1.03923	393.5	392.5	82	414	371.5	70	38.81	3.55	5965.1802	21176.38971	Indicated
							20	371.5	361	70	38.81	3.55	1454.922	5164.9731	Inferred
W-4	GSU-34	60°	2.42	2.09578	437	434.5	82	455	414	70	41	3.55	12029.7772	42705.70906	Indicated
W-4	GSU-24	60°	1.85	1.602146	478	476.5	83	498	455	140	35.84	3.55	18616.93652	66090.12465	Indicated
W-4	GSU-21	60°	2.6	2.251666	539	536.5	187.5	590	498	140	38.27	3.55	59106.2325	209827.1254	Indicated
													Total =	2182603.992	
Total tonnage is 2182603.992 tonne or 2.182 million tonne at 35.138% of Mn															

Table 12.3 Gross resource of material (ore + waste) estimated by LV section method.

Section Line	B.H. No.	App. Thickness	Angle of lode with core axis	Horizontal Width	Lode RL		Dip length	DIP RL (m)		Difference in Dip RL	Strike Length	Mn (in %)	Specific Gravity used based on weighted average of block	Volume	Tonnage	Category
					Upper	Lower		Upper	Lower							
W-1	GSU-30	1.55	70°	5	562	560.5	123	577	545	32	52.5	37.7	3.55	8400	29820	Indicated
							44	590	577	13	52.5	37.7	3.55	3412.5	12114.375	Inferred.
W-1	GSU-31	1.45	75°	3	529.5	528	196.5	545	467	78	52.5	34.13	3.55	12285	43611.75	Indicated
W-1	UKH-7	0.75	65°	3	404.5	403	266	467	341.5	125.5	52.5	33.91	3.55	19766.25	70170.1875	Indicated
							20	341.5	332	9.5	52.5	33.91	3.55	1496.25	5311.6875	Inferred.
W-2	UKH-4	2	59°	3	411.5	410	182	460	361	99	87.5	36.91	3.55	25987.5	92255.625	Indicated
							20	361	351	10	87.5	36.91	3.55	2625	9318.75	Inferred.
W-2	GUG-2	1.31	70°	3	510.5	509.5	182	541.5	460	81.5	87.5	40.63	3.55	21393.75	75947.8125	Indicated
W-2	GSU-29	0.9	72°	3	578.3	577.5	142	590	541.5	48.5	87.5	39.1	3.55	12731.25	45195.9375	Indicated
C-D	UKBH-4	1.8	70°	3	369	367.5	217	439	296.5	142.5	141.5	29.9035	3.55	60491.25	214743.9375	Indicated
							20	296.5	284	12.5	141.5	29.9035	3.55	5306.25	18837.1875	Inferred.
C-D	UKBH-1	2.05	85°	3	511.5	509.5	217	580	439	141	141.5	33.6151	3.55	59854.5	212483.475	Indicated
							15	590	580	10	141.5	33.615	3.55	4245	15069.75	Inferred.

											1					
W-3	UK H-14	5.4	54°	5	386.5	382.5	60	404	365	39	108.5	37.82	3.55	21157.5	75109.125	Indicated
							20	365	352	13	108.5	37.82	3.55	7052.5	25036.375	Inferred.
W-3	UK H-3	2.9	54°	4	425.5	423	60	441	404	37	108.5	43.63	3.55	16058	57005.9	Indicated
W-3	GSU -27	3.05	65°	4	505	502	165.5	528	441	87	140.5	34.36	3.55	48894	173573.7	Indicated
W-3	GSU -26	1.25	65°	3	554.5	553	111	577.5	528	49.5	140.5	45.85	3.55	20864.25	74068.087 5	Indicated
							32	590	577.5	12.5	140.5	45.85	3.55	5268.75	18704.062 5	Inferred.
E-F	UK BH- 3	3.645	75°	6	488.5	485.5	238	546	428	118	134	27.978 5	3.55	94872	336795.6	Indicated
							92	590	546	44	134	27.978 5	3.55	35376	125584.8	Inferred.
W-4	UK H-16	1.2	60°	2	393.5	392.5	82	414	371.5	42.5	70	38.81	3.55	5950	21122.5	Indicated
							20	371.5	361	10.5	70	38.81	3.55	1470	5218.5	Inferred.
W-4	GSU -34	2.42	60°	3	437	434.5	82	455	414	41	70	41	3.55	8610	30565.5	Indicated
W-4	GSU -24	1.85	60°	2	478	476.5	83	498	455	43	140	35.84	3.55	12040	42742	Indicated
W-4	GSU -21	2.6	60°	4	539	536.5	187.5	590	498	92	140	38.27	3.55	51520	182896	Indicated
														Total =	2013302.6 25	
Total tonnage is 2013302.625 tonne or 2.013 million tonne at 35.138% of Mn																

12.11 Category of resource as per UNFC classification

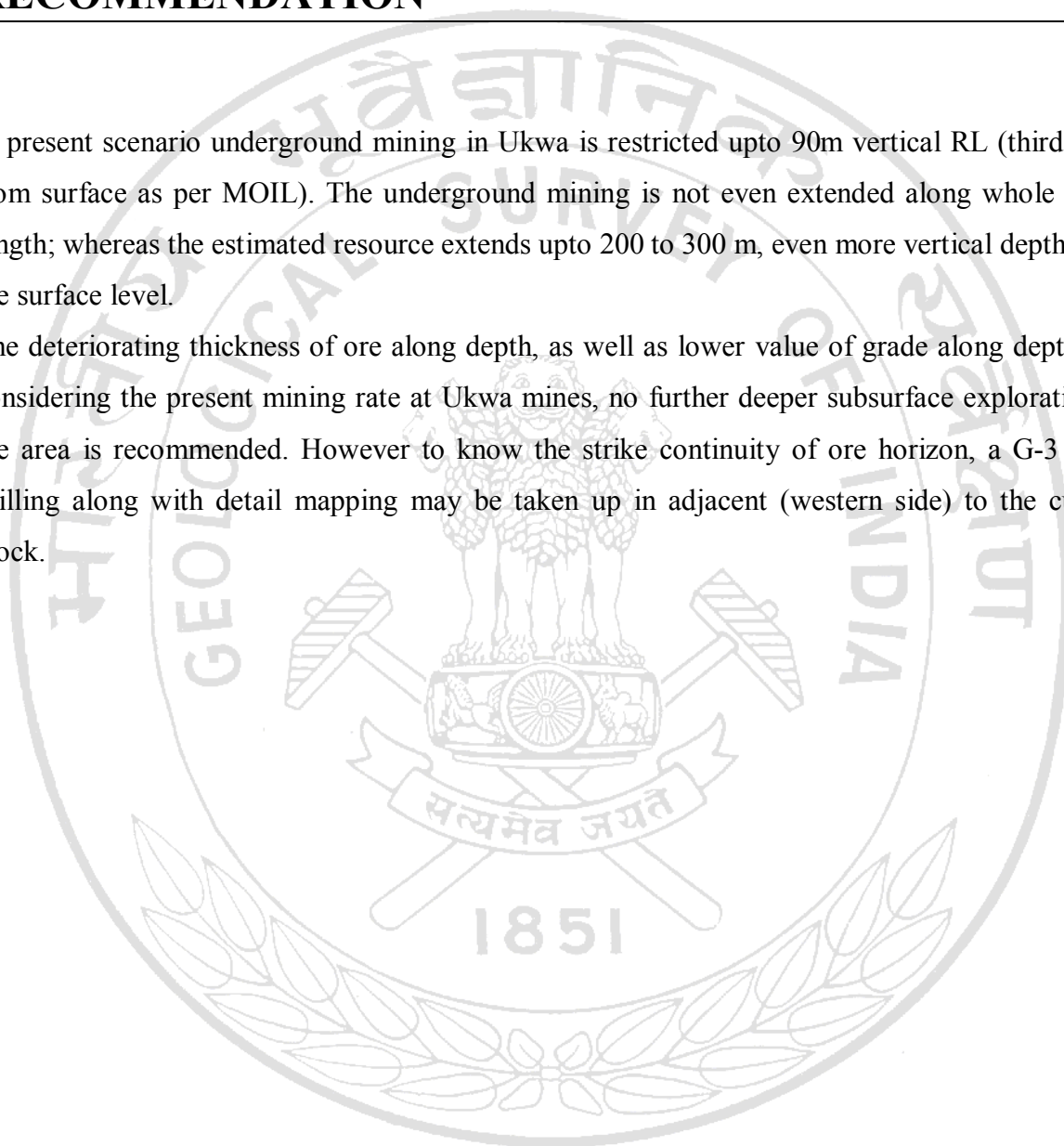
Only the geological axis is defined here by present authors as no study is being done for feasibility and economic axis. As per geological axis this resource falls under G-2 category. (See chapter 9, subheading 1 for mineral content rule as per UNFC.)



RECOMMENDATION

In present scenario underground mining in Ukwa is restricted upto 90m vertical RL (third level from surface as per MOIL). The underground mining is not even extended along whole strike length; whereas the estimated resource extends upto 200 to 300 m, even more vertical depth from the surface level.

The deteriorating thickness of ore along depth, as well as lower value of grade along depth and considering the present mining rate at Ukwa mines, no further deeper subsurface exploration in the area is recommended. However to know the strike continuity of ore horizon, a G-3 stage drilling along with detail mapping may be taken up in adjacent (western side) to the current block.



CONCLUSION

Drilling in western Ukwā block was started on 09.01.2017 and completed on 26.03.2017. A total of 1071.35 m of drilling were done and 6 nos. of boreholes were drilled at second (100m) and third (200m) level to know strike and depth continuity of manganese horizon. A total of 10.01 m of thick ore horizon was intersected in 6 different boreholes (UKBH-1, 2,3,4,5 & 6) with maximum thickness of ore horizon 3.61m (UKBH-3) and minimum thickness of ore horizon 0.15 m (UKBH-1). A total of 51 nos. core samples, 20 nos. BRS, 10 nos. REE, 10 nos. PS. 10 nos. PCS and 5 nos. of XRF samples were analysed. Open cast as well as underground mines were visited to know about sub-surface behaviour of body. The surrounding areas were also investigated during field work along with regional traverses to understand structure as well as regional geological setup of area. Core Logging along with geophysical logging of each borehole was done. 5 nos. of water samples were also collected and analysed. DGPS survey was carried out and demarcated borehole points along with block boundary. An attempt to augment the Mn ore in the area has been done. Here first series of 4 boreholes (Annexure-III) of previous workers lying in MOIL boundary are also included for resource estimation.

The area exposes rocks of Archaean age in the form of older metamorphics and of Precambrian age as Sausar Group. Both the sequences have a regional strike of ENE-WSW to NE-SW and dip at low to moderate angles towards NW. The area has undergone at least four phases of deformation. The contact between older metamorphics and Sausars is tectonic.

The manganese mineralisation in the area is synsedimentary associated with the Sausar Group of rocks. The ore horizon probably transgresses from Lohangi to Mansar formations. Ukwā deposit has been proved to extend over a strike length of 5400 m.

According to earlier workers, the continuity of the ore horizon has been tested for more than 300 m of vertical depth in the eastern part of the deposit and more than 200 m in rest of the deposit over major part of the strike length. In terms of dip length it measures up to 700 m from the

surface exposure of ore horizon in Ukwa mine. Bottoming of ore horizon has not been observed. However, grade and thickness of ore at depth are not very encouraging due to high content of phosphorous at greater depth. Thickness of ore horizon ranges between <15 cm to about 10 m, the average being 3.58 m.(Khan et. al.) .The aggregate thickness of the manganese ore bands ranges between 0.23 and 4.85 m, the average being 1.76 m. However during current drilling thickness of the ore horizon observed is ranging between 0.15 m to 3.64 m with maximum thickness of ore as of 3.61m and minimum as 0.15m. Lithologically the ore horizon constitutes a bed with sharp contacts with its hanging wall and footwall rocks.

Structurally the ore deposit is free from any complexities except pinch and swell structures. Pinching on the either ends of Ukwa deposit is recorded along the strike. At places, along dip direction also the ore horizon shows pinching at depth due to culmination and depression.

According to earlier workers data the major part of the ore in Ukwa deposit falls under medium grade ore having 35 to 45% Mn. During the course of augmentation average grade of ore is found to be 35.138%. This low grade ore is mixed with high grade for practical purposes. As hand picking is used by MOIL in the area for beneficiation thus ore of various grades can be selectively handpicked and mixed. Phosphorous content is observed to increase in Western block and at depth in other blocks also. The favourable geological and structural set up of the ore horizon may lead to easy mining of ore, but the high phosphorous content of the ore at depth is a discouraging factor. Such high phosphorous ores can only be used after they are blended with low phosphorous ore to get a desirable grade. Though the amount of silica was found to be high but as per IBM mineral book there are industries which are using this grade of ore (having high silica upto 15.86%).

Here gross (ore + waste) resource estimation of current area is also determined by previous workers sections using their LV section and previous chemical values; based on which resource was found to be 3.597 million tonne at 40.34% of Mn. (Annexure-VIII and IX and Plate-V). During current FSP total gross resource of thus calculated is 2.182 million tonne by cross section method and 2.013 million tonne by LV section method at 35.138% of Mn. As per UNFC category of the estimated resource is 332. IBM has recently taken up a new vision for Mn under Vision 2020 in which it is postulated to estimate resource till grade 10%; considering this block has potential.

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LOCALITY INDEX

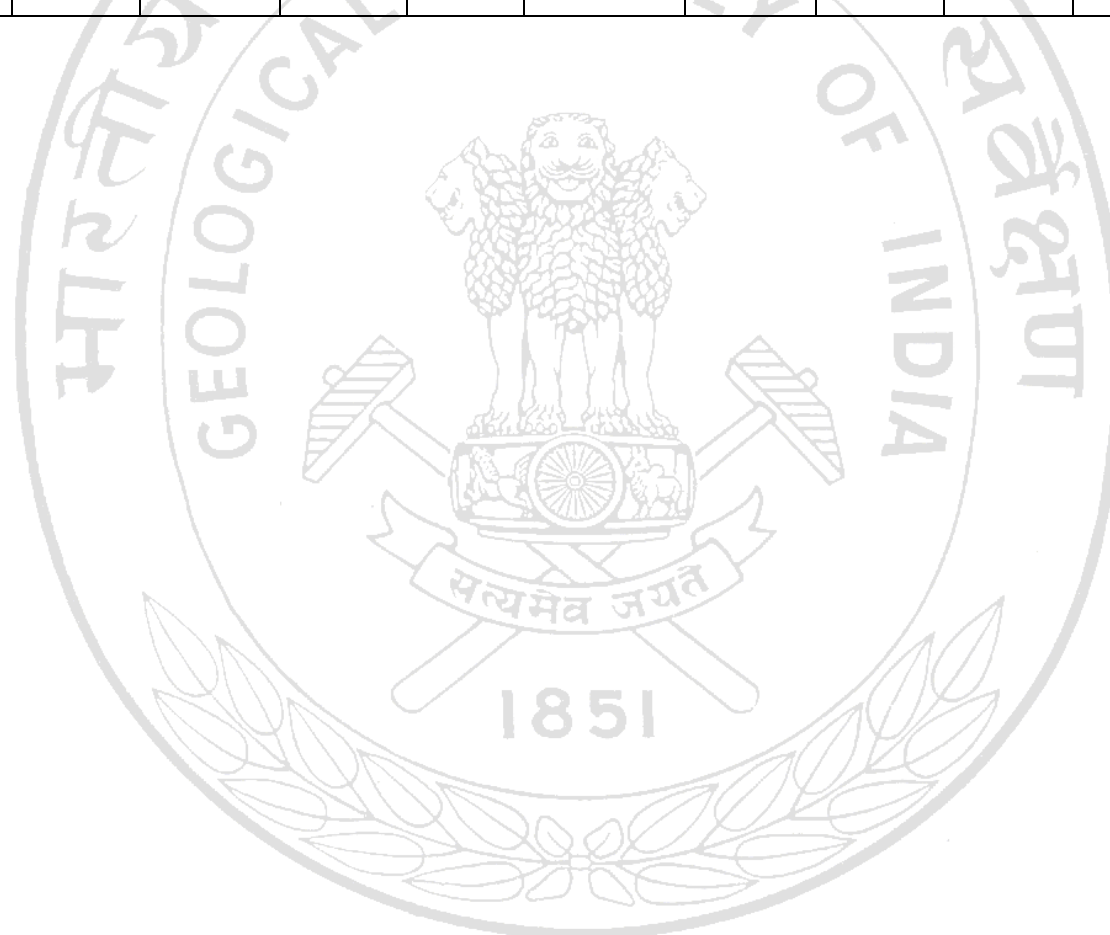
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Baihar	22	03	00	80	33	00	64B/012
Balaghat	21	49	00	80	11	00	64C/01
Baputola	21	53	00	79	44	00	64C/01
Bharweli	21	50	00	80	14	00	64C/01
Chhinditola	21	58	00	80	28	00	64C/05
Ghondi	21	55	00	79	25	00	64C/05
Gudma	21	58	00	80	27	00	64C/05
Jagantola	21	57	00	80	26	00	64C/05
Jagantola	21	57	00	79	16	00	64C/05
Kanhatola	21	52	30	80	21	00	64C/01
Laugur	21	56	00	79	21	00	64C/05
Rupjhar	21	57	00	80	25	00	64C/05
Samnapur	21	58	00	80	29	00	64C/05
Ukwa	21	58	00	80	26	00	64C/05

Annexure I:

Chemical analytical results of Bed Rock Samples

Sr. No.	Reference No.	SiO ₂ (%)	Fe ₂ O ₃ (%)	TiO ₂ (%)	P ₂ O ₅ (%)	MnO (%)	Cu (mg/Kg)	Pb (mg/Kg)	Zn (mg/Kg)	Co (mg/Kg)	Ni (mg/Kg)	Cr (mg/Kg)
1	SK/BRS/1 /FS/16-17	84.69	2.77	<0.1	0.22	6.97	70	40	80	70	30	40
2	SK/BRS/ 2/FS/16-17	87.94	2.23	<0.1	0.11	1.90	55	10	20	15	15	35
3	SK/BRS/ 3/FS/16-17	65.72	4.98	0.18	1.79	5.45	395	35	75	35	35	40
4	SK/BRS/4 /FS/16-17	6.07	1.93	<0.1	<0.01	72.62	10	15	20	30	30	10
5	SK/BRS/5 /FS/16-17	39.34	22.91	0.82	1.25	4.02	105	30	310	45	135	40
6	SK/BRS/ 6/FS/16-17	33.02	15.94	0.63	0.76	27.35	140	35	120	65	90	110
7	SK/BRS/ 7/FS/16-17	36.64	15.78	0.69	0.82	22.08	150	50	110	75	110	120
8	SK/BRS/ 8/FS/16-17	53.31	8.74	1.25	0.23	13.42	145	65	130	55	95	110
9	SK/BRS/ 9/FS/16-17	55.4	8.91	1.38	0.26	7.32	105	40	110	50	95	120
10	SK/BRS/ 10/FS/16-17	39.19	7.63	<0.1	1.72	33.46	0.10%	430	500	70	110	30
11	SK/BRS/ 11/FS/16-17	20.72	19.57	0.34	3.51	32.02	240	170	145	110	140	50
12	SK/BRS/ 12/FS/16-17	17.05	15.52	0.35	2.05	39.85	270	155	260	100	120	70
13	SK/BRS/ 13/FS/16-17	35.27	9.51	0.31	2.04	29.75	440	155	400	60	165	70
14	SK/BRS/14 /FS/16-17	30.23	20.28	0.41	2.54	26.20	320	310	215	85	100	85
15	SK/BRS/ 15/FS/16-17	24.97	18.27	0.12	3.85	26.14	850	800	100	55	100	100
16	SK/BRS/ 16/FS/16-17	11.69	8.88	0.04	0.76	51.8	75	65	155	110	120	85

17	SK/BRS/ 17/FS/16-17	36.15	9.8	0.17	1.62	28.38	290	195	135	85	90	170
18	SK/BRS/ 18/FS/16-17	41.49	6.59	0.06	1.19	29.04	300	95	85	80	90	130
19	SK/BRS/ 19/FS/16-17	39.33	5.22	0.14	0.75	31.76	110	40	100	65	80	245
20	SK/BRS/ 20/FS/16-17	35.19	7.78	0.19	1.29	26.7	225	140	75	95	80	150



Annexure II

Chemical analytical results of Core samples

Sr. No.	Reference No.	SiO ₂ (%)	Fe ₂ O ₃ (%)	MnO (%)	P ₂ O ₅ (%)	TiO ₂ (%)	Mn (%)	P(%)	Fe(%)	Si(%)
1	UKBH-1/CS/01/16-17#	55.81	18.09	7.59#	0.18	0.52	5.878#	0.0774	12.663	26.0447
2	UKBH-2/CS/01/16-17	59.01	10.64	16.27	2.39	0.22	12.600	1.0277	7.448	27.538
3	UKBH-2/CS/02/16-17	39.63	8.97	17.81	2.56	0.51	13.793	1.1008	6.279	18.494
4	UKBH-2/CS/03/16-17*	16.36	8.31	41.96	1.37	0.21	32.49*	0.5891	5.817	7.63467
5	UKBH-2/CS/04/16-17*	12.46	8.81	51.30	0.73	0.05	39.729*	0.3139	6.167	5.81467
6	UKBH-2/CS/05/16-17*	27.01	4.27	48.61	0.56	0.01	37.646*	0.2408	2.989	12.6047
7	UKBH-2/CS/06/16-17*	36.38	8.41	38.05	0.51	0.06	29.468*	0.2193	5.887	16.9773
8	UKBH-2/CS/07/16-17*	7.94	9.40	48.29	0.63	0.06	37.39*	0.2709	6.58	3.70533
9	UKBH-2/CS/08/16-17*	48.39	7.78	32.64	0.19	0.25	25.278*	0.0817	5.446	22.582
10	UKBH-2/CS/09/16-17	46.04	17.92	10.87	0.17	1.03	6.522	0.0731	12.544	21.4853
11	UKBH-2/CS/10/16-17*	36.67	9.55	41.47	0.20	0.28	32.116*	0.086	6.685	17.1127
12	UKBH-3/CS/01/16-17	41.15	8.09	13.25	1.62	0.42	10.2616	0.6966	5.663	19.2033
13	UKBH-3/CS/02/16-17	38.41	8.21	17.38	2.08	0.21	13.4601	0.8944	5.747	17.9247
14	UKBH-3/CS/03/16-17	42.81	6.49	27.45	1.23	0.06	21.2589	0.5289	4.543	19.978
15	UKBH-3/CS/04/16-17	55.22	3.76	19.36	0.29	0.03	14.9935	0.1247	2.632	25.7693
16	UKBH-3/CS/05/16-17*	22.30	8.56	40.05	1.02	0.04	31.0171*	0.4386	5.992	10.4067
17	UKBH-3/CS/06/16-17*	42.79	4.28	29.25	0.76	0.11	22.653*	0.3268	2.996	19.9687
18	UKBH-3/CS/07/16-17*	33.80	6.57	28.35	2.63	0.19	21.9559*	1.1309	4.599	15.7733
19	UKBH-3/CS/08/16-17*	25.46	7.67	39.47	2.39	0.18	30.5679*	1.0277	5.369	11.8813
20	UKBH-3/CS/09/16-17*	12.44	9.23	43.28	1.62	0.06	33.5186*	0.6966	6.461	5.80533
21	UKBH-3/CS/10/16-17*	29.75	8.04	37.19	0.79	0.12	28.8022*	0.3397	5.628	13.8833
22	UKBH-3/CS/11/16-17*	11.52	4.11	49.84	0.39	0.05	38.5991*	0.1677	2.877	5.376
23	UKBH-3/CS/12/16-17*	40.84	4.45	27.13	0.44	0.06	21.0111*	0.1892	3.115	19.0587
24	UKBH-3/CS/13/16-17*	17.00	3.99	44.86	0.30	0.02	34.7423*	0.129	2.793	7.93333
25	UKBH-3/CS/14/16-17*	23.07	6.01	29.98	0.56	0.04	23.2183*	0.2408	4.207	10.766
26	UKBH-3/CS/15/16-17*	22.39	4.62	30.61	0.67	0.01	23.7062*	0.2881	3.234	10.4487
27	UKBH-3/CS/16/16-17	32.15	7.45	18.79	2.46	0.19	14.5521	1.0578	5.215	15.0033
28	UKBH-3/CS/17/16-17*	17.08	4.54	31.35	0.66	0.05	24.2793*	0.2838	3.178	7.97067

29	UKBH-3/CS/18/16-17	39.20	7.25	13.31	1.42	0.33	10.3081	0.6106	5.075	18.2933
30	UKBH-4/CS/01/16-17	28.98	10.12	23.64	1.88	0.26	18.3082	0.8084	7.084	13.524
31	UKBH-4/CS/02/16-17*	41.48	5.08	31.13	0.59	0.13	24.1089*	0.2537	3.556	19.3573
32	UKBH-4/CS/03/16-17*	35.51	6.47	33.74	1.24	0.11	26.1303*	0.5332	4.529	16.5713
33	UKBH-4/CS/04/16-17*	17.03	6.32	43.38	1.18	0.04	33.5961*	0.5074	4.424	7.94733
34	UKBH-4/CS/05/16-17*	16.28	7.28	40.60	1.37	0.03	31.4431*	0.5891	5.096	7.59733
35	UKBH-4/CS/06/16-17*	16.68	4.92	43.45	0.71	0.03	33.6503*	0.3053	3.444	7.784
36	UKBH-4/CS/07/16-17*	17.64	4.20	38.22	0.72	0.03	29.5999*	0.3096	2.94	8.232
37	UKBH-4/CS/08/16-17	20.57	8.69	22.94	0.87	0.22	17.7661	0.3741	6.083	9.59933
38	UKBH-4/CS/09/16-17	40.49	5.25	15.73	0.56	0.20	12.1823	0.2408	3.675	18.8953
39	UKBH-5/CS/01/16-17	42.62	8.59	21.63	1.62	0.26	16.7516	0.6966	6.013	19.8893
40	UKBH-5/CS/02/16-17	34.67	12.30	13.20	2.79	0.47	10.2229	1.1997	8.61	16.1793
41	UKBH-5/CS/03/16-17*	37.31	9.86	27.73	1.50	0.17	21.4758*	0.645	6.902	17.4113
42	UKBH-5/CS/04/16-17	29.07	11.14	21.78	2.42	0.23	16.8677	1.0406	7.798	13.566
43	UKBH-5/CS/05/16-17*	27.33	7.47	31.40	1.13	0.08	24.318*	0.4859	5.229	12.754
44	UKBH-5/CS/06/16-17	29.51	4.77	14.49	0.85	0.05	11.2219	0.3655	3.339	13.7713
45	UKBH-5/CS/07/16-17	40.70	10.64	6.52	0.37	0.49	5.04948	0.1591	7.448	18.9933
46	UKBH-5/CS/08/16-17	41.05	14.69	13.60	0.83	0.60	10.5327	0.3569	10.283	19.1567
47	UKBH-5/CS/09/16-17	39.97	14.68	14.18	1.21	0.54	10.9818	0.5203	10.276	18.6527
48	UKBH-6/CS/01/16-17	37.79	7.48	23.64	1.25	0.21	18.3082	0.5375	5.236	17.6353
49	UKBH-6/CS/02/16-17*	36.55	6.37	30.18*	1.02	0.07	23.3732*	0.4386	4.459	17.0567
50	UKBH-6/CS/03/16-17	41.95	4.93	22.03	0.96	0.05	17.0614	0.4128	3.451	19.5767
51	UKBH-6/CS/04/16-17	54.82	5.18	1.47	0.28	0.37	1.13846	0.1204	3.626	25.5827

This value is diluted due to core loss so it will not be used in ore resource calculation.

* This indicates ore in ore horizon which is being used for calculations of weighted average.

Determination Of Mn from MnO: Here atomic weight of Mn (54.93809) is added with Atomic weight of O (15.9994) to determined molecular weight of MnO which comes to 70.937449. Now atomic weight of Mn is divided by molecular weight of MnO to determine conversion factor which comes to 0.77446. This conversion factor is multiplied with MnO values to determine Mn% in MnO.

Illustration: Here in sample UKBH-2/CS/04/16-17 MnO Value is 51.30% . This value when multiplied by 0.77446 gives Mn 39.729%.

Determination Of weighted average of Mn: Weighted average of Mn of ore in single borehole is determined by following steps.

1. Addition of multiplication of Mn values of ore with their sample length.
2. This value thus determined by total length of ore.

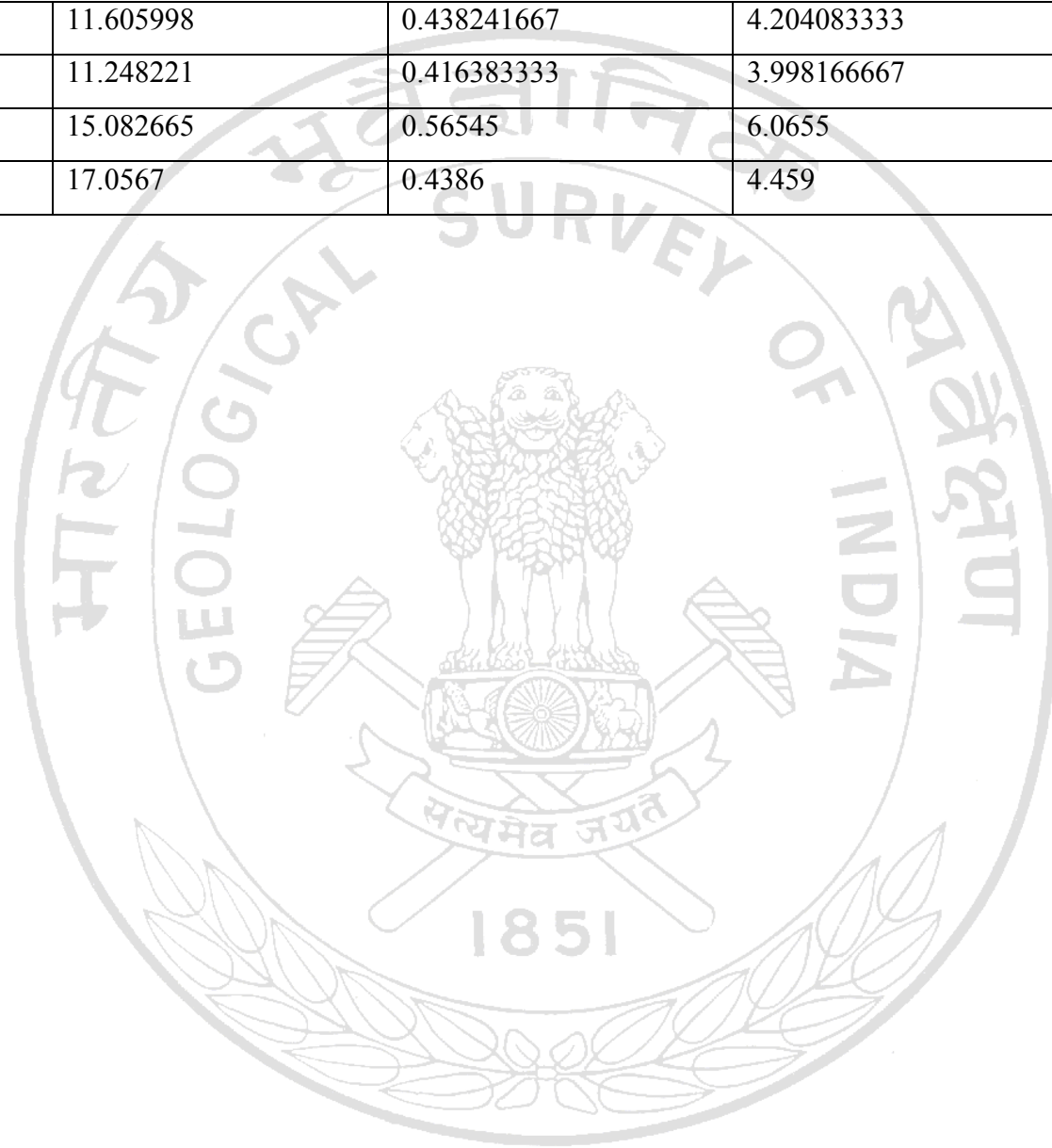
Illustration: In UKBH-5 sample number UKBH-5/CS/03/16-17 and UKBH-5/CS/05/16-17 are considered as ore. Therefore Their Mn values are multiplied with sample length and values thus obtained are added. $(21.4758 \times 0.201 + 24.318 \times 0.201)$. The value thus obtained is divided by total length of ore in borehole which is 0.40 which gives the value 23.0114.

Similarly for determination of Si, P and Fe from SiO_2 , P_2O_5 and Fe_2O_3 conversion factor was determined using atomic weights and is found to be 0.46666, 0.43 and 0.7 respectively.

Weighted average of Si, P, and Fe is determined in similar manner as mentioned above for Mn and is listed below in tabular form for ore in ore horizon along with Mn. For borehole no. UKBH-1 and UKBH-6 as there was single sample of ore so values have been used directly.

Borehole Number	Si (in %)	P (in %)	Fe (in %)	Mn (in %)
UKBH-1	26.044663	0.0774	12.663	5.878
UKBH-2	12.347332	0.257385714	5.653	33.6151

UKBH-3	11.605998	0.438241667	4.204083333	27.9785
UKBH-4	11.248221	0.416383333	3.998166667	29.9035
UKBH-5	15.082665	0.56545	6.0655	23.0114
UKBH-6	17.0567	0.4386	4.459	23.3732



Annexure-III

Borehole details of previous works and weighted average used in resource estimation

Borehole Number and section line	Mn (in %)	Angle of drilling	Azimuth	Date of initiation	Date of completion	Depth of intersection of ore along borehole	Vertical level of intersection of ore	Level of Borehole
*GSU-30, W-1	37.7	Vertical	-----	11/05/1978	17/05/1978	37.2 m	37.2 m	First level
GSU-31, W-1	34.13	Vertical	-----	21/05/1978	28/05/1978	68.6 m	68.6 m	First Level
UKH-7, W-1	33.91	65°	S40°E	17/05/1994	20/06/1994	208.3 m	188.78 m	Third Level
*GSU-29, W-2	39.1	Vertical	-----	04/05/1978	08/05/1978	23.8 m	23.8 m	First level
GUG-2, W-2	40.63	Vertical	-----	10/05/1972	08/06/1972	89.94 m	89.94 m	Second Level
UKH-4, W-2	36.91	Vertical	-----	13/02/1994	09/05/1994	189.7 m	189.7 m	Third Level
*GSU-26, W-3	45.85	Vertical	-----	09/04/1978	18/04/1978	48.85 m	48.85 m	First Level
GSU-27, W-3	34.36	Vertical	-----	24/04/1978	11/05/1978	96.6 m	96.6 m	Second Level
UKH-3, W-3	43.63	65°	S40°E	25/01/1994	24/03/1994	183.40 m	166.21 m	Third Level
UKH-14, W-3	37.82	65°	S40°E	07/01/1995	05/03/1995	221.8 m	201.01	Third Level
*GSU-21, W-4	38.27	Vertical	-----	09/02/1978	27/02/1978	67.8 m	67.8 m	First Level
GSU-24,	35.84	Vertical	-----	06/03/1978	01/04/1978	125.15 m	125.15 m	Second

W-4								Level
GSU-34, W-4	41.00	Vertical	-----	17/06/1978	19/07/1978	164.68 m	164.68 m	Third Level
UKH-16, W-4	38.81	65°	S40°E	26/02/1995	20/04/1995	219.70 m	199.11 m	Third Level

*Borehole written in red colour lies within MOIL boundary.

Weighted average calculations: Here weighted average of whole block is determined by adding multiplication of grade of Mn with true thickness calculated for individual boreholes; which is then divided by sum of true thickness. Here UKBH-1 is not considered for calculation.

Section Line	B.H. No.	True Thickness Calculated	Grade Of Mn	Weighted average of whole
W-1	GSU-30	0.6	37.7	35.138
W-1	GSU-31	0.27	34.13	
W-1	UKH-7	0.3	33.91	
A-B	UKBH-5	0.402	23.0117	
A-B	UKBH-1	0.14488	-----	
W-2	UKH-4	0.85	36.91	
W-2	GUG-2	0.62	40.63	
W-2	GSU-29	0.39	39.1	
C-D	UKBH-4	1.26	29.9035	
C-D	UKBH-2	1.4	33.6151	
W-3	UKH-14	1.54	37.82	
W-3	UKH-3	1.01	43.63	
W-3	GSU-27	0.93	34.36	
W-3	GSU-26	0.73	45.85	
E-F	UKBH-6	0.2	23.3732	
E-F	UKBH-3	2.41	27.9785	
W-4	UKH-16	0.3	38.81	
W-4	GSU-34	1.76	41	
W-4	GSU-24	0.95	35.84	
W-4	GSU-21	1.53	38.27	

ANNEXURE-IV

SUMMARISED LITHOLOGS OF PREVIOUS BOREHOLES IN UKWA MANGANESE DEPOSIT-WESTERN BLOCK

(Source- Choudhary et al; 1979; Khan et al;1993-96)

Note: All the UKH series boreholes are 70° South easterly inclined boreholes except UKH-4, which is vertical borehole. All other boreholes of GSU,GUG series are vertical boreholes.

BOREHOLE NO. UKH-3 :

(A) Location: Western Block, Section Line No. W-3.

(B) R.L at Collar: 599.484m.

(C) Date of commencement: 25/01/1994.

(D) Date of completion: 24/03/1994.

(E) Total Depth: 205.25m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0.00	3.20	Light yellow clayey soil.	
3.20	17.20	Laterized garnetiferous mica schist.	
17.20	39.92	Weathered garnetiferous muscovite schist.	
39.92	40.90	Milky white quartz vein.	
40.90	61.65	Garnetiferous muscovite schist.	
61.65	87.85	Interbanded sequence of garnetiferous quartz muscovite schist and quartzite.	
87.85	180.15	Quartz sericite muscovite schist with rare garnets.	
180.15	183.40	Quartz sericite schist/phyllite.	
183.40	186.30	Ore Horizon: Interbanded sequence of manganiferous quartzite and Mn ore at	1) Core recovery in ore horizon: 94%. 2) Cumulative true thickness of

		places traversed by quartzo-feldspathic veins, individual ore bands <1 to 14 cm thick.	manganese ore bands: 1.01m. 3) Cumulative true thickness of waste rock: 1.89m.
186.30	192.35	Muscovite-calcite schist.	
192.35	194.75	Calcareous phyllite with irregular bands and patches of granitic material.	
194.75	205.25	Augen gneiss with angular to rounded fragments of K-granite and pegmatite along with enclaves of biotite schist.	

BOREHOLE NO. UKH-4 :

(A) Location: Western Block, Section Line No. W-2.

(B) R.L at Collar: 602.489m.

(C) Date of commencement: 13/02/1994.

(D) Date of completion: 09/05/1994.

(E) Total Depth: 209.5m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	4	Yellow silty soil.	
4	40	Partly Laterized garnetiferous mica schist.	
40	65.65	Garnetiferous quartz-muscovite schist.	
65.65	102.65	Interbanded sequence of garnetiferous quartz-muscovite schist and quartzite.	
102.65	129.9	Garnetiferous quartz sericite muscovite schist.	
129.9	152.95	Phyllite.	
152.95	181.2	Garnetiferous quartz-sericite-muscovite	

		schist.	
181.2	189.7	Sericite phyllite.	
189.7	191.7	Ore horizon: Interbanded manganiferous quartzite and Mn ore traversed by thin quartz-carbonate veinlets, ore partly brecciated, individual ore bands <1 to 10 cm thick.	1) Core recovery in ore horizon: 100%. 2) Cumulative true thickness of manganese ore bands: 0.85m. 3) Cumulative true thickness of waste rock: 0.85m.
191.7	196.35	Muscovite calcite schist.	
196.35	203	Sericite phyllite.	
203	209.5	Augen gneiss traversed by quartz and pegmatite veins.	

BOREHOLE NO. UKH-7 :

(A) Location: Western Block, Section Line No. W-1.

(B) R.L at Collar: 601.668m.

(C) Date of commencement: 17/05/1994.

(D) Date of completion: 20/06/1994.

(E) Total Depth: 220.45m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	2	Yellow silty soil.	
2	53.8	Partly laterized weathered garnetiferous muscovite schist.	
53.8	69.4	Garnetiferous Muscovite schist.	
69.4	105.4	Interbanded garnetiferous quartz-muscovite schist and quartzite.	
105.4	198.65	Garnetiferous sericite muscovite schist.	

198.65	208.3	Muscovite schist.	
208.3	209.05	Ore horizon: Interbanded manganiferous quartzite and Mn ore traversed by thin quartz-carbonate veinlets, ore partly brecciated, individual ore bands <1 to 10 cm thick.	1) Core recovery in ore horizon: 100%. 2) Cumulative true thickness of manganese ore bands: 0.30m. 3) Cumulative true thickness of waste rock: 0.45m.
209.05	213.75	Muscovite calcite schist.	
213.75	216	Sericite phyllite.	
216	220.45	Augen gneiss traversed by pegmatite veins.	

BOREHOLE NO. UKH-14 :

- (A) Location: Western Block, Section Line No. W-3.
 (B) R.L at Collar: 598.709m.
 (C) Date of commencement: 07/01/1995.
 (D) Date of completion: 05/03/1995.
 (E) Total Depth: 243.65m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	3	Dark grey coloured clayey soil.	
3	56	Weathered garnetiferous mica schist.	
56	91.05	Garnetiferous quartz muscovite schist.	
91.05	110.75	Interbanded quartz muscovite schist and quartzite.	
110.75	219.1	Garnetiferous quartz muscovite schist.	
219.1	221.8	Quartz sericite muscovite schist.	
221.8	227.2	Ore horizon: Interbanded manganiferous	1) Core recovery in ore horizon: 100%.

		quartzite and Mn ore traversed by thin quartz-carbonate veinlets, ore partly brecciated, individual ore bands <1 to 18 cm thick.	2) Cumulative true thickness of manganese ore bands: 1.54m. 3) Cumulative true thickness of waste rock: 3.86m.
227.2	235.8	Sericite schist downward grading into sericite-calcite and chlorite-sericite schist.	
235.8	243.65	Augen gneiss traversed by thin quartz and pegmatite veins.	

BOREHOLE NO. UKH-16 :

(A) Location: Western Block, Section Line No. W-4.

(B) R.L at Collar: 601.176m.

(C) Date of commencement: 26/02/1995.

(D) Date of completion: 20/04/1995.

(E) Total Depth: 245.20m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	6	Dark grey to yellowish, clayey soil	
6	50.85	Weathered garnetiferous mica schist.	
50.85	100.5	Garnetiferous quartz muscovite schist.	
100.5	130	Banded garnetiferous quartz-muscovite schist.	
130	215.75	Garnetiferous quartz muscovite schist.	
215.75	219.7	Quartz sericite muscovite schist.	
219.7	220.9	Ore horizon: Interbanded sequence of manganiferous quartzite and Mn ore traversed by thin quartz-carbonate veinlets,	1) Core recovery in ore horizon: 99%. 2) Cumulative true thickness of manganese ore bands: 0.30m.

		brecciated, Single Mn ore band of 30 cm thick.	3) Cumulative true thickness of waste rock: 0.90m.
220.9	225	Muscovite calcite schist	
225	230.2	Chlorite-sericite schist/phyllite.	
230.2	245.2	Augen gneiss with enclaves of biotite schist and fragments of quartz and pegmatite traversed by quartz and pegmatite veins.	

BOREHOLE NO. GSU-21 :

- (A) Location: Western Block, Section Line No. W-4.
 (B) R.L at Collar: 607.17m.
 (C) Date of commencement: 09/02/1978.
 (D) Date of completion: 27/02/1978.
 (E) Total Depth: 71.60m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	12	Sericitic soil.	
12	23.8	Weathered sericite schist with muscovite.	
23.8	24.75	Vein quartz.	
24.75	67.8	Quartz muscovite schist.	
67.8	70.4	Ore horizon.	1) Core recovery in ore horizon: 94%. 2) Cumulative true thickness of manganese ore bands: 1.53m. 3) Cumulative true thickness of waste rock: 0.72m.
70.4	71.6	Muscovite calcite schist.	

BOREHOLE NO. GSU-24 :

(A) Location: Western Block, Section Line No. W-4.

(B) R.L at Collar: 604.10m.

(C) Date of commencement: 06/03/1978.

(D) Date of completion: 01/04/1978.

(E) Total Depth: 129.85m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	18.75	Laterite and quartz vein.	
18.75	29.15	Weathered muscovite schist with laterite.	
29.15	29.2	Vein quartz.	
29.2	125.15	Weathered muscovite schist.	
125.15	127	Weathered muscovite schist and ore horizon.	1) Core recovery in ore horizon: 97%. 2) Cumulative true thickness of manganese ore bands: 0.095m. 3) Cumulative true thickness of waste rock: 0.065m.
127	129.85	Muscovite calcite schist.	

BOREHOLE NO. GSU-26 :

(A) Location: Western Block, Section Line No. W-3.

(B) R.L at Collar: 602.99m.

(C) Date of commencement: 09/04/1978.

(D) Date of completion: 18/04/1978.

(E) Total Depth: 54m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	24.35	Clay.	
24.35	48.85	Weathered quartz muscovite schist.	
48.85	50.1	Ore horizon.	1) Core recovery in ore horizon: 65%. 2) Cumulative true thickness of manganese ore bands: 0.73m. 3) Cumulative true thickness of waste rock: 0.40m.
50.1	50.3	Sericite schist.	
50.3	54	Clay.	

BOREHOLE NO. GSU-27 :

- (A) Location: Western Block, Section Line No. W-3.
 (B) R.L at Collar: 602.19m.
 (C) Date of commencement: 24/04/1978.
 (D) Date of completion: 11/05/1978.
 (E) Total Depth: 101m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	29.85	Sericitic soil.	
29.85	37.05	Weathered quartz muscovite schist.	
37.05	96.6	Quartz muscovite schist.	
96.6	99.65	Ore horizon.	1) Core recovery in ore horizon: 89%. 2) Cumulative true thickness of manganese ore bands: 0.93m.

			3) Cumulative true thickness of waste rock: 1.83m.
99.65	101	Muscovite calcite schist.	

BOREHOLE NO. GSU-29 :

(A) Location: Western Block, Section Line No. W-2.

(B) R.L at Collar: 601.77m.

(C) Date of commencement: 04/05/1978.

(D) Date of completion: 08/05/1978.

(E) Total Depth: 28.50m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	6.1	Soil with sand.	
6.1	23.8	Weathered muscovite schist.	
23.8	24.7	Ore horizon.	1) Core recovery in ore horizon: 68%. 2) Cumulative true thickness of manganese ore bands: 0.39m. 3) Cumulative true thickness of waste rock: 0.48m.
24.7	26.15	Clay.	
26.15	28.5	Quartz muscovite schist.	

BOREHOLE NO. GSU-30 :

(A) Location: Western Block, Section Line No. W-1.

(B) R.L at Collar: 599.68m.

(C) Date of commencement: 11/05/1978.

(D) Date of completion: 17/05/1978.

(E) Total Depth: 43.0m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	16.5	Coarse sand with muscovite schist.	
16.5	16.6	Vein quartz.	
16.6	37.2	Quartz muscovite schist.	
37.2	38.75	Ore horizon.	1) Core recovery in ore horizon: 82%. 2) Cumulative true thickness of manganese ore bands: 0.60m. 3) Cumulative true thickness of waste rock: 0.86m.
38.75	43	Clay.	

BOREHOLE NO. GSU-31 :

(A) Location: Western Block, Section Line No. W-1.

(B) R.L at Collar: 598.83m.

(C) Date of commencement: 21/05/1978.

(D) Date of completion: 28/05/1978.

(E) Total Depth: 71.05m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	14	Sandy soil.	
14	19.75	Vein quartz and muscovite schist.	

19.75	68.6	Weathered quartz muscovite schist.	
68.6	70.05	Ore horizon.	1) Core recovery in ore horizon: 72%. 2) Cumulative true thickness of manganese ore bands: 0.27m. 3) Cumulative true thickness of waste rock: 1.13m.
70.05	71.05	Sand mixed with ore.	

BOREHOLE NO. GSU-34 :

(F) Location: Western Block, Section Line No. W-4.

(G) R.L at Collar: 601.74m.

(H) Date of commencement: 17/06/1978.

(I) Date of completion: 19/07/1978.

(J) Total Depth: 169.80m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	26.00	Laterite soil.	
26.00	28.70	Vein quartz.	
28.70	164.68	quartz muscovite schist.	
164.68	167.10	Ore horizon.	1) Core recovery in ore horizon: 97%. 2) Cumulative true thickness of manganese ore bands: 1.41m. 3) Cumulative true thickness of waste rock: 0.42m.
167.10	169.80	Muscovite calcite schist.	

BOREHOLE NO. GUG-2 :

(A) Location: Western Block, Section Line No. W-2.

(B) R.L at Collar: 600.81m.

(C) Date of commencement: 10/05/1972.

(D) Date of completion: 08/06/1972.

(E) Total Depth: 105.90m.

Depth along BH in meters		Summarized Lithology	Details of ore horizon
From	To		
0	21	Soil and subsoil.	
21	89.94	Phyllite/mica schist, weathered on top with thin bands and disseminations of altered garnet.	
89.94	91.25	Ore horizon.	1) Core recovery in ore horizon: 54%. 2) Cumulative true thickness of manganese ore bands: 0.62m. 3) Cumulative true thickness of waste rock: 0.59m.
91.25	92.15	Caving neither core nor sludge recovered.	
92.15	100.2	Quartzite felspathic and sericitic.	
100.2	105.9	Medium and fine grained biotite gneiss.	

Annexure-V:

Analytical Results of PCS samples

Sr. No .	Reference No.	SiO ₂ (%)	Al ₂ O ₃ (%)	Fe ₂ O ₃ (%)	CaO (%)	MgO (%)	Na ₂ O (%)	K ₂ O (%)	TiO ₂ (%)	P ₂ O ₅ (%)	MnO (%)	LOI (%)	Ba (mg/Kg)	Nb (mg/Kg)	Rb (mg/Kg)	Sr (mg/Kg)	Y (mg/Kg)	Zr (mg/Kg)
1	SK/PCS/1/FS/16-17	33.79	7.26	16.06	2.63	2.29	0.17	1.08	0.63	0.71	26.75	6.28	6767	6	34	394	28	64
2	SK/PCS/2/FS/16-17	36.35	7.47	15.93	3.20	2.59	0.18	1.25	0.69	0.79	22.40	6.75	8102	7	40	482	30	71
3	SK/PCS/3/FS/16-17	44.45	11.89	15.37	1.27	1.22	<0.05	1.73	0.97	0.15	15.02	7.05	7032	9	64	634	29	133
4	SK/PCS/4/FS/16-17	46.72	16.51	16.01	1.44	1.59	<0.05	1.95	1.31	0.18	6.73	6.96	5497	13	102	431	33	201
5	SK/PCS/5/FS/16-17	40.58	4.03	7.67	0.76	<0.1	0.07	1.44	0.09	1.87	32.72	6.83	17671	<5	36	496	59	22
6	SK/PCS/6/FS/16-17	19.23	12.60	13.87	0.95	<0.1	0.10	1.54	0.28	3.81	36.04	8.75	24373	<5	58	1120	61	58
7	SK/PCS/7/FS/16-17	17.10	13.82	12.57	0.89	<0.1	0.11	1.89	0.35	2.24	39.76	7.75	22837	6	52	877	70	57
8	SK/PCS/8/FS/16-17	34.71	6.59	9.37	1.41	0.49	0.30	1.21	0.31	2.07	29.06	8.63	17599	<5	37	602	63	37
9	SK/PCS/9/FS/16-17	42.79	5.50	15.10	0.86	<0.1	<0.05	1.17	0.25	1.90	25.22	5.82	11670	<5	43	856	53	46
10	SK/PCS/10/FS/16-17	21.33	9.84	17.06	0.97	<0.1	<0.05	0.98	0.19	5.87	24.08	9.87	39900	<5	37	1012	79	50

Annexure-VI:

Analytical Results of REE Samples

Sr. No.	Reference No.	La (mg/kg)	Ce (mg/kg)	Pr (mg/kg)	Nd (mg/kg)	Eu (mg/kg)	Sm (mg/kg)	Gd (mg/kg)	Tb (mg/kg)	Dy (mg/kg)	Ho (mg/kg)	Er (mg/kg)	Tm (mg/kg)	Yb (mg/kg)	Lu (mg/kg)
1	SK/REE/1/FS/16-17	95.11	162.48	15.93	48.86	2.11	5.43	4.26	0.39	1.39	0.21	0.46	0.07	0.41	0.06
2	SK/REE/2 /FS/16-17	11.52	21.83	2.50	8.34	0.38	2.38	2.29	0.61	4.23	0.78	2.56	0.40	2.88	0.42
3	SK/REE/ 3/FS/16-17	186.52	366.24	38.61	123.36	5.49	26.56	12.24	1.48	6.45	1.08	2.82	0.44	2.72	0.47
4	SK/REE/ 4/FS/16-17	26.96	47.84	5.28	18.32	1.03	4.13	2.73	0.43	2.28	0.37	1.03	0.16	1.22	0.20
5	SK/REE/5 /FS/16-17	318.92	503.68	51.45	161.33	3.63	19.10	15.77	1.88	8.07	1.29	3.30	0.48	2.82	0.41
6	SK/REE/6 /FS/16-17	125.00	216.85	25.60	83.05	1.81	11.88	8.73	1.30	6.52	1.09	3.14	0.50	3.49	0.55
7	SK/REE/ 7/FS/16-17	15.20	30.86	3.41	13.58	0.86	3.54	2.80	0.53	3.16	0.62	1.74	0.25	1.34	0.21
8	SK/REE/ 8/FS/16-17	48.76	117.01	12.01	46.77	2.84	12.47	9.68	1.73	10.46	2.10	6.26	0.91	5.48	0.81
9	SK/REE/9 /FS/16-17	2.87	8.72	0.60	2.35	0.15	0.58	0.45	0.09	0.60	0.11	0.26	0.05	0.29	0.05
10	SK/REE/ 10/FS/16-17	1.65	4.48	0.33	1.55	0.24	0.99	0.39	0.06	0.39	0.07	0.20	0.03	0.19	0.02

Annexure-VII:

Analytical Results of Water samples

Sr. No.	Reference No.	V (µg/K g)	Cr (µg/K g)	Mn (µg/K g)	Fe (µg/K g)	Co (µg/K g)	Ni (µg/K g)	Cu (µg/K g)	Zn (µg/K g)	As (µg/K g)	Ag (µg/K g)	Pb (µg/K g)	Cd (µg/K g)	Bi (µg/K g)	Hg (µg/K g)
1	64C/5/SK/UKW1/FS2 016-17	<5	<4	7.43	178.7	0.09	<1	<0.5	22.22	<1	1.89	<0.5	<0.01	0.045	<1
2	64C/5/SK/UKW2/FS2 016-17	<5	<4	265.24	145.82	0.22	<1	<0.5	2.00	<1	0.87	<0.5	<0.01	<0.02	<1
3	64C/5/SK/UKW3/FS2 016-17	<5	<4	7.17	125.44	0.05	<1	0.71	66.49	<1	1.97	<0.5	<0.01	<0.02	<1
4	64C/5/SK/UKW4/FS2 016-17	<5	<4	0.94	391.99	0.15	1.6	<0.5	24.14	<1	1.58	<0.5	<0.01	<0.02	<1
5	64C/5/SK/UKW5/FS2 016-17	<5	52.95	<0.5	<100	<0.05	<1	<0.5	4.75	1.18	1.89	<0.5	<0.01	<0.02	<1

ANNEXURE-VIII

ANALYTICAL RESULTS OF PREVIOUS BOREHOLES IN UKWA MANGANESE DEPOSIT-WESTERN BLOCK

(Source- Choudhary et al; 1979; Khan et al;1993-96

Borehole No: GSU-21

Drilling Run	% Core Rec	Sample No.	Depth along boreholes m.	Sample length cm.	Lithology	% Mn	% Fe	% SiO ₂	% P	Cumulative length of the sample (cm)
67.35-67.90	100	Not sampled	67.35-67.80 67.80-67.90	-	Sericite schist Manganiferous quartzite					
67.90-68.30	92.5	1/GSU-21	67.94-68.02 68.10-68.22	7 3	Quartzite with patch of Mn Quartzite with patch of Mn	24.92	3	30.45	0.427	10
		2/GSU-21	68.27-68.30			35.35	10.95	16.88	0.475	3
		Not sampled	67.90-67.94			-	-	-	-	-
		Not sampled	68.02-0.06			-	-	-	-	-
		Not sampled	68.06-68.10		Mn ore with quartzite Quartzite	-	-	-	-	-
		Not sampled	68.22-68.27		Sericite schist Quartzite with thin Mn band Quartzite	-	-	-	-	-
68.30-69.00	93	3/GSU-21	63.30-68.35 68.43-68.49 68.65-68.71	4 6 6	Mn ore with quartz vein Mn ore with quartz vein Mn ore with quartz vein	42	6.1	16.5	0.212	16
		4/GSU-21	68.71-68.84	12	Mn ore with quartz vein	39.6	6.1	14	0.285	12
		5/GSU-21	68.84-69.00	15	Mn ore +quartz vein	36.8	6.1	18.2	0.448	15
		Not sampled	68.35-68.43	-	+quartzite Quartzite Quartzite	-	-	-	-	-
		Not sampled	68.49-68.65	-	with thin Mn ores	-	-	-	-	-
69.00-69.30	100	67GSU-21	69.00-69.17	17	Mn ore with Calcite and quartz	39.15	5.2	21	0.227	17
		7/GSU-21	69.17-69.30	13	Mn ore with calcite and quartz	45.3	5	17	0.119	13
69.30-69.95	95	8/GSU-21	69.30-69.35 69.47-69.60	5 12	Mn ore with calcite & quartzite Mn ore with calcite &	36.1	3.7	30.9	0.123	17

					intercalation					
		9/GSU-21	69.37-69.47 69.38-69.95	10 11	Mn ore with calcite and quartzite band, do-		2.95	31.3	0.097	21
		10/GSU-21 Not sampled	69.60-69.93 69.73-69.83 69.35-69.37	13 10 -	Mn ore with calcite and quartz intercalation Quartzite	40.3 - -	4.8 - -	16.3 - -	0.147 - -	23 - -
69.95- 70.40	88	11/GSU-21	69.95-70.04	8	Mn ore with calcite	42.7	4.3	10.3	0.162	8
		12/GSU-21	70.09-70.18	8	Mn ore with calcite	31.8	2.4	9	0.059	8

Borehole No: GSU-24

Drilling Run	% Core Rec	Sample No.	Depth along boreholes m.	Sample length cm.	Lithology	% Mn	% Fe	% Sio2	% P	Cumulative length of the sample (cm)
124.20-125.70	94	Not sampled	124.20-125.15	-	Sericite schist					
		1/GSU-24 Not sampled Not sampled Not sampled	125.35-125.37 125.50-125.52 125.61-125.65 125.68-125.70 125.15-125.35 125.37-125.50 125.52-125.61 125.65-125.68	2 1 4 2 - - - -	Mn ore with calcite Mn ore with calcite Mn ore with calcite Mn ore with calcite Quartzite Quartzite Quartzite Quartzite	31.2 - - - - - -	4.6 - - - - - -	30.46 - - - - - -	0.317 - - - - -	09 - - - - - -
125.70-125.95	100	2/GSU-24	125.73-125.74 125.77-125.78 125.80-125.83 125.84-125.86 125.93-125.95	1 1 3 2 2	Mn ore Mn ore Mn ore with quartzite Mn ore with quartzite Mn ore	30	6.3	25.04	0.277	9
		3/GSU-24 Not sampled Not sampled Not sampled	125.86-125.93 125.70-125.73 125.74-125.77 125.78-125.80	7 - - -	Quartzite with Mn association Quartzite Quartzite Quartzite		-	-	-	-

		Not sampled Not sampled	125.83-125.84	-	Quartzite Mn ore with quartz					
125.95- 126.40	100	4/GSU-24	125.96-126.00 126.01-126.02 126.05-126.06 126.08-126.20	4 1 1 12	Mn ore -do- Mn ore with quartzite and calcite Mn ore with calcite quartz	40.72	5.5	18.3	0.162	18
		5/GSU-24 Not sampled Not sampled Not sampled Not sampled Not sampled	126.20-126.40 125.95-125.96 126.00-126.01 126.02-126.05 126.06-126.08	20 - - - -	Mn ore with quartzite & calcite Quartzite Quartzite Quartzite Quartzite	34.84	4.3	23	0.212	-
126.40- 126.60 j	100	6/GSU-24	126.40-126.60	20	Mn ore with quartzite calcite	37.73	4.3	16.8	0.199	20
126.60- 126.75	100	7/GSU-24	126.60-126.75	15	Mn ore with quartzite calcite	35.13	5.2	26.8	0.134	15
126.75- 127.00	92	8/GSU-24	126.75-126.30 126.97-126.99	5 2	Mn ore with calcite+quartzite Mn ore	38.53	4.4	18.9	0.267	7
		9/GSU-24 Not sampled Not sampled Not sampled	126.85-126.96 126.80-126.85 126.96-126.97 126.99-127.00	10 - - -	Mn ore with quartzite +calcite Quartzite Quartzite Calcite	34.33	13.8	11.1	0.247	10
125.70- 125.95	100	2/GSU-24	125.73-125.74 125.77-125.78 125.80-125.83 125.84-125.86 125.93-125.95	1 1 3 2 2	Mn ore Mn ore Mn ore with quartzite Mn ore with quartzite Mn ore	30	6.3	25.04	0.277	9

Borehole No: GSU-26

Drilling Run	% Core Rec	Sample No.	Depth along boreholes m.	Sample d length cm.	Lithology	% Mn	% Fe	% Sio2	% P	Cumulative length of the sample (cm)
48.80-49.80	84	Not sampled	48.80-48.82	1	Schist		-	-	-	-
		1/GSU-26	48.82-48.90	8	Manganiferous quartzite	6.4	-	-	-	-
		2/GSU-26	48.90-49.00	10	Mn ore	40.12	6.5	17.4	0.246	10
		3/GSU-26	49.00-49.30	22	Mn ore	49.1	8.5	1.5	0.189	22
		4/GSU-26	49.30-49.40	10	Mn ore powder	44.73	13.1	1.5	0.213	10
49.40-49.80	75	5/GSU-26	49.40-49.57	15	Mn ore	44.8	10.6	3.47	0.259	15
		6/GSU-26	49.57-49.65	5	Quartzite	2.7	-	-	-	-
		7/GSU-26	49.65-49.80	10	Mn ore with quartzite	47	6.8	5.31	0.215	10
49.80-50.10	50	8/GSU-26	49.80-49.90	5	Mn ore	48.1	5.2	4.3	0.194	5
		9/GSU-26	49.90-50.10	10	Mn ore with quartzite	43.4	6.4	13.2	0.182	10

Borehole No: GSU-27

Drilling Run	% Core Rec	Sample No.	Depth along boreholes m.	Sample d length cm.	Lithology	% Mn	% Fe	% Sio2	% P	Cumulative length of the sample (cm)
96.70-97.15	100	Not sampled	96.70-96.71		Quartzite					
		1/GSU-27	96.71-96.72	1	Mn ore	38.5	5.6	22	0.333	8
			96.74-96.76	2	Mn ore					
			96.80-96.81	1	Mn ore with quartz vein					
			96.84-96.88	4	Mn ore with quartz vein					

		2/GSU-27	96.89-96.95	6	Mn ore	36.9	6	23.25	0.244	15
			96.97-96.99	2	Mn ore with quartz vein					
			97.00-92.03	3	Mn ore with quartz vein					
			97.07-97.11	4	Mn ore					
			96.72-96.74	-	Quartzite					
		Not sampled	96.76-96.80	-	Quartzite					
			96.81-96.84	-	Quartzite					
			96.88-96.09	-	Quartzite					
			96.95-96.97	-	Quartzite					
			96.99-97.00	-	Quartzite					
			97.03-97.07	-	Quartzite					
			97.11-97.15	-	Quartzite					
97.15-97.35	85	3/GSU-27	97.17-97.10	1	Mn ore with quartz vein	34.6	4.2	24.64	0.494	5
			97.20-97.22	1	Mn ore					
			97.26-97.27	1	Mn ore with quartz vein					
			97.27-97.30	2	Mn ore with quartz vein					
			97.15-97.17	-	Quartzite					
		Not	97.19-97.20	-	Quartzite					
97.35-97.75	50	sampled Not sampled	97.22-97.26 97.30-97.35 97.35-97.75	-	Quartz vein Quartz vein & pegmatite egmatite					
97.75-98.35	93	4/GSU-27	98.20-98.35 98.35-98.50	11 -	Mn ore with quartz vein Quartz vein	25.15	12	27.68	0.315	11
98.35-98.75	100	5/GSU-27,	98.50-98.65	15	Mn ore soft	44.2	5.6	12.27	0.181	15
		6/GSU-27.	98.65-98.75	10	Mn ore	50.45	5.7	7.3	0.132	10
98.75-99.15	62	7/GSU-27.	98.75-98.95.	13	Mn ore powder with siliceous	7.6	-	-	-	-
		8/GSU-27	98.95-99.15	12	Mn ore powder with siliceous	18.29	-	-	-	-
99.15-99.55	87	9/GSU-27.	99.15-99.35	18	Mn ore powder with siliceous	26.5	8.4	34.5	0.181	18
		10/GSU-27	99.35-99.55,	17	Mn ore powder with siliceous matter	28.45	8.1	21.4	0.402	17
99.55-	100	11/GSU-27	99.55-99.65.	10	Mn ore powder with calcite and	22.85	6.3	27.7	0.623	10

99.65					calcite and siliceous matter					
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Borehole No: GSU-29

Drilling Run	% Core Rec	Sample No.	Depth along boreholes m.	Sample length cm.	Lithology	% Mn	% Fe	% SiO ₂	% P	Cumulative length of the sample (cm)
23.00-23.85	100	1/GSU-29	23.80-23.85	5	Manganiferous quartzite	13.74	-	-	-	-
23.85-24.15	100	2/GSU-29	23.85-23.90	5	Manganiferous quartzite	3.4	-	-	-	-
			23.97-24.05	8	Manganiferous quartzite					
		3/GSU-29	23.90-23.97	7	Mn ore with quartzite	43.28	7.1	10	0.431	7
		4/GSU-29	23.95-24.15	10	Mn ore with silica	19.73		•		.
24.14-24.45	100	5/GSU-29	24.15-24.22	7	Mn ore with quartz	32.82	7.5	27.5	0.734	7
		6/GSU-29	24.22-24.35	13	Mn ore with quartz	38	7.1	22	0.473	13
		7/GSU-29	24.35-24.45	10	Mn ore with silica	15.4	•	-	•	-
24.45-24.70	100	8/GSU-29	24.45-25.58	13	mn ore with quartzite	41.35	9.7	11	0.501	13
		9/GSU-29	24.50-24.70	12	Mn ore and quartzite powder	17.23	-	-	-	-

Borehole No: GSU-30

Drilling Run	% Core Rec	Sample No.	Depth along boreholes m.	Sample length cm.	Lithology	% Mn	% Fe	% SiO ₂	% P	Cumulative length of the sample (cm)
37.80-38.25	77	Not	37.80-37.83	-	Limonite	-	-	-	-	-
		sampeld								
		1/GSU-30	37.83-37.93	8	Mn ore with quartzite and	48.15	17.8	6.3	0.56	14

			38.01-38.08	6	limonite					
					Mn ore with quartzite and limonite					
		2/GSU-30	37.93-38.01	7	Mn ore with quartzite and limonite	43.3	12.3	5.2	0.465	12
			38.08-38.15	5						
					Mn ore with quartzite limonite					
		3/GSU-30	38.15-38.25	8	Mn ore with silicon powder	26.04	19.6	19	0.714	8
38.25-38.75	66	4/GSU-30	38.25-38.29	3	Mn ore with quartzite					
			38.33-38.35	1	Mn ore with quartzite	47.82	6	3.9	0.438	5
			38.35-38.37	1	Mn ore with quartzite					
		5/GSU-30	38.29-38.33	3	Manganiferous quartzite					
			38.40-38.42	2	Manganiferous quartzite	9.6				
			38.47-38.49	2	Manganiferous quartzite					
			38.53-38.54	1	Manganiferous quartzite					
		67GSU-30	38.37-38.40	2	Mn ore	48.51	6.6	4.8	0.313	5
			38.43-38.47	3	Mn ore					
		7/GSU-30	38.49-38.50	1	mn ore with silica					
			38.50-38.53	2	mn ore with silica	46.13	6	2.59	0.767	6
			30.54-38.58	3	mn ore with silica					
		8/GSU-30	38.50-38.05	10	mn ore with silica	30.28	12	21.63	0.568	10

Borehole No: GSU-31

Drilling Run	% Core Rec	Sample No.	Depth along boreholes m.	Sample length cm.	Lithology	% Mn	% Fe	% SiO ₂	% P	Cumulative length of the sample (cm)
68.60-69.30	100	Not sampled	68.60-69.30	-	Manganiferous quartzite					
69.30-69.85	45	1/GSU-31	69.30-69.34	2	Mn ore	48.87	8.2	0.7	0.24	2
		2/GSU-31	69.36-69.51	6	Mn ore with silica powder	24.72	10.1	32.8	0.142	6
		3/GSU-31	69.64-69.70	3	Mn ore with quartzite	51.55	6.1	3.4	0.146	3
		4/GSU-31 Not sampled	69.70-69.85 69.34-69.36 69.51-69.64	7	Mn ore with silica Quartzite Quartzite	25.91	10.7	30.1	0.146	7

69.85-70.05	50	5/GSU-31	69.85-69.95	5	Mn ore	49.9	5.3	2.06	0.275	5
		6/GSU-31	69.95-70.05	5	Mn ore with silica	24.77	10.7	32.94	0.157	5
70.05-70.30	100	7/GSU-31	70.05-70.30	25	Mn ore with schist (powdery)	21.21	9.9	39.04	0.163	25
70.30-70.55		Not sampled	70.30-70.55	-	Clay	-	-	-	•	-

Borehole No: GSU-34

Drilling Run	% Core Rec	Sample No.	Depth along boreholes m.	Sample length cm.	Lithology	% Mn	% Fe	% SiO ₂	% P	Cumulative length of the sample (cm)
164.38-165.00	100	Not sampled	16438-164.53	-						
		1/GSU-34	164.53-164.54 164.56-164.59 164.69-164.74 164.76-164.77 164.80-164.81 164.81-164.86	1 3 5 1 1 5	Manganiferous quartzite Mn ore with quartzite & schist Mn ore with quartzite & schist Mn ore with quartzite & schist Mn ore with quartzite & schist	46.81	6.4	6.4	0.172	16
		2/GSU-34	164.63-164.69 164.79-164.80	6 1	Mn ore with quartzite & schist Mn ore with quartzite & schist	41.84	4.8	16.4	0.118	7
		3/GSU-34 Not sampled	164.74-164.76 164.77-164.78 164.86-164.94 164.96-164.99 164.54-164.56 164.59-164.63 164.78-164.79 164.94-164.96 164.99-165.00	2 1 8 3 - - - - -	Mn ore with quartzite Mn ore with quartzite Mn ore with quartzite calcite Mn ore with quartzite calcite Quartzite Manganiferous quartzite Manganiferous quartzite Manganiferous quartzite Manganiferous quartzite	28.4	7.3	36.1	0.325	14
165.00-165.45	100	4/GSU-34	165.00-165.02 165.10-165.16	2 6	Mn ore with quartzite Mn ore with quartzite	32.8	10	17.1	0.458	13

			165.36-165.41	5	Mn ore with quartzite					
		5/GSU-34	165.02-165.12	8	Mn ore with quartzite+calcite					
			165.16-165.19	3	Mn ore with quartzite+calcite	48.1	3.9	8.4	0.1	15
			165.41-165.45	4	Mn ore with quartzite + calcite					
		6/GSU-34	165.19-165.36	17	Mn ore with quartz, quartzite and calcite	38.3	7.1	14.8	0.216	17
165.45-166.10	100	7/GSU-34	165.48-165.55	7	Manganiferous quartzite	14.6	-	-	-	-
		8/GSU-34	165.55-165.75	20	Mn ore with quartzite calcite and quartz	41.1	5	19.2	0.189	20
		9/GSU-34	165.45-165.48	3	Mn ore and calcite and quartz	49.4	5.3	0.9	0.162	15
			165.78-165.84	6	Mn ore and calcite and quartz					
			165.90-165.93	3	Mn ore and calcite and quartz	43.4	4.2	15.9	0.082	9
			166.07-166.10	3	Mn ore and calcite and quartz					
		10/GSU-34	165.75-165.78	3,6	Mn ore + quartz and calcite do					
			165.84-165.90							
		11/GSU-34	165.93-166.07	14	Mn ore with quartzite calcite	33.63	5.10	17.10	0.18	14.00
166.10-166.40		Not sampled	166.10-166.11	-	Mangniferous quartzite					
		12/GSU-34	166.11-166.24	12	Mn ore with quartzite calcite	49.60	5.00	10.30	0.12	12.00
		13/GSU-34	166.24-166.40	15	Mn ore with quartzite calcite	47.70	5.20	9.50	0.25	15.00
166.40-166.65	100	14/GSU-34	166.40-166.54	14	Calcite quartzite with Mn ore	31.70	3.60	20.90	0.21	14.00
		15/GSU-34	166.54-166.65	11	Mn ore with calcite and quartzite	47.91	5.30	8.00	0.11	11.00
166.65-166.80	100	16/GSU-34	166.65-166.66	1	Mn ore with quartzite	16.80	-	-	-	-
			166.70-166.72	2	Mn ore with quartzite					
			166.73-166.74	1	Mn ore with quartzite					
			166.79-166.80	1	Mn ore with quartzite					
		17/GSU-34	166.66-166.70	4	Mn ore with quartzite	34.93	8.60	10.70	0.45	10.00
			166.72-166.73	1	Calcite					
			166.74-166.79	5	Mn ore					
166.80-167.10	73	18/GSU-34	166.80-167.10	22	Mangniferous quartzite and schist	11.50	-	-	-	
167.10-169.80	100	Not sampled	167.10-169.80	-	Quartzite with Jasper and schist					

Borehole No: UKH-3

Sl. No.	From	To	True Width of horizon	Description of samples	Sp. Gravity	Mn (%)	Fe(%)	SiO ₂ (%)	P(%)	Remarks
UKH-3/M1	183.62	183.88	0.13	Mn ore	4.08	40.32	6.14	14.48	0.73	UKH-3/M1
M2	183.88	184.30	0.16	Mn ore	4.14	40.99	7.81	13.04	0.48	M2
M3	184.30	184.56	0.13	Mn ore	4.15	41.49	11.17	10.12	0.69	M3
M4	184.56	184.79	0.15	Mn ore	4.30	50.16	7.81	8.49	0.28	M4
M5	184.56	184.79	0.15	Mn ore	5.30	50.66	7.81	7.19	0.37	M5
M5	184.79	185.25	0.16	Mn ore	4.33	47.66	6.70	9.50	0.33	M5
M6	185.25	185.43	0.16	Mn ore	4.03	50.16	5.58	10.04	0.26	M6
M7	185.43	186.00	0.12	Mn ore	3.75	30.82	11.72	12.31	0.61	M7
			1.01	Total Mn ore	4.11	43.63	7.98	11.06	0.48	
Q1	183.40	186.30	1.90	Mn ore +Q	2.74	19.98	7.26	29.57	0.63	Q1

Borehole No: UKH-4

Sl. No.	From	To	True Width of horizon	Description of samples	Sp. Gravity	Mn (%)	Fe(%)	SiO ₂ (%)	P(%)	Remarks
UKH-4/M1	189.70	190.05	0.15	Mn ore+QV	4.11	35.80	7.54	14.70	0.92	UKH-4/M1
M2	190.05	190.45	0.15	Mn ore+CC3 vein	4.03	35.70	7.26	9.56	0.74	M2
M3	190.45	190.68	0.14	Mn ore+QCO3 vein	4.21	36.74	6.70	10.10	0.58	M3
M4	190.68	190.83	0.13	Mn ore+brecciatedCO3	4.27	45.85	5.86	10.72	0.45	M4
M7 Check Sample	190.68	190.83	0.13	Mn ore+brecciatedCO3	4.27	36.82	7.26	11.00	0.32	M7 Check Sample
M5	190.83	191.08	0.13	Mn ore+QCO3 vein	4.30	38.62	6.98	8.26	0.54	M5
M6	191.08	191.48	0.15	Mn ore+Q vein	4.21	30.14	6.00	27.84	0.65	M6
			0.85	Total Mn ore	4.19	36.91	6.74	13.75	0.66	
Q1	189.70	191.70	1.15	MnQ	2.76	15.07	4.32	36.36	0.96	Q1

Borehole No: UKH-7

Sl. No.	From	To	True Width of horizon	Description of samples	Sp. Gravity	Mn (%)	Fe(%)	SiO ₂ (%)	P(%)	Remarks
UKH-7/M1	208.30	209.05	0.30	Mn ore+QCO ₃ vein	4.24	33.91	7.54	14.36	0.62	UKH-7/M1
M2	208.30	209.05	0.30	Mn ore+QCO ₃ vein	-do-	41.16	6.70	14.94	0.37	M2
Q1	208.30	209.05	0.45	MnQ	2.81	13.50	3.91	42.92	0.87	Q1

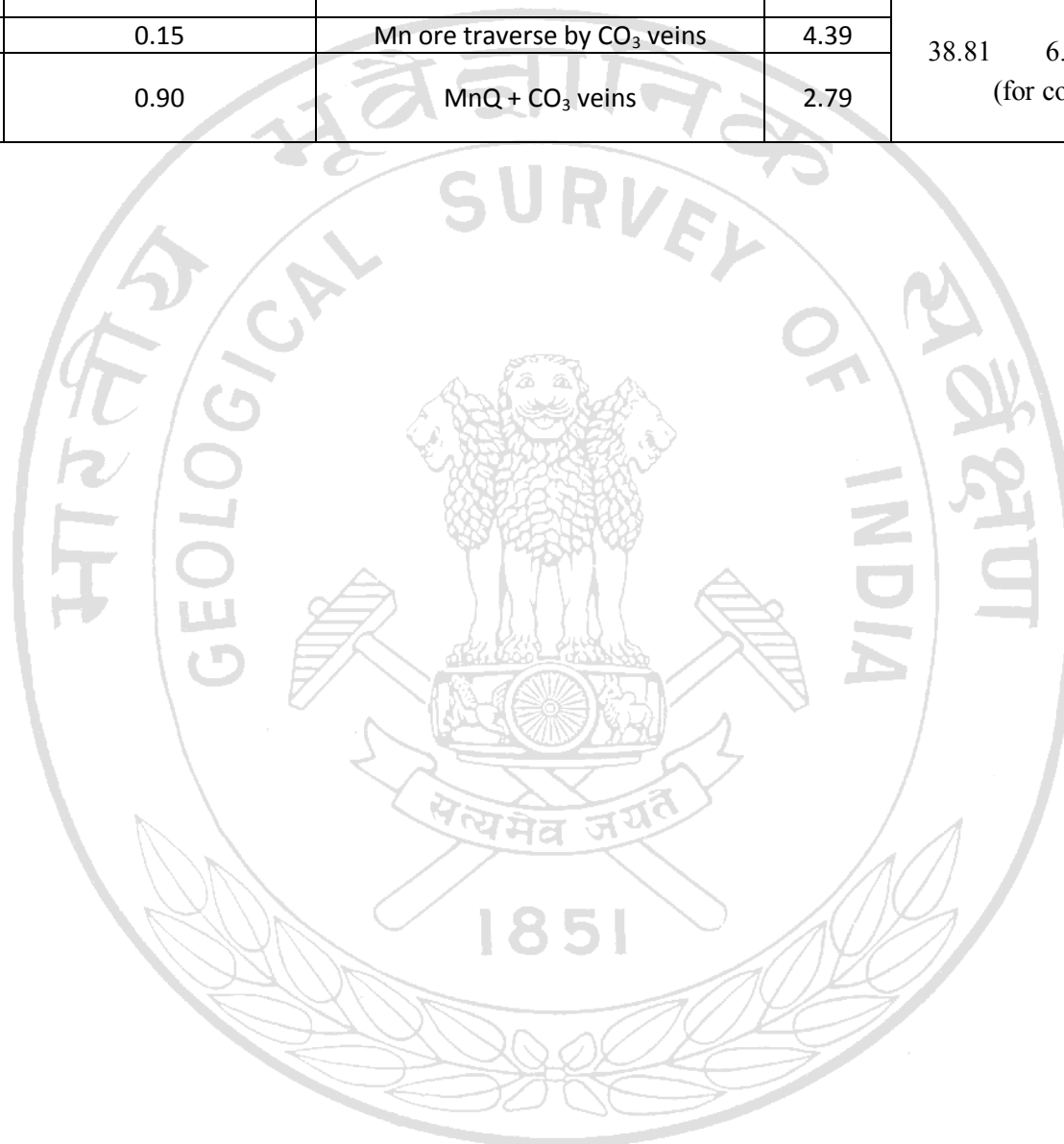
Borehole No: UKH-14

Sl. No.	From	To	True Width of horizon	Description of samples	Sp. Gravity	Mn (%)	Fe(%)	SiO ₂ (%)	P(%)	Remarks
UKH-14/M1	221.80	222.60	0.08	Mn ore & MnQ	3.97	37.82	7.54	8.99	0.64	(for composite sample)
M2	222.70	222.90	0.15	Mn Q+CQ ₃ vein	3.97					
M3	223.40	224.15	0.15	Banded Mn ore	4.39					
M4	224.15	224.35	0.15	Banded Mn ore	4.38					
M5	224.35	224.45	0.09	Banded Mn ore	4.48					
M6	224.80	225.03	0.11	Banded Mn ore CQ ₃	4.12					
M7	225.03	225.40	0.11	Banded Mn ore CO ₃	3.9					
M8	225.40	225.55	0.15	Banded Mn ore CO ₃	4.15					
M9	225.55	226.35	0.15	Banded Mn ore QCO ₃	3.8					
M10	226.35	226.55	0.15	Mn ore QCO ₃	4.13					
M11	226.55	227.00	0.15	Mn ore CO ₃	4.03					
M12	227.00	227.20	0.10	Mn ore CO ₃	3.95					
Q1	221.80	227.20	3.86	MnQ with Mn ore & CO ₃	2.95					

Borehole No: UKH-16

Sl. No.	From	To	True Width of horizon	Description of samples	Sp. Gravity	Mn (%)	Fe(%)	SiO ₂ (%)	P(%)	Remarks
UKH-	220.50	220.65	0.15	Mn ore traverse by CO ₃ veins	4.27					

16/M1						38.81 6.98 14.74 0.75 (for composite sample)	
M2	220.65	220.80	0.15	Mn ore traverse by CO ₃ veins	4.39		
Q1	219.70	220.90	0.90	MnQ + CO ₃ veins	2.79		



ANNEXURE-IX

GROSS RESOURCE ESTIMATION (Ore+waste) based on previous works IN UKWA MANGANESE DEPOSIT-WESTERN BLOCK

(Source- Choudhary et al; 1979; Khan et al;1993-96)

The resource presented below in tabular format (table-1 and 2) is part of western block. Here effective sp gravity used is 3.92 based on average value of Mn of western block (40.34% as reported by previous workers for western block of Ukwa).

Table for possible category is not given in previous reports, but resource estimated under possible category is given which is 0.66 mt.

Table 1: Probable A category for partial area of western block used in current FSP 2016-17.

Here strike influence has been changed for boreholes in section line W-1 and W-4 (see plate-V).

Section Line	B.H. No.	Strike length (m)	Upper level (RL) m	Level of Intersection (RL) m	Lower Level (RL) m	Difference (m)	Dip of Ore horizon	Correction factor	Dip length (m)	Thickn ess (m)	Volume (m3)
W-1	GSU-30	142.5	600	562	546	54	20	2.92	157.68	1.46	32805.324
	GSU-31	142.5	546	530	467	79	15	3.86	304.94	1.4	60835.53
	UKH-7	142.5	567	405	405	62	25	2.37	146.94	0.75	15704.212
W-2	GSU-29	230	600	578	544	56	15	3.86	216.16	0.87	43353.6
	GUG-2	230	544	511	460	84	20	2.92	245.28	1.6	90263
	UKH-4	230	460	410	410	50	31	1.92	96	1.7	37536
W-3	GSU-26	255	600	554	530	70	25	2.37	165.9	1.13	47804.1
	GSU-27	255	530	406	468	62	25	2.37	146.94	2.76	103416
	UKH-3	255	468	430	412	56	34	1.79	100.24	2.9	74127.5
	UKH-14	255	412	395	395	17	34	1.79	30.43	5.4	41902.1
W-4	GSU-21	183.75	600	539	509	91	30	2	182	2.25	75245.625
	GSU-24	183.75	509	479	458	51	30	2	102	1.6	29988

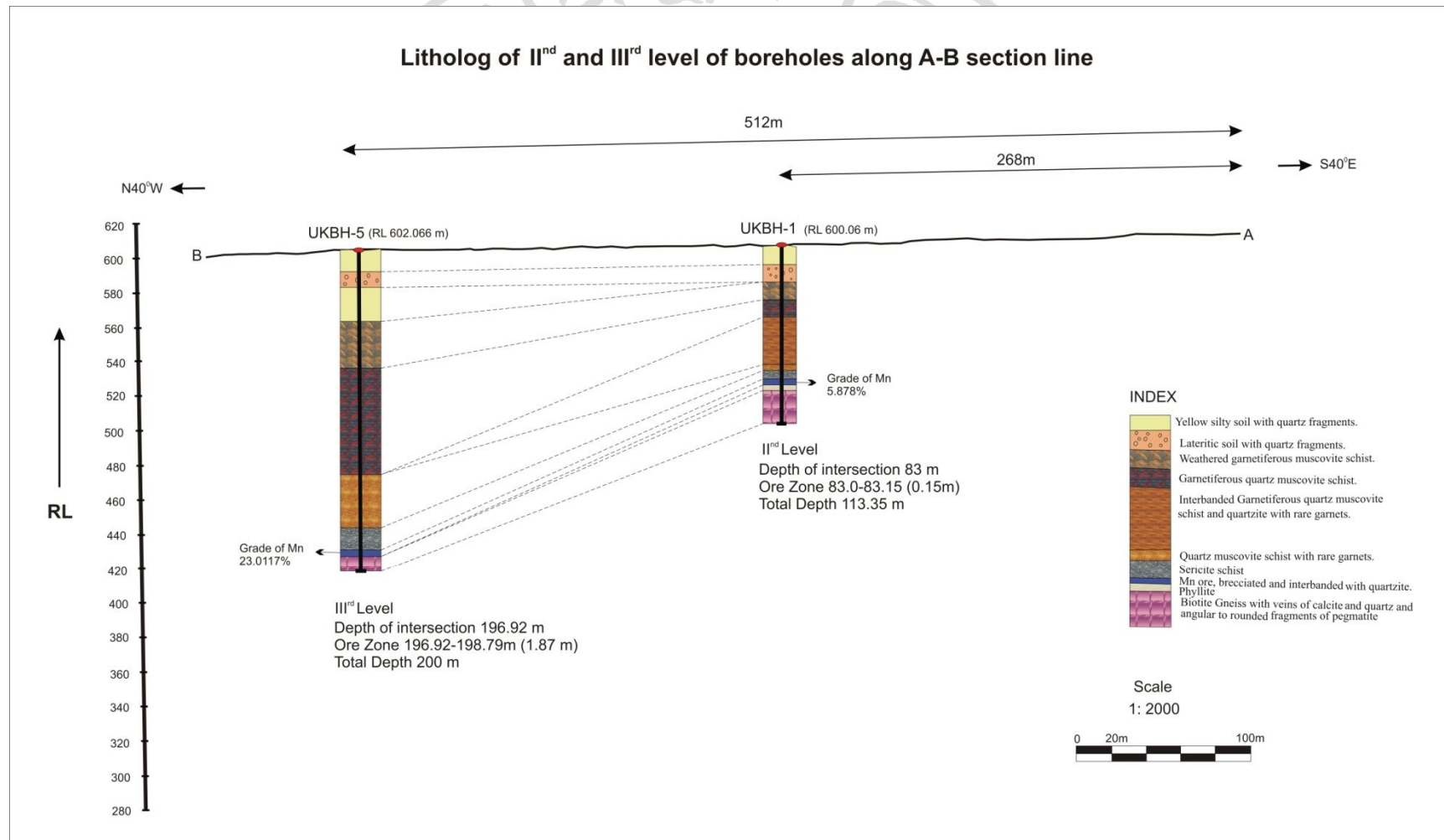
	GSU-34	183.75	458	437	416	42	30	2	84	2.18	33648.3
	UKH-16	183.75	416	395	395	21	30	2	42	1.2	9261
										Total=	695890.291
Total Gross Resource thus estimated is 695890.291*3.92= 2727889.941 or 2.727 million tonne											

Table 2: Probable category for partial area of western block used in current FSP 2016-17.

Section Line	B.H. No.	Strike length (m)	Upper level (RL) m	Lower Level (RL) m	Difference (m)	Dip of Ore horizon	Correction factor	Dip length (m)	Thickn ess (m)	Volume (m3)
W-I	UKH-7	142.5	405	374	31	25	3.37	73.47	0.75	7852.106
W-2	UKH-4	230	410	385	25	31	1.92	48	1.7	18768
W-3	UKH-14	255	395	386	9	34	1.73	16.11	5.4	22183.47
W-4	UKH-16	183.75	395	384	11	30	2	22	1.2	4851
									Total=	53654.576
Total Gross Resource thus estimated is 53654.576*3.92= 210325.93 or 0.21 million tonne										

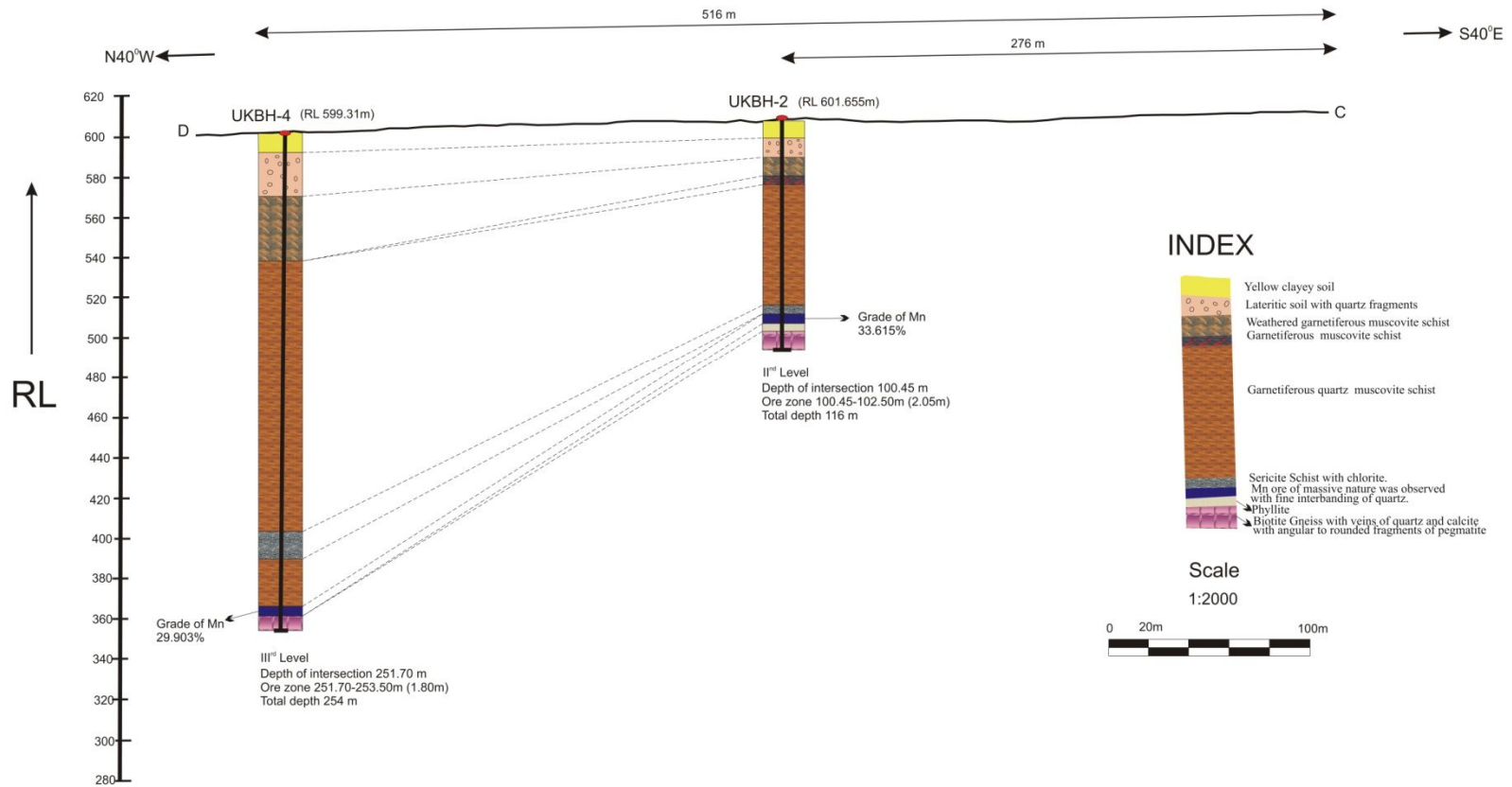
So total gross resource based on previous works for current block is 2.727+ 0.66 + 0.21 = 3.597 mt at 40.34%.

Graphical representation of lithologies of current boreholes along section lines



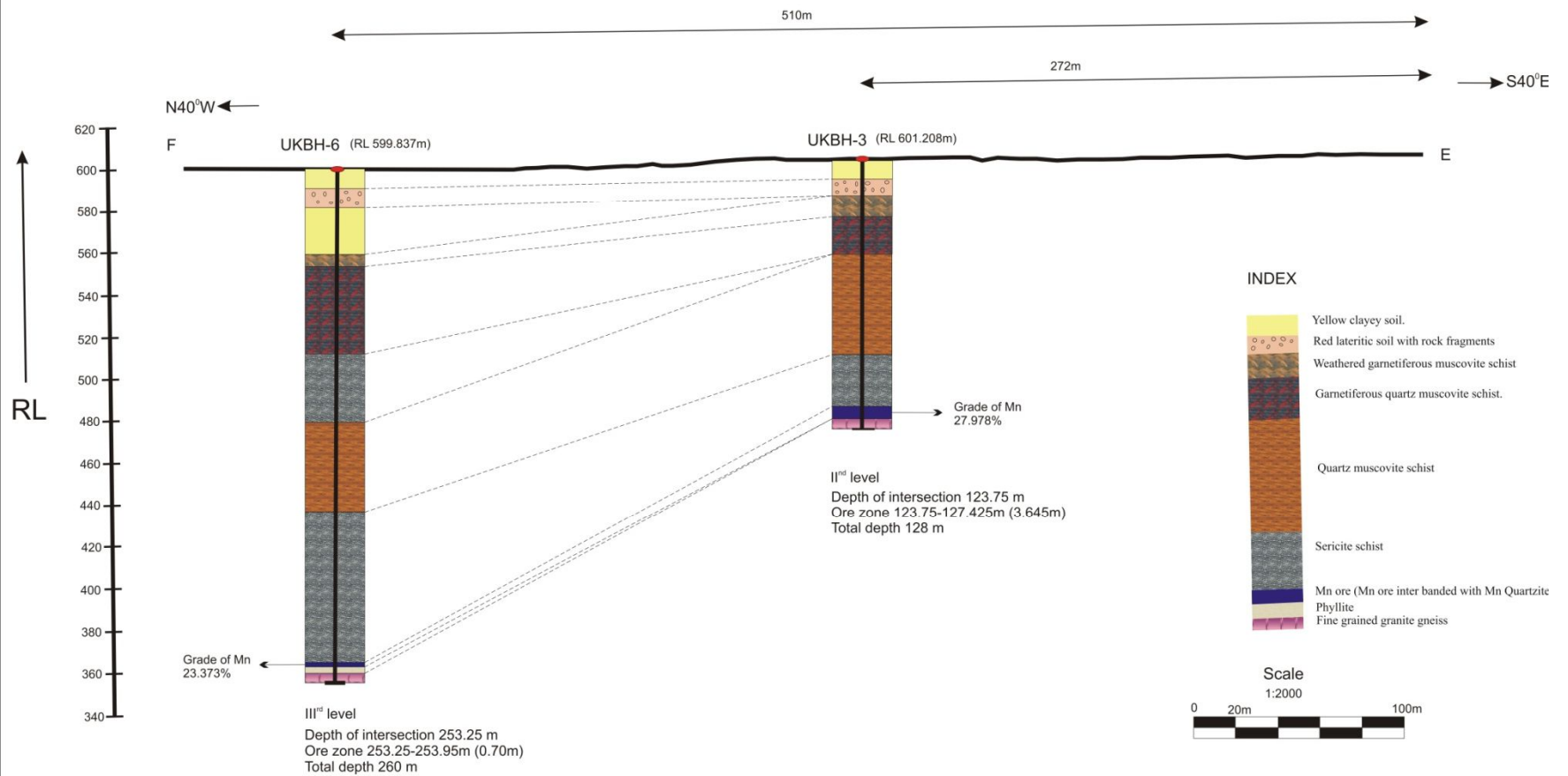
Litholog along AB section line

Litholog of IInd and IIIrd level boreholes along CD section line



Litholog along CD section line

Litholog of IInd and IIIrd level boreholes along E-F section line



Litholog along EF section line